In[164]:=
```
QMRITools`$Verbose = False;
<< QMRITools`
```

In[166]:=
```
QMRITools`$InstalledVersion
Grid[Transpose@{{"Packages", "Functions", "Options"},
   Length@Flatten[QMRIToolsFunctions[][[All, #]]] & /@ {1, 2, 3}}, Alignment → Left]
```

Out[166]=
```
5.1.1
```

Out[167]=
```
Packages  28
Functions 577
Options   516
```

In[168]:=
```
Column@QMRIToolsPackages[]
```

Out[168]=
```
AmaresTools
CardiacTools
CoilTools
DenoiseTools
DixonTools
ElastixTools
FasciculationTools
GeneralTools
GradientTools
IVIMTools
JcouplingTools
LoggingTools
MaskingTools
MuscleBidsTools
NeuralNetworkTools
NiftiTools
PlottingTools
ProcessingTools
ReconstructionTools
RelaxometryTools
ScientificColorData
SegmentationTools
ShapeTools
SimulationTools
SpectroTools
TaggingTools
TensorTools
TractographyTools
```

In[169]:=
```
QMRIToolsFunctions["All", 4]
```

Functions

ActivationLayer
ADCCalc
AddLossLayer
AddNoise
AddScientificColors
AddToLog
AmaresBasis
AmaresFitFidI
AmaresInitialValues
AmaresSignalCalc
AmaresSignalJacobianCalc
AnalyzeActivations
AnalyzeNetworkFeatures
AngleCalc
AngleMap
AnisoFilterData
AnisoFilterTensor
AnnalyzeTagging
ApodizeEcho
ApodizeFid
ApodizePadEcho
ApodizePadFid
ApodizePadSpectra
ApodizeSpectra
ApplyCrop
ApplySegmentationNetwork
ApplyShapeModel
AugmentImageData
AugmentTrainingData
AutoCropData
AverageKspace
B1MapCalc
B1Shimming
BidsDcmToNii
BlochSeries
Bmatrix
BmatrixCalc
BmatrixConv
BmatrixInv
BmatrixRot
BmatrixToggle
BSplineCurveFit
BullseyePlot
BVector
CalculateDisplacementParameters
CalculateGFactor
CalculateMoments
CalculateWallMap
CalibrateEPGT2Fit
CardiacCoordinateSystem
CardiacSegment
CardiacSegmentGUI
CELossFunction
ExportSparSdat
ExportTCK
ExportTracts
ExpTensor
ExtractColorData
ExtractDemoData
ExtractNiiFiles
FACalc
FiberTractography
FileSelect
FilterTractLength
FilterTracts
FinalGrads
FindActivations
FindCoilPosition
FindCrop
FindInPhaseEchos
FindMaxDimensions
FindMiddle
FindOrder
FindOutliers
FindPatchDim
FindTensorPermutation
FitData
FitGradientMap
FitRPBMDictionary
FitRPBMFunction
FitShapeModel
FitTracts
FixDixonFlips
FlipGradientOrientation
FlipTensorOrientation
FocalLossLayer
FourierKspace2D
FourierKspace3D
FourierKspaceCSI
FourierRescaleData
FourierShift
FourierShiftData
FourierShifted
FreezeEncoderLayers
FromFullAmaresMatrix
FullGrad
GenerateAmps
GenerateBidsFileName
GenerateBidsFolderName
GenerateBidsName
GenerateGradients
GenerateGradientsGUI
GenerateRotationFrames
GESignal
GetAssetLocation
GetCommonSegmentation
MakeClassImage
MakeCoilLayout
MakeCoordinates
MakeDistanceMap
MakeECVBloodMask
MakeFunctionGraph
MakeGradientDerivatives
MakeGradientMaps
MakeHammingFilter
MakeIntFunction
MakeLineImage
MakeMaskImage
MakeMuscleTemplate
MakeNiiOrientationQ
MakeNiiOrientationS
MakeNode
MakeNoisePlots
MakeRegionMesh
MakeSense
MakeShapeModel
MakeSliceImages
MakeSpinSystem
MakeTemplatePlot
MakeTrainData
MakeUnet
MakeWeightMask
ManageCheckFiles
Mask
MaskData
MaskFromDistanceMap
MaskHelix
MaskSegmentation
MaskToDistanceMap
MaskToLines
MaskVolume
MBCount
MeanBvalueSignal
MeanNoZero
MeanRange
MeanSignal
MeanStd
MeanStdRange
MeanType
MedCouple
MedFilter
MedianNoZero
MemoryUsage
MergeConfig
MergeJSON
MergeSegmentations
MeshGridPlot
MinMaxRange
MonitorFunction
RegisterTensorData
RemoveIsoImages
RemoveMaskOverlaps
ReplaceSegmentations
ResampleTracts
RescaleData
RescaleSegmentation
RescaleTracts
ResetLink3DGraphic
ResetLog
ResidualCalc
ReverseCrop
ReverseDimensions
RMSNoZero
RotateData
RotateDimensionsLeft
RotateDimensionsRight
RotateTensor
RotationMatrixToQuaternion
RotationMatrixToQuaternionVector
RPBMFunction
RunMuscleMap
SagittalTranspose
SaveImage
ScaleToVolume
SeedDensityMap
SegmentationCrossSection
SegmentationVolume
SegmentData
SegmentDataGUI
SegmentLinesToMask
SegmentMask
SegmentsPerSlice
SegmentTracts
SelectActivations
SelectBidsFolders
SelectBidsSessions
SelectBidsSubjects
SelectBvalueData
SelectMaskComponents
SelectSegmentations
SelectSubjects
SequencePulseAcquire
SequenceSpaceEcho
SequenceSpinEcho
SequenceSteam
SequenceTSE
SetDemoDirectory
SetLogFile
ShiftedFourier
ShiftedInverseFourier
ShiftPulseProfile
ShiftSpectra

Options

AcquisitionMethod
ActivationBackground
ActivationIterations
ActivationOutput
ActivationSize
ActivationThreshold
ActivationType
AffineDirections
AllowSelfDependencies
AmaresUseB1
AnisoFilterSteps
AnisoIterations
AnisoKappa
AnisoKernel
AnisoStepTime
AnisoWeightType
ApodizationFunction
AspectRatio
Augment2D
AugmentData
Axes
AxesLabel
AxesMethod
B1EqualPower
B1FilterData
B1Masking
B1MaxPower
B1Output
B1Scaling
B1ShimMethod
BackgroundValue
Bandwidth
BasisSequence
BatchSize
BidsIncludeSession
BidsOutputImages
BidsTractographyMethod
BlockType
BloodMaskRange
Boxed
BsplineDirections
BsplineSpacing
BullPlotMethod
CatenateMethod
CenterFrequency
CenterRange
CenterVoxel
Channels
Classes
CleanUpSegmentations
ClippingStyle
DixonTolerance
DownsampleSchedule
DropoutRate
DropSamples
DropSlices
EchoShiftData
EPGCalibrate
EPGFatShift
EPGFitFat
EPGFitPoints
EPGMethod
EPGMethodCal
EPGOutput
EPGRefocussingPhase
EPGRelaxPars
EPGSmoothB1
FatFieldStrength
FeatureSchedule
FiberAngle
FiberLengthRange
FieldStrength
FileType
Filling
FilterInternal
FilterMaps
FilterShape
FilterSize
FilterType
FindTransform
FitFunction
FitKernel
FitOrder
FitOutput
FittingOrder
FitTractSegments
FlipAxes
FlipBvec
FlipGrad
Frame
FrameStyle
FreezeEncoderDepth
FullOutput
FullSphere
GetMaskOnly
GetMaskOutput
GOutput
GradientStepSize
GradType
GRegularization
GridLines
GridLineSpacing
MeshPointSize
Method
MethodReg
MethodRegA
Monitor
MonitorCalc
MonitorInterval
MonitorTagging
MonitorUnwrap
MotionCorrectSets
MultiChannel
MuscleMapLabels
MuscleMapPath
MuscleMapPythonEnvironment
NetworkArchitecture
NetworkDepth
NetworkOutput
NiiDataType
NiiLegacy
NiiMethod
NiiOffset
NiiScaling
NiiSliceCode
NoisePrewhiten
NoiseSize
NoiseType
NormalizationType
NormalizeDensity
NormalizeIVIM
NormalizeMethod
NormalizeOutputSpectra
NormalizeOverlap
NormalizeSets
NormalizeSignal
NumberSamples
NumberSamplesA
OrderSpan
OutlierCheckNormality
OutlierIncludeZero
OutlierIterations
OutlierMethod
OutlierOutput
OutlierRange
OutputCalibration
OutputCheckImage
OutputCoilSurface
OutputForm
OutputImage
OutputLabels
OutputPlot
OutputSense
Resolutions
ResolutionsA
RestartLearningCycle
ReverseData
ReverseDirection
ReversePoints
ReverseSets
RobustFit
RobustFitParameters
RotateGradients
RotationCorrect
RoundLength
RowSize
RPBMDictionarySize
RPBMRangeTau
RPBMRangeZeta
Runs
SaveLogFile
Scaling
SegmentationDimension
SegmentationMethod
SegmentationsPerSlice
SenseSmoothing
SenseWeight
SettingSchedule
ShowMetric
ShowOutliers
ShowPlot
SimNucleus
SizeMethod
SliceRange
SliceRangeSamples
SmartMaskOutput
SmartMethod
SmoothHelix
SmoothIterations
SmoothSNR
SortVecs
SparID
SparName
SparOrientation
SpectraBandwidth
SpectraFieldStrength
SpectraNucleus
SpectraPpmShift
SpectraSamples
SpectraSpacing
SphereColor
SphereSize
SplineDegree
SplineKnotsNumber

In[170]:=
```
QMRIToolsFunctions[4]
```

AmaresTools

Functions

AmaresBasis	AmaresSignalCalc	FromFullAmaresMatrix	$AmaresB1Regularization
AmaresFitFidI	AmaresSignalJacobianCalc	ShowParameterMatrix	$AmaresMatrixDefaults
AmaresInitialValues	EditAmaresMatrix	ToFullAmaresMatrix	$AmaresT1Values

Options

AmaresUseB1

CardiacTools

Functions

BullseyePlot	ECVCalc	MakeECVBloodMask	RadialSample
CalculateWallMap	ExcludeSlices	MakeLineImage	SegmentLinesToMask
CardiacCoordinateSystem	GetMaskSegmentPoints	MakeMaskImage	SegmentsPerSlice
CardiacSegment	GetSegmentLines	MaskHelix	TransmuralPlot
CardiacSegmentGUI	GetSegmentSlices	MaskToLines	
CentralAxes	HelixAngleCalc	PlotSegmentMask	
CreateHeart	LinesToSegmentIndex	PlotSegments	

Options

AxesMethod	GridLineSpacing	PlotLabel	ShowOutliers
BackgroundValue	GroupPerSegment	PlotRange	ShowPlot
BloodMaskRange	ImageSize	PlotStyle	SmoothHelix
BullPlotMethod	LCMMethod	RadialSamples	StartPoints
ColorFunction	MakeSegmentPlots	ReverseDirection	StartSlices
CutOffMethod	MaskWallMap	ReversePoints	TextNumberForm
DistanceMeasure	Method	RowSize	TextOffset
DropSamples	OutputCheckImage	SegmentationMethod	TextSize

CoilTools

Functions

CoilSNRCalc	LoadCoilSetup	MakeCoilLayout	MakeWeightMask
FindCoilPosition	LoadCoilTarget	MakeNoisePlots	

Options

CoilArrayPlot	ColorFunction	OutputCoilSurface
CoilSurfaceVoxelSize	ImageSize	PlotRange

DenoiseTools

Functions

AnisoFilterData	DenoiseCSIdata	NNThreshold	WeightMapCalc
AnisoFilterTensor	DenoiseDynamicSpectraData	P2SDenoise	$debugDenoise
DeNoise	HarmonicDenoiseTensor	PCADeNoise	

Options

AnisoFilterSteps	DeNoiseMonitor	PCAClipping	RangeFA
AnisoIterations	GradientStepSize	PCAComplex	RangeMD
AnisoKappa	Iterations	PCAKernel	TensorFlips
AnisoKernel	MaxIterations	PCANoiseSigma	TensorPermutations
AnisoStepTime	Method	PCAOutput	Threshold
AnisoWeightType	Monitor	PCATolerance	Tolerance
DeNoiseIterations	MonitorCalc	PCAWeighting	
DeNoiseKernel	Parallelize	RadialBasisKernel	

DixonTools

Functions

DixonPhase	FixDixonFlips	Unwrap	Wrap
DixonReconstruct	GenerateAmps	UnwrapDCT	
DixonToPercent	OptimizeDixonEcho	UnwrapList	
FindInPhaseEchos	SimulateDixonSignal	UnwrapSplit	

Options

DixonAmplitudes	DixonFilterOutput	DixonMaskThreshold	MonitorCalc
DixonBipolar	DixonFilterSize	DixonNucleus	MonitorUnwrap
DixonClipFraction	DixonFilterType	DixonPhases	Parallelize
DixonConstrainPhase	DixonFitPhase	DixonPrecessions	PhaseEchos
DixonCorrectT1	DixonFixT2	DixonRelaxivity	UnwrapDimension
DixonFieldStrength	DixonFrequencies	DixonTolerance	UnwrapMask
DixonFilterInput	DixonIterations	MaxIterations	UnwrapThresh

ElastixTools

Functions

ClearQMRIToolsTemp
ReadTransformParameters
RegisterCardiacData
RegisterData
RegisterDataSplit
RegisterDataTransform
RegisterDataTransformSplit
RegisterDiffusionData
RegisterDiffusionDataSplit
RegisterTensorData
TransformData
$debugElastix
$lastElastixTemp

Options

AffineDirections
BsplineDirections
BsplineSpacing
DeleteTempDirectory
FindTransform
HistogramBins
HistogramBinsA
ImportDeformation
InterpolationOrderReg
InterpolationOrderRegA
Iterations
IterationsA
Method
MethodReg
MethodRegA
NumberSamples
NumberSamplesA
OutputImage
OutputTransformation
PCAComponents
PrintTempDirectory
RegistrationTarget
Resolutions
ResolutionsA
ShowMetric
SplitMethod
TempDirectory
TransformMethod
UseGPU

FasciculationTools

Functions

AnalyzeActivations
EvaluateActivation
FindActivations
NormalizeFasciculationData
SelectActivations

Options

ActivationBackground
ActivationIterations
ActivationOutput
ActivationSize
ActivationThreshold
IgnoreSlices
MaskDilation
SizeMethod
ThresholdMethod

GeneralTools

Functions

ApplyCrop
AutoCropData
BSplineCurveFit
ClearTemporaryVariables
CompilableFunctions
ConvertExtension
CropData
CutData
DataToVector
DateName
DecomposeAffineMatrix
DecomposeScaleMatrix
DivideNoZero
DynamicPartition
EmptyDirectoryQ
ExpNoZero
ExtractDemoData
FileSelect
FindCrop
FindMaxDimensions
FindMiddle
FitGradientMap
GetAssetLocation
GridData
GridData3D
GyromagneticRatio
LapFilter
LightDarkV
LLeastSquares
LogNoZero
MADNoZero
MakeCoordinates
MakeFunctionGraph
MakeIntFunction
MBCount
MeanNoZero
MedFilter
MedianNoZero
MemoryUsage
MonitorFunction
NiiFileExistQ
NNLeastSquares
OpenDemonstrationNotebook
OpenQMRIToolsLocation
PadToDimensions
ParseCommandLine
QMRIToolsFuncPrint
QMRIToolsFunctions
QMRIToolsPackages
QuaternionToRotationMatrix
QuaternionVectorToRotationMatrix
RandomString
RescaleData
ReverseCrop
ReverseDimensions
RMSNoZero
RotateDimensionsLeft
RotateDimensionsRight
RotationMatrixToQuaternion
RotationMatrixToQuaternionVector
SaveImage
SetDemoDirectory
SignNoZero
Squeeze
StandardDeviationNoZero
StdFilter
StichData
StringPadInteger
SumOfSquares
TensMat
TensVec
VectorToData

Options

AllowSelfDependencies
CenterRange
CenterVoxel
CropAlways
CropInit
CropOutput
CropPadding
FileType
FilterInternal
ImageResolution
ImageSize
InterpolationOrder
LabelPlacement
Monitor
OutputWeights
Padding
PadDirection
PadValue
SplineDegree
SplineKnotsNumber
SplineRegularization
WindowTitle

GradientTools

Functions

Bmatrix
BmatrixCalc
BmatrixConv
BmatrixInv
BmatrixRot
BmatrixToggle
BVector
CalculateMoments
ConditionNumberCalc
ConvertGrads
CorrectBmatrix
CorrectGradients
EnergyCalc
FinalGrads
FindOrder
FullGrad
GenerateGradients
GenerateGradientsGUI
GetGradientScanOrder
GetSliceNormal
GetSliceNormalDir
GradBmatrix
GradientCoilTensor
GradSeq
ImportGradObj
MakeGradientDerivatives
MakeGradientMaps
OverPlusCalc
SelectBvalueData
UniqueBvalPosition

Options

ConditionCalc
FlipAxes
FlipGrad
FullSphere
GradType
Method
MethodReg
OrderSpan
OutputPlot
OutputType
PhaseEncoding
Runs
Steps
StepSizeI
SwitchAxes
UnitMulti
UseGrad
VisualOpt

IVIMTools

Functions

IVIMCalc
IVIMCorrectData
IVIMFunction
IVIMResiduals
IVIMTensFit
MeanBvalueSignal

Options

FilterMaps
FilterSize
FilterType
IVIMComponents
IVIMConstrained
IVIMConstrains
IVIMFixed
Method
Parallelize

JcouplingTools

Functions

GetSpinSystem
MakeSpinSystem
SequencePulseAcquire
SequenceSpaceEcho
SequenceSpinEcho
SequenceSteam
SequenceTSE
SimAddPhase
SimEvolve
SimHamiltonian
SimReadout
SimRotate
SimSignal
SimSpoil
SysTable

Options

CenterFrequency
FieldStrength
Linewidth
LinewidthShape
ReadoutBandwidth
ReadoutMethod
ReadoutOutput
ReadoutPhase
ReadoutSamples
SimNucleus

LoggingTools

Functions

AddToLog
CheckFile
DirectoryTree
ExportDirectoryTree
ExportLog
ImportLog
MakeCheckFile
PrintDirectoryTree
ResetLog
SetLogFile
ShowLog

Options

SaveLogFile

MaskingTools

Functions

DilateMask
GetCommonSegmentation
GetSegmentationLabels
HomogenizeData
JoinSegmentations
Mask
MaskData
MaskSegmentation
MaskVolume
MergeSegmentations
NormalizeData
NormalizeMeanData
RemoveMaskOverlaps
ReplaceSegmentations
RescaleSegmentation
SegmentationCrossSection
SegmentationVolume
SegmentMask
SelectMaskComponents
SelectSegmentations
SmoothMask
SmoothSegmentation
SplitSegmentations

Options

FitOrder
MaskClosing
MaskComponents
MaskDilation
MaskFiltKernel
MaskSmoothing
Method
NormalizeMethod
SmoothIterations

MuscleBidsTools

Functions

BidsDcmToNii
CheckBidsOutput
CheckConfigLabels
CheckDataDescription
GenerateBidsFileName
GenerateBidsFolderName
GenerateBidsName
GetConfig
GetJSONPosition
GetProtocolNames
ImportJSON
ManageCheckFiles
MergeConfig
MergeJSON
MuscleBidsAnalysis
MuscleBidsCheck
MuscleBidsConvert
MuscleBidsGUI
MuscleBidsMerge
MuscleBidsProcess
MuscleBidsReport
MuscleBidsSegment
MuscleBidsTractography
PartitionBidsFolderName
PartitionBidsName
SelectBidsFolders
SelectBidsSessions
SelectBidsSubjects
SelectSubjects
ViewConfig
ViewProtocolNames
$debugBids

Options

BidsIncludeSession
BidsOutputImages
BidsTractographyMethod
DeleteAfterConversion
ProcessSubjects
VersionCheck

NeuralNetworkTools

Functions

ActivationLayer
AddLossLayer
AnalyzeNetworkFeatures
CELossFunction
CELossLayer
ChangeNetDimensions
ClassConfidence
ClassDecoder
ClassEncoder
DiceLossLayer
FocalLossLayer
GetNetNodes
JaccardLossLayer
MakeClassifyImage
MakeClassifyNetwork
MakeNode
MakeUnet
NetDimensions
NetSummary
OverlapLossFunction
TopKLossLayer
TverskyLossLayer
$debugUnet

Options

Channels
Classes
Dimensions
ActivationType
BlockType
CatenateMethod
DownsampleSchedule
DropoutRate
FeatureSchedule
ImageSize
Monitor
NetworkArchitecture
NetworkDepth
NormalizationType
RescaleMethod
SettingSchedule

NiftiTools

Functions

CompressNiiFiles
CorrectNiiOrientation
DcmToNii
ExportBmat
ExportBval
ExportBvalvec
ExportBvec
ExportNii
ExtractNiiFiles
GetNiiOrientation
ImportBmat
ImportBval
ImportBvalvec
ImportBvec
ImportExploreDTITens
ImportNii
ImportNiiDiff
ImportNiiDix
ImportNiiT1
ImportNiiT2
MakeNiiOrientationQ
MakeNiiOrientationS

Options

CompressNii
DecompressDcm
DeleteOutputFolder
FlipBvec
MergeEchos
Method
MonitorCalc
NiiDataType
NiiLegacy
NiiMethod
NiiOffset
NiiScaling
NiiSliceCode
PositiveZ
RotateGradients
UseSubFolders
UseVersion

PlottingTools

Functions

ColorFAPlot
GenerateRotationFrames
GetSliceData
GetSlicePositions
GradientPlot
LegendImage
Link3DGraphic
ListSpherePlot
LoessPlot
MakeSliceImages
PlotContour
PlotCorrection
PlotData
PlotData3D
PlotDuty
PlotIVIM
PlotMoments
PlotSegmentations
PlotSequence
ResetLink3DGraphic
ShowLink3DGraphic
$debugPlot
$plotOptions

Options

Axes
Bandwidth
ClippingStyle
ColorFunction
ColorLabels
ConfidenceLevel
ContourColor
ContourColorRange
ContourOpacity
ContourResolution
ContourScaling
ContourSmoothRadius
ContourSpecularity
DropSlices
FitKernel
FitOrder
Frame
FrameStyle
ImageLegend
ImageOrientation
ImageSize
LabelStyle
MakeCheckPlot
MaskOpacity
MaxPlotPoints
Method
NormalizeIVIM
PeakNumber
PerformanceGoal
PlotColor
PlotHighlighting
PlotRange
PlotSpace
PlotStyle
PositiveZ
PredictionInterval
RandomizeColor
SphereColor
SphereSize

ProcessingTools

Functions

B1MapCalc
B1Shimming
CombineB1
CorrectJoinSetMotion
DataTransformation
DatTot
DatTotXLS
ErrorPlot
FindOutliers
FitData
GetMaskData
GetMaskMeans
GetTractMeans
Hist
Hist2
InvertDataset
JoinSets
MeanRange
MeanSignal
MeanStd
MeanStdRange
MedCouple
MinMaxRange
NumberTableForm
ParameterFit
ParameterFit2
RotateData
RotateTensor
SmartMask
SNRCalc
SNRMapCalc
SplitSets

Options

AxesLabel
B1EqualPower
B1FilterData
B1Masking
B1MaxPower
B1Output
B1Scaling
B1ShimMethod
ColorValue
FitFunction
FitOutput
GetMaskOnly
GetMaskOutput
ImageSize
InterpolationOrder
JoinSetSplit
MaskCompartment
MeanMethod
Method
MonitorCalc
MotionCorrectSets
NormalizeOverlap
NormalizeSets
OutlierCheckNormality
OutlierIncludeZero
OutlierIterations
OutlierMethod
OutlierOutput
OutlierRange
OutputSNR
PadOutputDimensions
PadOverlap
PlotLabel
ReferenceB1
ReverseData
ReverseSets
Scaling
SmartMaskOutput
SmartMethod
SmoothSNR
Strictness
TableAlignments
TableDepth
TableDirections
TableHeadings
TableMethod
TableSpacing
UseMask

ReconstructionTools

Functions

AverageKspace
CoilCombine
CoilWeightedRecon
CoilWeightedReconCSI
DeconvolveCSIdata
FourierKspace2D
FourierKspace3D
FourierKspaceCSI
FourierRescaleData
FourierShift
FourierShiftData
FourierShifted
HammingFilterCSI
HammingFilterData
InverseFourierShift
InverseFourierShifted
MakeHammingFilter
MakeSense
MeanType
NoiseCorrelation
NoiseCovariance
NoisePrewhitening
NormalizeSpectra
OrderKspace
ReadListData
SagittalTranspose
ShiftedFourier
ShiftedInverseFourier
TotalType
ZeroPad

Options

AcquisitionMethod
CoilSamples
DeconvolutionMethod
DenoiseCSI
EchoShiftData
HammingFilter
Method
NoisePrewhiten
NormalizeOutputSpectra
OutputSense
ReconFilter
RescaleRecon
SenseSmoothing
SenseWeight
WienerRegularization

RelaxometryTools

Functions

CalibrateEPGT2Fit	EPGSignal	ShiftPulseProfile	T2Fit
CreateT2Dictionary	EPGT2Fit	T1Fit	TriExponentialT2Fit
DictionaryMinSearch	NonLinearEPGFit	T1rhoFit	

Options

DictB1Range	EPGCalibrate	EPGMethodCal	Method
DictT2fRange	EPGFatShift	EPGOutput	MonitorCalc
DictT2fValue	EPGFitFat	EPGRefocussingPhase	OutputCalibration
DictT2IncludeWater	EPGFitPoints	EPGRelaxPars	WaterFatShift
DictT2Range	EPGMethod	EPGSmoothB1	WaterFatShiftDirection

ScientificColorData

Functions

AddScientificColors	ExtractColorData

Options

SegmentationTools

Functions

ApplySegmentationNetwork
AugmentImageData
AugmentTrainingData
CheckSegmentation
ClassifyData
CommonLabels
CompareSegmentations
CopyTrainedNetwork
DataToPatches
DiceSimilarity
FindPatchDim
FreezeEncoderLayers
GetNeuralNet
GetTrainData
ImportITKLabels
JaccardSimilarity
MakeChannelClassGrid
MakeChannelClassImage
MakeChannelGrid
MakeChannelImage
MakeClassImage
MakeDistanceMap
MakeTrainData
MuscleLabelToName
MuscleNameToLabel
PatchesToData
PrepareTrainingData
RunMuscleMap
SegmentData
SegmentDataGUI
ShowTrainLog
SplitDataForSegmentation
SurfaceDistance
TrainSegmentationNetwork
$SegmentationLabels

Options

Augment2D
MaskedPretraining
ActivationType
AugmentData
BatchSize
BlockType
CatenateMethod
CleanUpSegmentations
DataPadding
DataTag
DistanceRange
DownsampleSchedule
DropoutRate
FeatureSchedule
FreezeEncoderDepth
InputLabels
L2Regularization
LabelTag
LearningRate
LoadTrainingData
LossFunction
MaskedPretraining
MaxMemorySize
MaxTrainingRounds
Method
Monitor
MonitorInterval
MultiChannel
MuscleMapLabels
MuscleMapPath
MuscleMapPythonEnvironment
NetworkArchitecture
NetworkDepth
NetworkOutput
NormalizationType
OutputLabels
OutputTag
PadData
PatchesPerSet
PatchNumber
PatchPadding
PatchSize
RescaleMethod
RestartLearningCycle
RoundLength
SegmentationDimension
SegmentationsPerSlice
SettingSchedule
SplitOverlap
TargetDevice
TestRun
TrainVoxelSize
UseParallelKernels

ShapeTools

Functions

ApplyShapeModel
EvaluateModel
FitShapeModel
MakeMuscleTemplate
MakeRegionMesh
MakeShapeModel
MakeTemplatePlot
MaskFromDistanceMap
MaskToDistanceMap
MeshGridPlot
PlotMesh
PlotShapeVariation
ScaleToVolume
SplitRegionMesh
TemplateToVolume

Options

MeshColor
MeshEdgeColor
MeshesPerRow
MeshOpacity
MeshOutput
MeshPointColor
MeshPoints
MeshPointSize
Monitor
SplineSpacing
TemplateDilation
TemplateOutput
TemplatePadding
VolumeRescale

SimulationTools

Functions

AddNoise
BlochSeries
CalculateGFactor
CreateDiffData
ErnstAngle
GESignal
GetPulseProfile
GFactorSimulation
PlotSimulation
PlotSimulationAngle
PlotSimulationAngleHist
PlotSimulationHist
PlotSimulationVec
PlotSincProfile
Pulses
Signal
SimAngleParameters
SimParameters
SimulateDualTR
SimulateSliceEPG
Tensor

Options

FatFieldStrength
GOutput
GRegularization
MagnetizationVector
NoiseSize
NoiseType
PlotRange
Reject
ReportFits
SliceRange
SliceRangeSamples
SortVecs
TensOutput

SpectroTools

Functions

ApodizeEcho
ApodizeFid
ApodizePadEcho
ApodizePadFid
ApodizePadSpectra
ApodizeSpectra
ChangeDwellTimeFid
CorrectTEFid
CorrectTESpec
ExportSparSdat
GetGyro
GetPpmRange
GetSpectraBasisFunctions
GetTimePpmRange
GetTimeRange
ImportSparSdat
PadEcho
PadFid
PadSpectra
PhaseCorrectSpectra
PhaseShiftFid
PhaseShiftSpectra
PlotCSIData
PlotFid
PlotSpectra
ReadjMRUI
ShiftSpectra
SpectraSimulator
TimeShiftEcho
TimeShiftFid

Options

ApodizationFunction
AspectRatio
BasisSequence
CenterFrequency
Filling
GridLines
GridLineSpacing
ImageSize
Method
PaddingFactor
PlotColor
PlotLabel
PlotLabels
PlotRange
ReadoutType
SparID
SparName
SparOrientation
SpectraBandwidth
SpectraFieldStrength
SpectraNucleus
SpectraPpmShift
SpectraSamples
SpectraSpacing

TaggingTools

Functions

AnnalyzeTagging
CalculateDisplacementParameters
TaggingParPlot

Options

HistoryWeighting
MonitorTagging

TensorTools

Functions

ADCCalc	ECalc	FlipGradientOrientation	SigmaCalc
AngleCalc	EigensysCalc	FlipTensorOrientation	SortDiffusionData
AngleMap	EigenvalCalc	GetRPBMValues	TensorCalc
ConcatenateDiffusionData	EigenvecCalc	LogTensor	TensorCorrect
Correct	ExpTensor	ParameterCalc	TransformTensor
CreateRPBMDictionary	FACalc	RemoveIsoImages	WestinMeasures
Deriv	FitRPBMDictionary	ResidualCalc	
DriftCorrect	FitRPBMFunction	RPBMFunction	

Options

Distribution	MonitorCalc	RejectMap	RPBMRangeTau
FilterShape	NormalizeSignal	RobustFit	RPBMRangeZeta
FullOutput	Parallelize	RobustFitParameters	UseMask
MeanRes	PerformanceGoal	RotationCorrect	
Method	Reject	RPBMDictionarySize	

TractographyTools

Functions

EndpointDensityMap	FitTracts	ResampleTracts	TractCurvatureMap
ExportTCK	GetTractValues	RescaleTracts	TractDensityMap
ExportTracts	ImportTCK	SeedDensityMap	TractLength
FiberTractography	ImportTracts	SegmentTracts	TractLengthMap
FilterTractLength	MoveTracts	TractAngle	
FilterTracts	PlotSegmentedTracts	TractAngleMap	
FindTensorPermutation	PlotTracts	TractCurvature	

Options

Boxed	ImageSize	OutputForm	TractColoring
ColorFunction	InterpolationOrder	Parallelization	TractMethod
ContourOpacity	MaxSeedPoints	PerformanceGoal	TractMonitor
FiberAngle	MaxTracts	StepSize	TractReduction
FiberLengthRange	Method	StopThreshold	TractScaling
FittingOrder	Monitor	TensorFlips	TractSize
FitTractSegments	NormalizeDensity	TensorPermutations	

In[171]:=

```
QMRIToolsFuncPrint[]
```

AmaresTools

Functions

Symbol

AmaresBasis[settings, {metabolites, split}] generates basis function needed for Amares fitting.

The settings are defined as {nSamp, bw, field, nuc, te, f0} and the metabolites are a member of

"PE", "PC", "Piin", "Piex", "GPE", "GPC", "PTDC", "PCr", "ATP", "NAD" and "UDPG".

When "ATP" is defined in split the 3 ATP peaks will be separate spectra to fit.

The output is {basisFunc, {time, ppm, gyro}} where basisFunc is an Association.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`AmaresBasis

Symbol

AmaresFitFidl[fid, time, gyro, basis, matrix, cons] fits the fid with using the basis spectra.

AmaresFitFidl[fid, time, gyro, basis, matrix, {cons, init}] fits the fid using initial values init which can be obtained from a previous fit or from AmaresInitialValues.

AmaresFitFidl[fid, time, gyro, basis, matrix, b1pars, cons] the b1pars are needed if in the matrix B1 correction is set True.

The b1pars are defined as {tau, fa, TR} with tau and TR in ms and fa in degrees.

AmaresFitFidl[fid, time, gyro, basis, matrix, b1pars, {cons, init}] fits using initial values using b1pars for B1 correction.

The time, basis and gyro can be obtained from AmaresBasis and the matrix can be obtained form EditMaresMatrix.

The cons have to be defined for {"amp", "sigma", "gamma", "shift", "phase", "B1"} as {min, max} for each.

Output is the {fitted parameters, the SE, the SE in %, the standard deviation of the residuals.}.

Documentation Local »

Options AmaresUseB1 → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`AmaresFitFidl

Symbol

AmaresInitialValues[fid, time, {gyro, dw}, basis, matrix] makes a guess for the initial fit values to initialize AmaresFitFidI.

The time, basis and gyro can be obtained from AmaresBasis and the matrix can be obtained form EditMaresMatrix.

AmaresInitialValues[fid, time, {gyro, dw}, basis, matrix, plot] also exports the calibration plots if plot is set to True.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`AmaresInitialValues

Symbol

AmaresSignalCalc[basis, matrix, time, pars, gyro] generates a fid from the basis functions according to matrix and pars.

AmaresSignalCalc[basis, matrix, time, pars, b1Pars, gyro] also generates a fid but also using the B1 effect if set True in matrix.

The values of b1Pars are {tau, fa, TR} with tau and TR in ms and fa in degrees.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`AmaresSignalCalc

Symbol

AmaresSignalJacobianCalc[basis, matrix, time, pars, gyro] generates a fid and the Jacobian from the basis functions according to matrix and pars.

AmaresSignalJacobianCalc[basis, matrix, time, pars, b1Pars, gyro] also generates a fid but also using the B1 effect if set True in matrix.

The values of b1Pars are {tau, fa, TR} with tau and TR in ms and fa in degrees.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`AmaresSignalJacobianCalc

Symbol

EditAmaresMatrix[matrix] is used to edit the Amares fitting matrix which can be compressed or extended using

ToFullAmaresMatrix or FromFullAmaresMatrix.

The default properties are "amp": –1, "sigma": 0, "gamma": –1, "shift": –1, "phase": 0, "B1": False, "init": False.

The matrix has one row per metabolite and one column per fit property. For "amp", "sigma", "gamma", "shift" and "phase" the value sets how the parameter is shared between metabolites:

–1 (unique): the metabolite has its own free parameter.

0 (same): all metabolites with 0 share one common parameter.

1, 2, ... (group–01, group–02, ...): all metabolites with the same number share one parameter, different numbers are different parameters.

"B1" is True or False and switches the B1 (Ernst–angle) amplitude correction on or off for that metabolite.

"init" is True or False and selects the metabolites that are summed to the base spectrum used by AmaresInitialValues for the initial shift and linewidth calibration.

EditAmaresMatrix[matrix, names] will do the same but will only do it for the values in the list names. If values are not in matrix

they will be added, if values are names but not in matrix they will be removed from matrix.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`EditAmaresMatrix

Symbol

FromFullAmaresMatrix[matrix] compresses the Amares fitting matrix from its full form by removing all defaults from the defined.

To edit the matrix the function EditAmaresMatrix can be used and to extend it ToFullAmaresMatrix can be used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`FromFullAmaresMatrix

Symbol

ShowParameterMatrix[fit, metabolites] shows the fitted or initial fit parameter matrix from AmaresFitFid and AmaresInitialValues using met as labels.

ShowParameterMatrix[fit, true, metabolites] shows the relative difference between the true and fitted parameters.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`ShowParameterMatrix

Symbol

ToFullAmaresMatrix[names_List] completes the Amares fitting matrix to its full form by using defaults for the values not defined.

To edit the matrix the function EditAmaresMatrix can be used and to compress it FromFullAmaresMatrix can be used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`ToFullAmaresMatrix

Symbol

$AmaresB1Regularization is de regularization factor used for b1 regularization during Amares fitting.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`AmaresTools`$AmaresB1Regularization

Symbol

$AmaresMatrixDefaults is an association that holds the default values for the Amares fit matrix.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`AmaresTools`$AmaresMatrixDefaults

Symbol

$AmaresT1Values is an association that holds the T1 values of the metabolites used in Amares fitting.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`AmaresTools`$AmaresT1Values

Options

Symbol

AmaresUseB1 is an option for AmaresFitFidI. When True, the B1 value supplied through init is used to apply the Ernst–angle amplitude correction even if the fitting matrix has "B1"–>False for all metabolites (i.e. B1 is applied but not fitted). Default is False.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`AmaresTools`AmaresUseB1

CardiacTools

Functions

Symbol

BullseyePlot[data, segmask] generates a AHA–17 segment bullseye plot.

BullseyePlot[list] generates a AHA–17 segment bullseye plot of the lists (which needs to have 17 values) provide.

data is a 3D volume used for the plot.

segmask is the AHA–17 segmentation resulting form the CardiacSegment function when AHA17 is selected.

Output is a bullseye plot or a plotWindow, depending on the Method which can be "Dynamic" else it will be static.

BullseyePlot[] is based on DOI: 10.1161/hc0402.102975.

Documentation Local »
Default Definitions SyntaxInformation[BullseyePlot] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › TextOffset → 0.5 … (7 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`BullseyePlot

Symbol

CalculateWallMap[mask,vox] calculates the wall distance map and the wall derivative.

Output is {wallmap, wallDerivative}.

Documentation	Local »
Default Definitions	SyntaxInformation[CalculateWallMap] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options	{ShowPlot → True, MaskWallMap → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`CalculateWallMap

Symbol

CardiacCoordinateSystem[mask, vox] creates the cardiac coordinate system within the mask and is used in HelixAngleCalc.

Output is a set of vectors {radvecn, norvecc, cirvec}, being the radial, normal and circular axes of each voxel respectively.

If the option showPlot is true the output is {{radvecn, norvecc, cirvec}, plots}.

Documentation	Local »
Default Definitions	SyntaxInformation[CardiacCoordinateSystem] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options	{ShowPlot → False, LCMMethod → WallMap, AxesMethod → Quadratic}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`CardiacCoordinateSystem

Symbol

CardiacSegment[mask, vox, pts] segments the mask in the AHA17 segmentation using pts to indicate the attachments.

CardiacSegment[mask, back, vox, pts] the same where back is used for image generation.

CardiacSegment[mask, vox, pts, seg] does the same but seg can be an alternate segmentation to the AHA17.

CardiacSegment[mask, back, vox, pts, seg] does the same but seg can be an alternate segmentation to the AHA17 where back is used for image generation.

Documentation	Local »
Default Definitions	SyntaxInformation[CardiacSegment] = {ArgumentsPattern → {_, _, _, _., _., OptionsPattern[]}}
Options	{ReversePoints → True, ReverseDirection → False, MakeSegmentPlots → True}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`CardiacSegment

Symbol

CardiacSegmentGUI[data, mask, vox] allows to segment the heart in 1, 4, 6 or AHA–17 segments for each slice 360 radial samples are generated.

data is a background image on which all overlays are projected.

mask is the mask of the left ventricle (same as used for CentralAxes) and defines the area in which the data is sampled.

off is the centerpoints generated by CentralAxes.

Output is {points, slices , {rev, flip}}.

Documentation Local »

Default Definitions SyntaxInformation[CardiacSegmentGUI] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options ❯ StartPoints → Default … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`CardiacSegmentGUI

Symbol

CentralAxes[mask, vox] calculates the center of the lumen from a mask of the left ventricle. vox is the voxels size, {slice, x, y}.

CentralAxes[mask, maskp, vox] allows for fancy visualization of the other structures using maskp.

Output is {centerpoints, normalvecs, inout} or {centerpoints, normalvecs, inout, fit}.

Documentation Local »

Default Definitions SyntaxInformation[CentralAxes] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options {ShowPlot → False, RowSize → Automatic, AxesMethod → Cubic}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`CentralAxes

Symbol

CreateHeart[] creates a simulated left ventricle shape.

CreateHeart[pars] creates a simulated left ventricle shape with predifined parameters pars.

Output is the heart shape, the voxel size and the parameters needed to generate the heart, {mask, vox, pars}.

Documentation Local »
Default Definitions SyntaxInformation[CreateHeart] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`CreateHeart

Symbol

ECVCalc[T1pre, T1post, hema] calculates the ECVmap using MakeECVBloodMask.

ECVCalc[T1pre, T1post, bloodMask, hema] calculates the ECVmap using bloodMask.

The T1pre and T1post maps are assumed to be in ms.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`ECVCalc

Symbol

ExcludeSlices[data] excludes slices that do not look like the others based on various distance measures.

Output is an array with 1 or 0 with the dimensions {slices, diff dirs}.

Documentation Local »
Default Definitions SyntaxInformation[ExcludeSlices] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {CutOffMethod → Auto, DistanceMeasure → 5, ShowOutliers → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`ExcludeSlices

Symbol

GetMaskSegmentPoints[mask] get the attachment points from a cardiac segmentation where the heart has label 1, and the attachment points have label 2 and 3.

Output is {maks, pts} where now in mask the points are removed.

Documentation Local »

Default Definitions SyntaxInformation[GetMaskSegmentPoints] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GetMaskSegmentPoints

Symbol

GetSegmentLines[lines, lineIndex, segments] groups the transmural lines per segment.

Documentation Local »

Default Definitions SyntaxInformation[GetSegmentLines] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GetSegmentLines

Symbol

GetSegmentSlices[mask] based on the mask it gives back the slice numbers of the apex, apical, mid–vertical, and basal slices.

GetSegmentSlices[points] does the same but then based on the points obtained form GetMaskSegmentPoints.

Documentation Local »

Default Definitions SyntaxInformation[GetSegmentSlices] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GetSegmentSlices

Symbol

HelixAngleCalc[eigenvectors, mask, vox] calculates the helix angle matrix of cardiac data using only a left ventricle mask.

HelixAngleCalc[eigenvectors, mask, maskp, vox] calculates the helix angle matrix of cardiac data using only a left ventricle mask, and a maskp for visualization.

HelixAngleCalc[eigenvectors, mask, centerpoint, vec, inout, vox] calculates the helix angle matrix of cardiac data using only a left ventricle mask.

HelixAngleCalc[eigenvectors, mask, maskp, centerpoint, vec, inout, vox] calculates the helix angle matrix of cardiac data using a left ventricle mask and a maskp for visualization.

eigenvectors are the tensor eigenvectors calculated with EigenvecCalc.

mask is a mask of the left ventricle.

maskp is a mask used for visualization.

vox is the voxels size, {slice, x, y}.

The following values are calculated automatically Using CentralAxes but can also be provided as an input.

centerpoint is the center of each slice calculated with CentralAxes.

inout is the inner and outer radius calculated with CentralAxes.

vec is the vector describin the central axes of the heart, calculated with CentralAxes.

Output is het fiber angle matrix FAM = {9, slice, x, y} or {FAM, plot}.

The angles are in degrees.

HelixAngleCalc[] is based on DOI: 10.1186/1532–429X–17–S1–P15.

Documentation	Local »
Default Definitions	SyntaxInformation[HelixAngleCalc] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}
Options	{ShowPlot → True, LCMMethod → WallMap, AxesMethod → Quadratic}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`HelixAngleCalc

Symbol

LinesToSegmentIndex[lines, points, segments] finds the lines indices corresponding to the points and the segments borders. Additionally it finds all the lines indices for all lines within each segment.

The lines are computed by MaskToLines, the points are cumputed by GetMaskSegmentPoints, and the segments is the output of SegmentsPerSlices.

Output {pointIndex, segmentIndex, lineIndex}.

Documentation	Local »
Default Definitions	SyntaxInformation[LinesToSegmentIndex] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options	{ReversePoints → True, ReverseDirection → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`LinesToSegmentIndex

Symbol

MakeECVBloodMask[T1pre, T1post] makes a bloodpool mask based on the T1pre and T1post images. It assumes that the hart is cropped with the blood in the center.

The T1pre and T1post maps are assumed to be in ms.

Documentation	Local »
Default Definitions	SyntaxInformation[MakeECVBloodMask] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options	{BloodMaskRange → {1400, {0, 700}}, OutputCheckImage → True}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`MakeECVBloodMask

Symbol

MakeLineImage[back, segLines, pts] makes an image of the cardiac segmentation lines.

Documentation	Local »
Default Definitions	SyntaxInformation[MakeLineImage] = {ArgumentsPattern → {_, _, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`MakeLineImage

Symbol

MakeMaskImage[back, mask] maskes an image of the cardiac segmentation mask.

Documentation Local »
Default Definitions SyntaxInformation[MakeMaskImage] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`MakeMaskImage

Symbol

MaskHelix[helix, mask] masks helix angle data, sets the background to −100 and allows for Median filter of the helix mask.

helix can be a singel map or the FAM.

Output is the masked helix angle data.

Documentation Local »
Default Definitions SyntaxInformation[MaskHelix] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options {BackgroundValue → −100, SmoothHelix → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`MaskHelix

Symbol

MaskToLines[mask, vox] calculates lines perpendicular to the heart wall per slice within the mask. Internally it uses CalculateWallMap and CentralAxes to obtain the cardiac geometry from mask.

MaskToLines[mask, wall, cent] where mask is the first output of CalculateWallMap and cent is the first output of CentralAxes.

Documentation Local »
Default Definitions SyntaxInformation[MaskToLines] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`MaskToLines

Symbol

PlotSegmentMask[mask, segmask, vox] plots the mask segments created by CardiacSegment.

mask is a mask the left ventricle that was used in the CardiacSegment.

segmask is the output of CardiacSegment.

vox is the voxels size, {slice, x, y}.

Output is a plot window.

Documentation Local »

Default Definitions SyntaxInformation[PlotSegmentMask] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`PlotSegmentMask

Symbol

PlotSegments[mask, data, segang] shows how the heart wil be sampled by RadialSample.

mask is a mask the left ventricle that was used in the CardiacSegment.

function and the segang is the output of the cardiac SegmentFunction.

Output is a plot window.

Documentation Local »

Default Definitions SyntaxInformation[PlotSegments] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options RadialSamples → 10

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`PlotSegments

Symbol

RadialSample[mask, data, segang] radialy samples the provided parameter map data.

The mask should be a mask of the left ventricle that was used in the CardiacSegment.

segang is the output of the cardiac SegmentFunction.

Output is {points, vals} which are orderd as indicated by the user.

Documentation	Local »
Default Definitions	SyntaxInformation[RadialSample] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options	{RadialSamples → 10, DropSamples → 0}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`RadialSample

Symbol

SegmentLinesToMask[mask, segLines] cuts the mask based one the tranmural lines per segments which can be obtained by GetSegmentLines.

Documentation	Local »
Default Definitions	SyntaxInformation[SegmentLinesToMask] = {ArgumentsPattern → {_, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`SegmentLinesToMask

Symbol

SegmentsPerSlice[points] gives the number of segments per slice where the slice distribution is determined by GetSegmentSlices.

SegmentsPerSlice[slices, points] does the same but the slices are given manually.

Documentation	Local »
Default Definitions	SyntaxInformation[SegmentsPerSlice] = {ArgumentsPattern → {_, OptionsPattern[]}}
Options	{GroupPerSegment → True, SegmentationMethod → AHA}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`CardiacTools`SegmentsPerSlice

Symbol

TransmuralPlot[data] plots transmural profiles of the data which are created by RadialSample.

data can be a single profile or a list of profiles. In the second case the mean and standard deviations are plotted.

Output is a plot of the transmural profile.

Documentation Local »
Default Definitions SyntaxInformation[TransmuralPlot] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options › GridLineSpacing → 10 … (6 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`TransmuralPlot

Options

Symbol

AxesMethod is an option for HelixAngleCalc and CentralAxes and CardiacCoordinateSystem. Can be "Linear", "Quadratic", "Cubic".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`AxesMethod

Symbol

BackgroundValue is an option for MaskHelix. Sets the background value (default is –100).

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`BackgroundValue

Symbol

BloodMaskRange is an option for MakeECVBloodMask.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`BloodMaskRange

Symbol

BullPlotMethod is an option for BullseyePlot. Can be "Dynamic" of "Normal".

"Dynamic" allows to change plotting parameters in Manipulation window.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`BullPlotMethod

Symbol

ColorFunction is an option for graphics functions that specifies a function to apply to determine colors of elements.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ColorFunction

Symbol

CutOffMethod is an option for ExcludeSlices. Default value is "Auto" or it can be a fixed percentage (value between 0 and .5).

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`CutOffMethod

Symbol

DistanceMeasure is an option for ExcludeSlices. Defaul value is 5. (1 ManhattanDistance, 2 SquaredEuclideanDistance, 3 EuclideanDistance, 4 Correlation, 5 SpearmanRho.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`DistanceMeasure

Symbol

DropSamples is an option for RadialSample and PlotSegments. Defines how manny samples are droped form star and end. Can be an number or set (strat, end) of numbers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`DropSamples

Symbol

GridLineSpacing is an option of TransmuralPlot. It defines the spacing of the grid lines.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GridLineSpacing

Symbol

GroupPerSegment is an option for SegmentsPerSlice. If set False segments are grouped per slice and not per segment.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GroupPerSegment

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageSize

Symbol

LCMMethod is an option for HelixAngleCalc and LMCSystemCalc. Can be "CentralAxes" or "WallMap".

"CentralAxes" uses wall distance calculation using projection of the central axes and circular approximation of the ventricle. This method is fairly fast and uses CentralAxes internally.

"WallMap" uses wall distance interpolation and subsequential gradient calculation. Can take long for high res datasets but is most accurate. Uses CalculateWallMap internally.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`LCMMethod

Symbol

MakeSegmentPlots is an option for CardiacSegment. If True plots of the segmentation are made.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`MakeSegmentPlots

Symbol

MaskWallMap is an option for CalculateWallMap. if True or False.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`MaskWallMap

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

OutputCheckImage is an option for MakeECVBloodMask.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`OutputCheckImage

Symbol

PlotLabel is an option for graphics functions that specifies an overall label for a plot.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PlotLabel

Symbol

PlotRange is an option for graphics functions that specifies what range of coordinates to include in a plot.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotRange

Symbol

PlotStyle is an option for plotting and related functions that specifies styles in which objects are to be drawn.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PlotStyle

Symbol

RadialSamples is an option for RadialSample and PlotSegments. Defines how manny transmural samples are taken.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`RadialSamples

Symbol

ReverseDirection is an option for LinesToSegmentIndex, CardiacSegment. Defines the direction of rotation, clockwise or anti−clockwise, can be True of False.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`ReverseDirection

Symbol

ReversePoints is an option for LinesToSegmentIndex, CardiacSegment. Defines at which point to start, can be True or False.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`ReversePoints

Symbol

RowSize is an option for CentralAxes. defines the number or images per showing the segmentation.

Can be "Automatic" of an integer.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CardiacTools`RowSize

Symbol

SegmentationMethod is an option for SegmentsPerSlice. Values can be "AHA", "AHA+", 1, 2, 3, 6 or 8.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`SegmentationMethod

Symbol

ShowOutliers is an option for ExcludeSlices.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`ShowOutliers

Symbol

ShowPlot is an option for CentralAxes, HelixAngleCalc and CardiacCoordinateSystem. True shows the fit of the central axes and outpu the plot as extra output.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`ShowPlot

Symbol

SmoothHelix is an option for MaskHelix, sets the kernel size for the MedianFilter.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`SmoothHelix

Symbol

StartPoints is an option for CardiacSegmentGUI. Value is "Default" or the point list given by CardiacSegment.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`StartPoints

Symbol

StartSlices is an option for CardiacSegmentGUI. Value is "Default" or the list given by CardiacSegment.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`StartSlices

Symbol

TextNumberForm is an option for BullseyePlot. Specifies how many number and decimals to use like in NumberForm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`TextNumberForm

Symbol

TextOffset is an option for BullseyePlot. Determines where the text is placed, can be 0 to 1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`TextOffset

Symbol

TextSize is an option for BullseyePlot. Determines the text size.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`TextSize

CoilTools

Functions

Symbol

CoilSNRCalc[coils, noise] calculates the sensitivity weighted snr of multiple coil elements using magnitude signal and noise.

Output is {data, noise, sos, snr, sigmap, weights}.

Documentation Local »
Default Definitions SyntaxInformation[CoilSNRCalc] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`CoilSNRCalc

Symbol

FindCoilPosition[weights] finds the coil posision by locating the highest intensity location in the coil weight map, which can be obtianed by LoadCoilSetup or SumOfSquares.

Internally it uses MakeWeightMask to remove the noise of the weightmasks.

FindCoilPosition[weights, mask] limits the search region to the provided mask.

Documentation Local »
Default Definitions SyntaxInformation[FindCoilPosition] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {OutputCoilSurface → False, CoilSurfaceVoxelSize → {1, 1, 1}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`FindCoilPosition

Symbol

LoadCoilSetup[file] load a very specific type of coil experiment, a dynmic scan with a setup of which the second dynamic is a noise measurement.

The input file is the Nii file that conatins the individually reconstruted coil images and the noise data.

Internally it uses CoilSNRCalc and SumOfSquares.

Output is the coil data with coil noise data and snrmap based on the SumOfSquares addition, the SOS reconstruction and the SOS weights.

{dataC, noiseC, sosC, snrC, sigmapC, weights, vox}.

Documentation Local »
Default Definitions SyntaxInformation[LoadCoilSetup] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`LoadCoilSetup

Symbol

LoadCoilTarget[file] loads a very specific typ of experiment, a dynamic scan with with the second dynmaic is a noise measurement.

The input file is the Nii file that conatins the scanner reconstruction and the noise data.

Internally it uses SNRMapCalc,

Output is the reconstructed data with noise data and snrMap {dataC, noiseC, sosC, snrC, sigmapC, weights, vox}.

Documentation Local »

Default Definitions SyntaxInformation[LoadCoilTarget] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CoilTools`LoadCoilTarget

Symbol

MakeCoilLayout[{name, size, number}] makes a coil grid with label name, partioned in size rows and with label number.

MakeCoilLayout[{name, size, number}, val] makes a coil grid with label name, partioned in size rows and with label the val at location number.

MakeCoilLayout[{coils..}] same but for multile coils grids. Each coil grid is defined as {name, size, number}.

MakeCoilLayout[{coils..}, val] savem but for multiple coil grids.

Documentation Local »

Options ❯ PlotRange → Automatic … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`CoilTools`MakeCoilLayout

Symbol

MakeNoisePlots[noise] returns a grid of plots of the noise per channel

MakeNoisePlots[noise, {met, prt}] met can be "Grid" with prt a number or Automatic. Else all plots will be returend as a list of plots.

MakeNoisePlots[noise, {met, prt}, sub] sub defines how much the noise is subsampled, default is 40 (every 40th sample is used in plot).

Documentation Local »

Default Definitions SyntaxInformation[MakeNoisePlots] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`CoilTools`MakeNoisePlots

Symbol

MakeWeightMask[weights] creates a mask of homogeneous regions of weightmaps removing the noise.

Documentation Local »
Default Definitions SyntaxInformation[MakeWeightMask] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`MakeWeightMask

Options

Symbol

CoilArrayPlot is an option for MakeCoilLayout. If True and values are provided it makes an arrayplot of the coil layouts.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`CoilArrayPlot

Symbol

CoilSurfaceVoxelSize is an option for FindCoilPosition. Specifies the voxel size used for OutputCoilSurface.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`CoilTools`CoilSurfaceVoxelSize

Symbol

ColorFunction is an option for graphics functions that specifies a function to apply to determine colors of elements.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ColorFunction

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ImageSize

Symbol

OutputCoilSurface is an option for FindCoilPosition. If set true it will also output a SurfacePlot of the coil location volume.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CoilTools`OutputCoilSurface

Symbol

PlotRange is an option for graphics functions that specifies what range of coordinates to include in a plot.

Documentation Local » | Web »

Attributes {Protected, ReadProtected}

Full Name System`PlotRange

DenoiseTools

Functions

Symbol

AnisoFilterData[data] Filter the diffusion tensor data using an anisotropic filter based on the structure tensor of the data.

Output is the smoothed data.

AnisoFilterData[] is based on DOI: 10.1016/j.jbiomech.2021.110540 and 10.1016/j.mri.2009.10.001 and 10.1371/journal.pone.0126953.

Documentation Local »

Default Definitions SyntaxInformation[AnisoFilterData] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {AnisoStepTime → 1, AnisoIterations → 1, AnisoKernel → {0.25, 0.5}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoFilterData

Symbol

AnisoFilterTensor[tens, data] Filter the tensor tens using an anisotropic diffusion filter (Perona–Malik).

It uses the diffusion weighted data data to find edges that are not visible in the tensor.

Edge weights based on the diffusion data are averaged over all normalized diffusion direction.

AnisoFilterTensor[tens] Same but does not use the data for edge identification.

Output is the smoothed tensor.

AnisoFilterTensor[] is based on DOI: 10.1109/ISBI.2006.1624856.

Documentation	Local »
Default Definitions	SyntaxInformation[AnisoFilterTensor] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options ❯	AnisoWeightType → 2 … (4 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`DenoiseTools`AnisoFilterTensor

Symbol

DeNoise[data, sigma, filterSize] removes Rician noise with standard deviation "sigma" from the given dataset using a kernel with size "filterSize" a gaussian kernel.

Output is data denoised.

DeNoise[] is based on DOI: 10.1109/TMI.2008.920609.

Documentation	Local »
Default Definitions	SyntaxInformation[DeNoise] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options	{DeNoiseKernel → Gaussian, DeNoiseMonitor → False, DeNoiseIterations → 1}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`DenoiseTools`DeNoise

Symbol

DenoiseCSIdata[spectra] performs PCA denoising of the complex values spectra, data has to be 3D and the spectral dimensions is last, {x,y,z,spectra}.

Documentation Local »
Default Definitions SyntaxInformation[DenoiseCSIdata] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ❯ PCAKernel → 5 … (4 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`DenoiseCSIdata

Symbol

DenoiseDynamicSpectraData[spectra] performs PCA denoising of the complex values spectra, The data is given as a list of dynamically acquired spectra {dynamic ,spectra}.

Documentation Local »
Default Definitions SyntaxInformation[DenoiseDynamicSpectraData] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`DenoiseDynamicSpectraData

Symbol

HarmonicDenoiseTensor[tens, mask, vox] uses the harmonic denoising method to denoise the tensor within the mask.

Values for which the tensor is not defined but are within the mask will be filled in.

HarmonicDenoiseTensor[tens, seg, vox] will do the same for each segmentation label in seg.

HarmonicDenoiseTensor[tens, seg, vox, labs] will do the same for each segmentation label in seg that is specified in labs.

HarmonicDenoiseTensor[] is based on 10.1016/j.media.2011.01.005.

Documentation Local »
Default Definitions SyntaxInformation[HarmonicDenoiseTensor] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options ❯ RadialBasisKernel → 12 … (10 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`HarmonicDenoiseTensor

Symbol

NNThreshold is an options for P2SDenoise and specifies the automated back ground masking value.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`NNThreshold

Symbol

P2SDenoise[data] removes noise from the data using self–supervised linear regression, where each volume is predicted from all other volumes.

P2SDenoise[data, mask] does the same but only within the mask.

The regression is fitted on Iterations random row sketches of 10% of the voxels and the predictions are averaged.

The option Method sets the sketch and can be "Leverage" (default, leverage score sampling) or "Count" (count sketch).

P2SDenoise[] is based on DOI:10.48550/arXiv.2011.01355 (Patch2Self) and Patch2Self2 (CVPR 2024).

Documentation Local »

Default Definitions SyntaxInformation[P2SDenoise] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {Threshold → 2, Iterations → 10, Method → Leverage}

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`P2SDenoise

Symbol

PCADeNoise[data] removes Rician noise from the data with PCA.

PCADeNoise[data, mask] removes Rician noise from the data with PCA only withing the mask.

PCADeNoise[data, mask, sig] removes Rician noise from the data with PCA only withing the mask using sig as prior knowledge or fixed value.

Output is de {data denoise, sigma map} by default if PCAOutput is Full then fitted {data

denoise , {sigma fit, average sigma}, {number components, number of fitted voxesl, number of max fits}, total fit –time per 500 ittt}.

PCADeNoise[] is based on DOI: 10.1016/j.neuroimage.2016.08.016 and 10.1002/mrm.26059.

Documentation Local »
Default Definitions SyntaxInformation[PCADeNoise] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options › PCAKernel → 5 … (9 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`PCADeNoise

Symbol

WeightMapCalc[data] calculates a weight map which is used in AnisoFilterTensor.

Output is a weight map of the data which is high in isotropic regions and low at edges.

Documentation Local »
Default Definitions SyntaxInformation[WeightMapCalc] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {AnisoWeightType → 2, AnisoKappa → 10.}
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`WeightMapCalc

Symbol

$debugDenoise is a debug flag for Denoise functionality within DenoiseTools currently only for HarmonicDenoiseTensor.

Documentation Local »
Attributes {ReadProtected}
Full Name QMRITools`DenoiseTools`$debugDenoise

Options

Symbol

AnisoFilterSteps is an option for AnisoFilterTensor and defines the amount of diffusin steps taken. Higher is more smoothing.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoFilterSteps

Symbol

AnisoIterations is an options for AnisoFilterData. It specifies the amount of denoising iterations.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoIterations

Symbol

AnisoKappa is an option for AnisoFilterTensor and WeightMapCalc and defines the weighting strength, all data is normalize to 100 before filetering.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoKappa

Symbol

AnisoKernel is an options for AnisoFilterData. It defines the kernel size.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoKernel

Symbol

AnisoStepTime is an option for AnisoFilterTensor and defines the diffusion time, when small more step are needed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoStepTime

Symbol

AnisoWeightType is an option for AnisoFilterTensor and WeightMapCalc and defines the weighting, eigher 1, the exponent of (–g/kappa) or 2, 1/(1+g/kappa).

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`AnisoWeightType

Symbol

DeNoiseIterations is and option for DeNoise. Specifies the number of the denoising iterations.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`DeNoiseIterations

Symbol

DeNoiseKernel is and option for DeNoise. Values can be "Disk", "Box" or "Gaussian".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`DeNoiseKernel

Symbol

DeNoiseMonitor is and option for DeNoise. Monitor the denoising progress.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`DeNoiseMonitor

Symbol

GradientStepSize is an option for HarmonicDenoiseTensor. It defines the step size of the gradient descent for the harmonic and radial parts.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`GradientStepSize

Symbol

Iterations is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the number of iterations used by the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`Iterations

Symbol

MaxIterations is an option that specifies the maximum number of iterations that should be tried in various built–in functions and algorithms.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`MaxIterations

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »

Attributes {HoldAll, Protected, ReadProtected}

Full Name System`Monitor

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

Parallelize[*expr*] evaluates *expr* using automatic parallelization.

Definitions

```
Parallelize[Parallel`Kernels`Private`args$___] := (Parallel`Protected`doAutolaunch[TrueQ[Parallel`Static`$enableLaunchFeedback]];
Parallelize[Parallel`Kernels`Private`args$])
```

Documentation Local » | Web »
Options {DistributedContexts :→ $Context, Method → Automatic, ProgressReporting :→ $ProgressReporting}
Attributes {HoldFirst, Protected}
Full Name System`Parallelize

Symbol

PCAClipping is an option of PCADeNoise and can be True of False. If True the output is clipped between 0 and the max absolute value of the input data.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`PCAClipping

Symbol

PCAComplex is an option of PCADeNoise and can be True of False. If set true the input data is expected to be {real, imag}.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`PCAComplex

Symbol

PCAKernel is an option of PCADeNoise. It sets the kernel size.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`DenoiseTools`PCAKernel

Symbol

PCANoiseSigma is an option of DenoiseCSIdata and can be "Corners" or "Automatic".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`PCANoiseSigma

Symbol

PCAOutput is an option of PCADeNoise. If output is full the output is {datao, {output[[1]], sigmat}, {output[[2]], output[[3]], j}, timetot}.

Else the output is {datao, sigmat}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`PCAOutput

Symbol

PCATolerance is an option of PCADeNoise and should be an integer > 0. Default value is 0. When increased the denoise method removes less noise.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`PCATolerance

Symbol

PCAWeighting is an option of PCADeNoise and can be True of False. Default

value is False. When True the weights of the per voxel result are calculated based on the number of non noise components.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`PCAWeighting

Symbol

RadialBasisKernel is an option for HarmonicDenoiseTensor. It defines the kernel size of the radial basis functions in mm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`RadialBasisKernel

Symbol

RangeFA is an option for HarmonicDenoiseTensor. It defines the range of the FA values of voxels to include in the minimization.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`RangeFA

Symbol

RangeMD is an option for HarmonicDenoiseTensor. It defines the range of the MD values of voxels to include in the minimization.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DenoiseTools`RangeMD

Symbol

TensorFlips is an option for FiberTractography and specifies if the tensor orientation is flipped, see FlipTensorOrientation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TensorFlips

Symbol

TensorPermutations is an option for FiberTractography and specifies if the tensor orientation is permuted, see FlipTensorOrientation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TensorPermutations

Symbol

System`Threshold

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Threshold

Symbol

System`Tolerance

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Tolerance

DixonTools

Functions

Symbol

DixonPhase[real, imag, echos] calculates the b0 and ph0 maps.

Documentation Local »

Default Definitions SyntaxInformation[DixonPhase] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options › DixonPrecessions → 1 ... (10 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonPhase

Symbol

DixonReconstruct[{real, imag}, echo] reconstruxt Dixon data with initial guess b0 = 0 and T2star = 0.

DixonReconstruct[{real, imag}, echo, {b0}] reconstructs Dixon data with initial guess T2star = 0.

DixonReconstruct[{real, imag}, echo, {b0, t2}] reconstructs Dixon data with t2star and B0.

DixonReconstruct[{real, imag}, echo, {b0, t2, ph0}] reconstructs Dixon data with initial phase.

DixonReconstruct[{real, imag}, echo, {b0, t2, ph0, phb}] reconstructs Dixon data with bipolar phase .

Output is {{watF,fatF},{watSig,fatSig},{inphase,outphase},{{b0, ph0, phb},{r2, t2}},iterations}.

The fractions are between 0 and 1, the B0 field map is in Hz and the T2start map is in ms.

DixonReconstruct[] is based on DOI: 10.1002/mrm.20624 and 10.1002/mrm.21737 (10.1002/nbm.3766).

Documentation Local »

Default Definitions SyntaxInformation[DixonReconstruct] = {ArgumentsPattern → {_, _, _, _., _., OptionsPattern[]}}

Options ❯ DixonPrecessions → 1 … (21 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonReconstruct

Symbol

DixonToPercent[water, fat] converts the dixon water and fat data to percent maps.

Output is {waterFraction, fatFraction}.

The values of water and fat are arbitrary units and the output fractions are between 0 and 1.

DixonToPercent[] is based on DOI: 10.1002/mrm.21301.

Documentation Local »

Default Definitions SyntaxInformation[DixonToPercent] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonToPercent

Symbol

FindInPhaseEchos[echos, iop] finds the two nearest echos to inphase which are best used for unwrapping using the iop time.

Documentation Local »
Default Definitions SyntaxInformation[FindInPhaseEchos] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options DixonBipolar → False
Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`FindInPhaseEchos

Symbol

FixDixonFlips[{mag, phase, real, imag}] checks if any volumes are 180 degrees out of phase and corrects them.

Documentation Local »
Default Definitions SyntaxInformation[FixDixonFlips] = {ArgumentsPattern → {{_, _, _, _}}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`FixDixonFlips

Symbol

GenerateAmps[amp] generates the fat based chemical species amplitudes needed for DixonReconstruct.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`GenerateAmps

Symbol

OptimizeDixonEcho[] shows a manipulate panel which allows to optimize the dixon echos.

OptimizeDixonEcho[echos] shows a manipulate panel which allows to optimize the predefined dixon echos.

Documentation Local »
Options {DixonNucleus → 1H, DixonFrequencies → {{0.}, {−3.81, −3.4, −3.12, −2.67, −2.45, −1.94, −0.63, −0.4, 0.52, 0.62}}, DixonAmplitudes → {{1}, {0.089, 0.577, 0.059, 0.093, 0.059, 0.013, 0.02, 0.02, 0.01, 0.059}}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`OptimizeDixonEcho

Symbol

SimulateDixonSignal[echo, fr, B0, T2] simulates an Dixon gradient echo sequence with echotimes.

Echotimes echo in ms, fat fraction fr between 0 and 1, field of resonance B0 in Hz and relaxation T2 in ms.

Documentation Local »

Default Definitions SyntaxInformation[SimulateDixonSignal] = {ArgumentsPattern → {_, _, _, _, OptionsPattern[]}}

Options › DixonNucleus → 1H … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`SimulateDixonSignal

Symbol

Unwrap[data] unwraps the given dataset. The data should be between −Pi and Pi.

Unwrap[] is based on DOI: 10.1364/AO.46.006623 and 10.1364/AO.41.007437.

Documentation Local »

Default Definitions SyntaxInformation[Unwrap] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {MonitorUnwrap → False, UnwrapDimension → 2D, UnwrapThresh → 0.5}

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`Unwrap

Symbol

UnwrapDCT[data] unwraps the given dataset using DCT transform . The data should be between −Pi and Pi.

UnwrapDCT[data, weights] unwraps using weighted least squares, weights can be Automatic or an array with the dimensions of data.

UnwrapDCT[] is based on DOI: 10.1364/JOSAA.11.000107.

Documentation Local »

Default Definitions SyntaxInformation[UnwrapDCT] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`UnwrapDCT

Symbol

UnwrapList[list] unwraps a 1D list of values between –Pi and Pi.

Documentation Local »
Default Definitions SyntaxInformation[UnwrapList] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`UnwrapList

Symbol

UnwrapSplit[phase, data] unwarps the give phase dataset but splits the data into

left and right using SplitData based in the data and performs the unwrapping separately. The data should be between –Pi and Pi.

UnwrapSplit[] is based on DOI: 10.1364/AO.46.006623 and 10.1364/AO.41.007437.

Documentation Local »
Default Definitions SyntaxInformation[UnwrapSplit] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options {MonitorUnwrap → False, UnwrapDimension → 2D, UnwrapThresh → 0.5}
Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`UnwrapSplit

Symbol

Wrap[data] wraps phase values between –Pi and Pi.

Documentation Local »
Default Definitions SyntaxInformation[Wrap] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`Wrap

Options

Symbol

DixonAmplitudes is an option for DixonReconstruct. Defines the relative amplitudes of the fat peaks being used.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`DixonTools`DixonAmplitudes

Symbol

DixonBipolar is an option for FindInPhaseEchos. If set True the found echos will always be of the same polaritiy.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonBipolar

Symbol

DixonClipFraction is an option for DixonReconstruct. If set True the fat fraction is clipped between 0 and 1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonClipFraction

Symbol

DixonConstrainPhase is an option for DixonReconstruct. If set True everything is fitted as real values.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonConstrainPhase

Symbol

DixonCorrectT1 is an option for DixonReconstruct. To perform T1 correction provide the TR and FA as a list, {TR, FA}. TR is in ms and FA in degrees.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonCorrectT1

Symbol

DixonFieldStrength is an option for DixonReconstruct. Defines the fieldstrengths in Tesla on which the data was acquired.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFieldStrength

Symbol

DixonFilterInput is an option for DixonReconstruct. If True the input b0 and T2star values are smoothed using a gaussian kernel.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFilterInput

Symbol

DixonFilterOutput is an option for DixonReconstruct. If True the out b0 and T2star values are smoothed Median filter and lowpassfiltering after which the water and fat maps are recomputed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFilterOutput

Symbol

DixonFilterSize is an option for DixonReconstruct. Defines the number of voxel with which the input b0 and T2star values are smoothed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFilterSize

Symbol

DixonFilterType is an option for DixonReconstruct. FilterType can me "Median" or "Laplacian".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFilterType

Symbol

DixonFitPhase is an option for DixonReconstruct. If set to True the phase maps are fitted using polynomial functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFitPhase

Symbol

DixonFixT2 is an option for DixonReconstruct. If set to true the R2' is fitted rather then the R2*. This is done by fixing T2–water to 30ms and T2–fat to 100ms.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFixT2

Symbol

DixonFrequencies is an option for DixonReconstruct. Defines the frequencies in ppm of the fat peaks being used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonFrequencies

Symbol

DixonIterations is an option for DixonReconstruct. Defines the maximum iterations the fit can use.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonIterations

Symbol

DixonMaskThreshold is an option for DixonReconstruct. Defines at which Threshold the dixon reconstruction considers a voxel to be background noise. default values is 0.05.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonMaskThreshold

Symbol

DixonNucleus is an option for DixonReconstruct. Defines the nucleus for which the reconstruction is performed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonNucleus

Symbol

DixonPhases is an option for DixonReconstruct. It defines which phases to fit within the model.

The order is {T2*, B0, bipolar, initial, bipolar}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonPhases

Symbol

DixonPrecessions is an option for DixonReconstruct. Defines the rotation of the signal {−1,1} default is −1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonPrecessions

Symbol

DixonRelaxivity is an option for DixonReconstruct. Defines the T2 relaxation times of the fat peaks being used. Is only used if DixonFixT2 is set to True.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonRelaxivity

Symbol

DixonTolerance is an option for DixonReconstruct. Defines at which change per iteration of b0 and R2star the itarative methods stops. Default value is 0.1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`DixonTolerance

Symbol

MaxIterations is an option that specifies the maximum number of iterations that should be tried in various built−in functions and algorithms.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`MaxIterations

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

MonitorUnwrap is an option for Unwrap. Monitor the unwrapping progress.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`MonitorUnwrap

Symbol

Parallelize[*expr*] evaluates *expr* using automatic parallelization.

Definitions

```
Parallelize[Parallel`Kernels`Private`args$___] := (Parallel`Protected`doAutolaunch[TrueQ[Parallel`Static`$enableLaunchFeedback]];
Parallelize[Parallel`Kernels`Private`args$])
```

Documentation Local » | Web »

Options {DistributedContexts :→ $Context, Method → Automatic, ProgressReporting :→ $ProgressReporting}

Attributes {HoldFirst, Protected}

Full Name System`Parallelize

Symbol

PhaseEchos is an option for DixonPhase. Can be Automatic or a list of two integers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`PhaseEchos

Symbol

UnwrapDimension is an option for Unwrap. Can be "2D" or "3D". 2D is for unwarpping 2D images

or unwrapping the individual images from a 3D dataset (does not unwrap in the slice direction). 3D unwraps a 3D dataset in all dimensions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`UnwrapDimension

Symbol

UnwrapMask is an option for DixonPhase. If True the mask is used as weights in UnwrapDCT, which removes artefacts at the mask edges but is slower.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`UnwrapMask

Symbol

UnwrapThresh is an option for Unwrap. Is a value between 0.6 and 0.9, and defines when to unwrap, the higher the value the less unwrapping will be done.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`DixonTools`UnwrapThresh

ElastixTools

Functions

Symbol

ClearQMRIToolsTemp[] clears the temporary directory used by QMRITools for registration.

Documentation Local »

Default Definitions SyntaxInformation[ClearQMRIToolsTemp] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`ClearQMRIToolsTemp

Symbol

ReadTransformParameters[directory] reads the transformation parameters generated

by RegisterData. The directory should be the TempDirectory were the registration is stored. DeleteTempDirectory should be False.

Output is the affine transformation vector per volume.

Documentation Local »

Default Definitions SyntaxInformation[ReadTransformParameters] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`ReadTransformParameters

Symbol

RegisterCardiacData[data] registers the data using a 2D algorithm. data can be 3D or 4D.

RegisterCardiacData[{data,vox}] registers the data series using the given voxel size.

RegisterCardiacData[{data,mask}] registers the data series only using data within the mask.

RegisterCardiacData[{data,mask,vox}] registers the data series using the given voxel size only using data within the mask.

Output is the registered data.

Documentation Local »

Default Definitions SyntaxInformation[RegisterCardiacData] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options › RegistrationTarget → Mean … (18 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`RegisterCardiacData

Symbol

RegisterData[data] registers the data series. If data is 3D it performs multiple 2D registration, if data is 4D it performs multiple 3D registration.

The input data can be in the forms: data, {data, vox}, {data, mask} or {data, mask, vox}.

RegisterData[target, moving] registers the moving data to the target data. target can be 2D or 3D. moving can be the same of one dimension higher than the target.

The inputs target and moving can be in the forms: data, {data, vox}, {data, mask} or {data, mask, vox}.

Output is the registered data with the dimensions of the moving data.

If OutputTransformation is True it also outputs the translation, rotation scale and skew of all images or volumes.

RegisterData[] is based on DOI: 10.1109/TMI.2009.2035616 and 10.3389/fninf.2013.00050.

Documentation	Local »
Default Definitions	SyntaxInformation[RegisterData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options ❯	Iterations → 250 … (17 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`ElastixTools`RegisterData

Symbol

RegisterDataSplit[target, moving] is identical to RegisterData data however left and right side of the data are registered separately.

Splitting the data is done using the function CutData and merged wit Stich data.

Output is the registered data.

Documentation	Local »
Default Definitions	SyntaxInformation[RegisterDataSplit] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options ❯	Iterations → 250 … (18 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`ElastixTools`RegisterDataSplit

Symbol

RegisterDataTransform[target, moving, {moving2nd, vox}] performs the registration exactly as RegisterData. target and moving are the inputs for RegisterData, which can be {data,mask,vox}.

After the registration is done the moving2nd data is deformed according to the output of the registration of moving.

moving2nd can have the same dimensions of moving or one dimension higher (e.g. 3D and 3D or 3D and 4D).

Output is {registered moving, deformed moving2nd}.

Documentation	Local »
Default Definitions	SyntaxInformation[RegisterDataTransform] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options ❯	Iterations → 250 … (19 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`ElastixTools`RegisterDataTransform

Symbol

RegisterDataTransformSplit[target, moving, {moving2nd, vox}] is identical to RegisterDataTransform with the same functionality as RegisterDataSplit.

This means the data is split in two using the function CutData and merged wit Stich data.

Output is {registered moving, deformed moving2nd}.

Documentation	Local »
Default Definitions	SyntaxInformation[RegisterDataTransformSplit] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options ❯	Iterations → 250 … (19 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`ElastixTools`RegisterDataTransformSplit

Symbol

RegisterDiffusionData[{dtiData, vox}] registers a diffusion dataset. dtiData should be 4D {slice, diff, x, y}. vox is the voxel size of the data.

RegisterDiffusionData[{dtiData, dtiMask, vox}] registers the data series using the given voxel size only using data within the mask.

RegisterDiffusionData[{dtiData ,vox}, {anatData, voxAnat}] registers a diffusion dataset. The diffusion data is also registered to the anatData.

RegisterDiffusionData[{dtiData, dtiMask, vox}, {anatData, voxAnat}] registers the data series using the given voxel size only using data within the mask.

RegisterDiffusionData[{dtiData,vox}, {anatData, anatMast, voxAnat}] registers the data series using the given voxel size only using data within the mask.

RegisterDiffusionData[{dtiData, dtiMask, vox}, {anatData, anatMast, voxAnat}] registers the data series using the given voxel size only using data within the mask.

Output is the registered dtiData and, if anatData is given, the registered dtiData in

anatomical space. If OutputTransformation is True it also outputs the translation, rotation scale and skew of all images or volumes.

Documentation Local »

Default Definitions SyntaxInformation[RegisterDiffusionData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options ❯ Iterations → 250 … (24 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`RegisterDiffusionData

Symbol

RegisterDiffusionDataSplit[dtiData, vox] is identical to Register diffusion data however left and right side of the data are registered separately.

RegisterDiffusionDataSplit[{dtiData, vox}, {anatData, voxAnat}] is identical to Register diffusion data however left and right side of the data are registered separately.

RegisterDiffusionDataSplit[{dtiData, dtiMask, vox}, {anatData, anatMast, voxAnat}] is identical to Register diffusion data however left and right side of the data are registered separately.

Splitting the data is done using the function CutData and merged wit Stich data.

Output is the registered data.

Documentation Local »

Default Definitions Options[RegisterDiffusionDataSplit] := Options[RegisterDiffusionData]

SyntaxInformation[RegisterDiffusionDataSplit] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`RegisterDiffusionDataSplit

Symbol

RegisterTensorData[target, moving] registers the moving tensor to the target tensor. After registration the tensor is reorientated appropriately.

The inputs target and moving can be in the forms: tensor, {tensor, vox}, {tensor, mask} or {tensor, mask, vox}.

RegisterTensorData[target, moving, tensor] registers the moving data to the target data and applies the deformation to the tensor.

The inputs target and moving can be in the form: data, {data, vox}, {data, mask} or {data, mask, vox}.

The input tensor can be in the form: tensor or {tensor, vox}.

only "rigid", "affine" and "bspline" can be used as MetReg.

Documentation Local »

Default Definitions SyntaxInformation[RegisterTensorData] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › Iterations → 250 … (17 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`RegisterTensorData

Symbol

TransformData[{data,vox}] deforms the data according to the last output of register data.

The directory should be the TempDirectory were the registration is stored. DeleteTempDirectory should be False.

Documentation Local »

Default Definitions SyntaxInformation[TransformData] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options › TempDirectory → Default … (6 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`TransformData

Symbol

$debugElastix is a debug flag for Elastix functionality.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`ElastixTools`$debugElastix

Symbol

$lastElastixTemp is the last temporary directory used by Elastix.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`ElastixTools`$lastElastixTemp

Options

Symbol

AffineDirections is an option for RegisterData ad RegisterDiffusionData.

It gives the directions in which data can be moved when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`AffineDirections

Symbol

BsplineDirections is an option for RegisterData ad RegisterDiffusionData.

It gives the direction in which the bspline are allowed to move when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`BsplineDirections

Symbol

BsplineSpacing is an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies the spacing of the bspline if the method is "bspline".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`BsplineSpacing

Symbol

DeleteTempDirectory an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies if the temp directory should be deleted after the registration is finished.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`DeleteTempDirectory

Symbol

FindTransform is an option for TransformData and RegisterTransformData. It specifies where to find the transform file.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`FindTransform

Symbol

HistogramBins is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the number of bins of the joined histogram used by the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`HistogramBins

Symbol

HistogramBinsA is an option for RegisterDiffusionData.

It specifies the number of bins of the joined histogram used when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`HistogramBinsA

Symbol

ImportDeformation is an option for TransformData and RegisterTransformData. It specifies if the transformation field which is generated by

Transformix should be imported along side the transformation result.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`ImportDeformation

Symbol

InterpolationOrderReg is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the interpolation order used in the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`InterpolationOrderReg

Symbol

InterpolationOrderRegA is an option for RegisterDiffusionData.

It specifies the interpolation order used in the registration functions when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`InterpolationOrderRegA

Symbol

Iterations is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the number of iterations used by the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`Iterations

Symbol

IterationsA is an option for RegisterDiffusionData.

It specifies the number of iterations used when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`IterationsA

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

MethodReg is an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies which registration method to use.

Methods can be be "translation", "rigid", "affine", "bspline", "rigidDTI", "affineDTI", "PCAtranslation", "PCArigid", "PCAaffine", or "PCAbspline".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`MethodReg

Symbol

MethodRegA is an option for RegisterDiffusionData.

It specifies which registration method to use when registering diffusion data to anatomical space. Methods can be be "rigid","affine" or "bspline".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`MethodRegA

Symbol

NumberSamples is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the number of random samples that are taken each iteration used by the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`NumberSamples

Symbol

NumberSamplesA is an option for RegisterDiffusionData.

It specifies the number of random samples that are taken each iteration when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`NumberSamplesA

Symbol

OutputImage is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies if the result image should be written in the TempDirectory as nii file.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`OutputImage

Symbol

OutputTransformation is an option for RegisterData ad RegisterDiffusionData.

It specifies if the transformation parameters (translation, rotation, scale and skew) should be given as output in the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`OutputTransformation

Symbol

PCAComponents is an option for RegisterData. It specifies how many PCA components are used if method is set to "PCA".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`PCAComponents

Symbol

PrintTempDirectory is an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies if the location of the temp directory should be displayed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`PrintTempDirectory

Symbol

RegistrationTarget is an option for RegisterDiffusionData and RegisterCardiacData. Specifies which target to uses for registration if using "rigid", "affine" or "bspline" as MethodReg.

If the MethodReg is "PCA" based it does not need a target and this options does nothing.

Values can be "First", "Mean" or "Median".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`RegistrationTarget

Symbol

Resolutions is an options for RegisterData, RegisterDiffusionData, and RegisterDataTransform.

It specifies the number of scale space resolutions used by the registration functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`Resolutions

Symbol

ResolutionsA is an option for RegisterDiffusionData.

It specifies the number of scale space resolutions used when registering diffusion data to anatomical space.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`ResolutionsA

Symbol

ShowMetric is an option for RegisterData. If set to True it will print a plot of the Metric.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`ShowMetric

Symbol

SplitMethod is an option for RegisterDataSplit and RegisterDataTransformSplit. values can be "mean", "moving", "target".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`SplitMethod

Symbol

TempDirectory is an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies the temporary directory used to perform and output the registration.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`TempDirectory

Symbol

TransformMethod is an option for RegisterDataSplit and RegisterDataTransformSplit. values can be "Data", "Mask", "Segmentation".

If set to "Mask" a binary mask is expected as the second moving input.

If set to "Segmentation" a multi label segmentation is expected as the second moving input.

Uses SplitSegmentations internally then.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`TransformMethod

Symbol

UseGPU is an option for RegisterData. The value is {bool, gpu} where bool is True or False, and gpu is the gpu ID which is an integer or Automatic.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`UseGPU

FasciculationTools

Functions

Symbol

AnalyzeActivations[actMap, mask] Analysis of the activation map generated from the mask.

AnalyzeActivations[actMap, mask, lab] the same but the output is an association with the name lab, or one name per mask for a 4D mask.

AnalyzeActivations[actMap, mask, vox] the same but with the voxel size vox = {z, y, x} in mm, which adds the columns "ROI Vol [dm3]", "Chance [%/dm3]", "Chance Obs [%/dm3]" and the sizes as area in cm^2 "Mean Area [cm2]", "StDv Area [cm2]", "Median Area [cm2]", "5% Area [cm2]" and "95% Area [cm2]".

AnalyzeActivations[actMap, mask, lab, vox] the same with the name lab and the voxel size vox.

"Amount" is the number of activations and "Amount Obs" the number of observations (a slice in one volume) that contain at least one activation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`FasciculationTools`AnalyzeActivations

Symbol

EvaluateActivation[out] allows to evaluate the activation detection using FindActivations, where out is the output of that function with the option ActivationOutput set to True.

EvaluateActivation[out, actS] The same with the extra analysis of the SelectActivations function output given as actS.

Documentation Local »
Default Definitions SyntaxInformation[EvaluateActivation] = {ArgumentsPattern → {_, _., _., _., _., OptionsPattern[]}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`FasciculationTools`EvaluateActivation

Symbol

FindActivations[data] Finds the activation in MUMRI or DTI data after data normalization.

FindActivations[data, mask] Finds the activation in MUMRI or DTI data after data normalization within the mask.

Documentation Local »
Default Definitions SyntaxInformation[FindActivations] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options › ActivationThreshold → {3., 0.75} … (7 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`FasciculationTools`FindActivations

Symbol

NormalizeFasciculationData[data_, mask, {tens, grad, val}] will normalize DWI data based on the tensor fit.

For each diffusion direction and b−value the estimated unweighted image is calculated.

Documentation Local »

Default Definitions SyntaxInformation[NormalizeFasciculationData] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`NormalizeFasciculationData

Symbol

SelectActivations[act] selects the activations above the given ActivationSize.

SelectActivations[act, vox] selects the activations above the given ActivationSize, the voxel size vox = {z, y, x} in mm is needed when SizeMethod is "Area".

SelectActivations[act, mask] selects the activations above the given ActivationSize within the given mask or masks. The mask can be 3D or 4D.

SelectActivations[act, {mask, back}] selects the activations above the given ActivationSize within the given mask or masks. All voxels outside the back are ignored.

Output is {actSelected, actTotal} is mask is 3D.

Output is {{actSelected, Total[actSelected]}, {actTotal, Total[actTotal]}} is mask is 4D where actSelected and actTotal are per mask.

Documentation Local »

Default Definitions SyntaxInformation[SelectActivations] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ActivationSize → Automatic, SizeMethod → Voxels, IgnoreSlices → {0, 0}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`SelectActivations

Options

Symbol

ActivationBackground is an option for FindActivations. If all normalized signals, which range between 0−150, are below this value the algorithm does nothing.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ActivationBackground

Symbol

ActivationIterations is an option for FindActivations. The maximum number of iteration that can be used for activation detection.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ActivationIterations

Symbol

ActivationOutput is an option for ActivationOutput. If set to All also the mn and threshold values are returned.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ActivationOutput

Symbol

ActivationSize is an option for SelectActivations. Its the minimal size of the

activations selected, in number of connected voxels within a slice if SizeMethod is "Voxels" and in mm^2 if SizeMethod is "Area".

The default is Automatic which is 4 voxels for "Voxels" and 30 mm^2 for "Area". For "Area" the size is rounded up to whole voxels.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ActivationSize

Symbol

ActivationThreshold is an option for FindActivations. Fist value is the number of standard deviations second is the percentage threshold.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ActivationThreshold

Symbol

IgnoreSlices is an option for FindActivations and SelectActivations. determines how many slices of the start and end of the dataset are ignored.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`IgnoreSlices

Symbol

MaskDilation is an option for Mask, SmoothMask and SmoothSegmentation. If the value is greater than 0 it will dilate the mask, if the value is smaller than 0 it will erode the mask.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskDilation

Symbol

SizeMethod is an option for SelectActivations. Values can be "Voxels" or "Area". With "Voxels"

ActivationSize is the number of connected voxels within a slice, with "Area" it is the area in mm^2 for which the voxel size is needed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`SizeMethod

Symbol

ThresholdMethod is an option for FindActivations. Values can be "StandardDeviation", "Fraction" or "Both".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`FasciculationTools`ThresholdMethod

GeneralTools

Functions

Symbol

ApplyCrop[data, crop] applies the cropped region obtained form CropData to the data.

ApplyCrop[data, crop, {voxOrig, voxNew}] applies the cropped region obtained form CropData to the data.

Documentation Local »

Default Definitions SyntaxInformation[ApplyCrop] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options CropAlways → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`ApplyCrop

Symbol

AutoCropData[data] crops the data by removing all background zeros.

Documentation Local »
Default Definitions SyntaxInformation[AutoCropData] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options CropPadding → 5
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`AutoCropData

Symbol

BSplineCurveFit[points] fits a b–spline to the points. Output is a list of same size as points.

Documentation Local »
Default Definitions SyntaxInformation[BSplineCurveFit] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {SplineDegree → 2, SplineKnotsNumber → 50, SplineRegularization → 0}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`BSplineCurveFit

Symbol

ClearTemporaryVariables[] Clear temporary variables.

Documentation Local »
Default Definitions SyntaxInformation[ClearTemporaryVariables] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`ClearTemporaryVariables

Symbol

CompilableFunctions[] generates a formatted table of all compilable functions generated by Compile`CompilerFunctions.

Documentation Local »
Default Definitions SyntaxInformation[CompilableFunctions] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`CompilableFunctions

Symbol

ConvertExtension[filename, extension] checks if file has correct extension. Removes .gz or changes the extension or adds extension if not present.

Documentation Local »
Default Definitions SyntaxInformation[ConvertExtension] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`ConvertExtension

Symbol

CropData[data] creates a dialog window to crop the data (assumes voxel size (1,1,1)).

CropData[data, vox] creates a dialog window to crop the data.

Documentation Local »
Default Definitions SyntaxInformation[CropData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {CropOutput → All, CropInit → Automatic}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`CropData

Symbol

CutData[data] splits the data in two sets left and right base on a automated algorithm assuming leg data.

CutData[data, cut] splits the data in two sets left and right at location cut.

CutData[data, {cut, over}] splits the data in two sets left and right at location cut but with overlap defined by over.

Documentation Local »
Default Definitions SyntaxInformation[CutData] = {ArgumentsPattern → {_, _.}}

Options Monitor → False
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`CutData

Symbol

DataToVector[data] converts the non zero data to vector.

DataToVector[data, mask] converts the data within the mask to vector.

The data can be reconstructed using VectorToData.

Output is the vectorized data and a list containing the original data dimensions and a list with the data coordinates. {vec, {dim,pos}}.

Documentation Local »
Default Definitions SyntaxInformation[DataToVector] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DataToVector

Symbol

DateName[] gives the current date and time in the format "YYYYMMDD-HH".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DateName

Symbol

DecomposeAffineMatrix[mat] decomposes the affine matrix in T, R, S and Q.

Documentation Local »
Default Definitions SyntaxInformation[DecomposeAffineMatrix] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DecomposeAffineMatrix

Symbol

DecomposeScaleMatrix[S] decomposes the scale matrix in S1, S2 and S3.

Documentation Local »
Default Definitions SyntaxInformation[DecomposeScaleMatrix] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DecomposeScaleMatrix

Symbol

DivideNoZero[a, b] divides a/b but when b=0 the result is 0. a can be a number or vector.

Documentation Local »
Default Definitions SyntaxInformation[DivideNoZero] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DivideNoZero

Symbol

DynamicPartition[data, {part}] partitions the data into parts which is a list of integers. The remainders is los.

DynamicPartition[data,part,last] partitions the data into parts which is a list of integers. The remainders is partitioned into equal parts defined by last.

If last is All, the remainders is just one partition.

Documentation Local »
Default Definitions SyntaxInformation[DynamicPartition] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`DynamicPartition

Symbol

EmptyDirectoryQ[dir] checks if directory dir is empty.

Documentation Local »
Default Definitions SyntaxInformation[EmptyDirectoryQ] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`EmptyDirectoryQ

Symbol

ExpNoZero[val] return the Exp of the val which can be any dimension array. if val=0 the output is 0.

Documentation Local »
Default Definitions SyntaxInformation[ExpNoZero] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`ExpNoZero

Symbol

ExtractDemoData[] Extracts the demo data archive.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`ExtractDemoData

Symbol

FileSelect[action] creates a system dialog which returns file/foldername action can be "FileOpen", "FileSave" or "Directory".

FileSelect[action, {type}] same but allows the definition of filetypes for "FileOpen" and "FileSave" e.g. "jpg" or "pdf".

Documentation Local »

Default Definitions SyntaxInformation[FileSelect] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options WindowTitle → Automatic

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`FileSelect

Symbol

FindCrop[data] finds the crop values of the data by removing all zeros surrounding the data.

Documentation Local »

Default Definitions SyntaxInformation[FindCrop] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options CropPadding → 5

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`FindCrop

Symbol

FindMaxDimensions[{data1, data2, ..}] finds the maximal dimensions of all datasets. Each dataset is 3D.

Documentation Local »

Default Definitions SyntaxInformation[FindMaxDimensions] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`FindMaxDimensions

Symbol

FindMiddle[data] find the center of a two leg data, assuming leg data.

FindMiddle[data, "Legs"] same as FindMiddle[data], the only anatomy currently supported.

Documentation	Local »
Default Definitions	SyntaxInformation[FindMiddle] = {ArgumentsPattern → {_, _.}}
Options	Monitor → False
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`GeneralTools`FindMiddle

Symbol

FitGradientMap[data] fits a polynomial gradient field of order 2 to data, sampling every voxel.

FitGradientMap[data, ord] fits a polynomial gradient field of order ord to data, sampling every voxel.

FitGradientMap[data, ord, smp] fits a polynomial gradient field of order ord to data, sampling every smp voxels.

FitGradientMap[{data, mask}, ord, smp] same but only fits within mask.

Documentation	Local »
Default Definitions	SyntaxInformation[FitGradientMap] = {ArgumentsPattern → {_, _., _.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`GeneralTools`FitGradientMap

Symbol

GetAssetLocation[name] Gets the location of the executable assets of the package for the highest installed version.

Current assess are "Elastix", "Transformix" and "DcmToNii".

Documentation	Local »
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`GeneralTools`GetAssetLocation

Symbol

GridData[{data1,data2,...}, part] makes a grid of multiple datasets with part sets on each row.

Documentation Local »
Default Definitions SyntaxInformation[GridData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options Padding → None
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`GridData

Symbol

GridData3D[{data1,data2,...}, part] same as grid data, but only works on 4D data where the data is gridded in axial, coronal and sagittal.

Documentation Local »
Default Definitions SyntaxInformation[GridData3D] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`GridData3D

Symbol

GyromagneticRatio[nuclei] gives the gyromagnetic ratio for the nuclei, e.g. "31P" or "1H", in MHz/T.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`GyromagneticRatio

Symbol

LapFilter[data] Laplacian filter of data with kernel size 0.5.

LapFilter[data, ker] Laplacian filter of data with kernel ker.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`LapFilter

Symbol

LightDarkV[] gives Black for light mode and White for dark mode in version 14.3+ else it gives White.

LightDarkV[light, dark] does the same as LightDarkSwitched but only for version 14.3 or higher.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`LightDarkV

Symbol

LLeastSquares[A, y] = performs a Linear Linear Least Squares fit.

It uses a compiled version of the Pseudo inverse of A.

Documentation Local »

Default Definitions SyntaxInformation[LLeastSquares] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`LLeastSquares

Symbol

LogNoZero[val] return the log of the val which can be any dimension array. if val=0 the output is 0.

Documentation Local »

Default Definitions SyntaxInformation[LogNoZero] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`LogNoZero

Symbol

MADNoZero[vec] return the MAD error of the vec which can be any dimension array. if vec={0...} the output is 0. Zeros are ignored.

Documentation Local »

Default Definitions SyntaxInformation[MADNoZero] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`MADNoZero

Symbol

MakeCoordinates[data, vox] gives the coordinates of every voxel.

MakeCoordinates[dim, vox] gives the coordinates of every voxel for a dataset with dimensions dim.

Documentation Local »
Default Definitions SyntaxInformation[MakeCoordinates] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MakeCoordinates

Symbol

MakeFunctionGraph[function] makes a function dependency graph of the function.

Documentation Local »
Default Definitions SyntaxInformation[MakeFunctionGraph] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {LabelPlacement → Tooltip, AllowSelfDependencies → False, FilterInternal → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MakeFunctionGraph

Symbol

MakeIntFunction[data] makes an interpolation function of the data using voxel size {1, 1, 1} and interpolation order 1

MakeIntFunction[data, int] makes an interpolation function of the data using voxel size {1, 1, 1} and interpolation order int.

MakeIntFunction[data, vox ,int] makes an interpolation function of the data using voxel size vox and interpolation order int.

Documentation Local »
Default Definitions SyntaxInformation[MakeIntFunction] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {CenterVoxel → True, CenterRange → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MakeIntFunction

Symbol

MBCount[expr] gives the memory usage of the expression in MB.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MBCount

Symbol

MeanNoZero[data] calculates the mean of the data ignoring the zeros.

Documentation Local »
Default Definitions SyntaxInformation[MeanNoZero] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MeanNoZero

Symbol

MedFilter[data] Median filter of data with kernel size 1.

MedFilter[data, ker] Median filter of data with kernel ker.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MedFilter

Symbol

MedianNoZero[data] calculates the Median of the data ignoring the zeros.

Documentation Local »
Default Definitions SyntaxInformation[MedianNoZero] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MedianNoZero

Symbol

MemoryUsage[] gives a table of which definitions use up memory.

MemoryUsage[n] gives a table of which definitions use up memory, where n is the amount of definitions to show.

Documentation Local »
Default Definitions SyntaxInformation[MemoryUsage] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MemoryUsage

Symbol

MonitorFunction[x] echos x in a notebook, outside a notebook it prints x as text prefixed with ">> ".

MonitorFunction[x, label] does the same and adds the string label before x.

Any other input is printed as a list. MonitorFunction[x] and MonitorFunction[x, label] return x.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`MonitorFunction

Symbol

NiiFileExistQ[file] checks if the *.nii or *.nii.gz file exists.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`NiiFileExistQ

Symbol

NNLeastSquares[A, y] performs a Non Negative Linear Least Squares fit.

finds an x that solves the linear least−squares problem for the matrix equation A.x==y.

output is the solution x.

Documentation Local »
Default Definitions SyntaxInformation[NNLeastSquares] = {ArgumentsPattern → {_, _}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`NNLeastSquares

Symbol

OpenDemonstrationNotebook[] Opens the demonstration notebook.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`OpenDemonstrationNotebook

Symbol

OpenQMRIToolsLocation[] Opens the QMRITools location in the file explorer.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`OpenQMRIToolsLocation

Symbol

PadToDimensions[data] pads the data to the max dimensions of data, using FindMaxDimensions.

PadToDimensions[data, dim] pads the data to dimensions dim.

Documentation Local »

Default Definitions SyntaxInformation[PadToDimensions] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {PadValue → 0., PadDirection → Center}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`PadToDimensions

Symbol

ParseCommandLine[arguments] parses command line arguments and in QmriTools is used for using arguments given with command line scripts for processing.

Flags start with "--" and the result is an association with the flags as keys. Values following a flag are joined with spaces, a flag without a value gets the value True.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`ParseCommandLine

Symbol

QMRIToolsFuncPrint[] gives a list of all the QMRITools functions with their usage information.

Documentation Local »

Default Definitions SyntaxInformation[QMRIToolsFuncPrint] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`QMRIToolsFuncPrint

Symbol

QMRIToolsFunctions[] give list of all the QMRITools packages, functions and options.

QMRIToolsFunctions[p] print a table with length p of all the QMRITools functions and options.

QMRIToolsFunctions["toolbox"] gives a list of all the functions and options in toolbox.

QMRIToolsFunctions["toolbox", p] gives a table op length p of all the functions and options in toolbox. If toolbox is "All" it will list all toolboxes.

Documentation Local »
Default Definitions SyntaxInformation[QMRIToolsFunctions] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`QMRIToolsFunctions

Symbol

QMRIToolsPackages[] give list of all the QMRITools packages.

Documentation Local »
Default Definitions SyntaxInformation[QMRIToolsPackages] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`QMRIToolsPackages

Symbol

QuaternionToRotationMatrix[{a, b,c,d}] converts quaternion to rotation matrix R.

Documentation Local »
Default Definitions SyntaxInformation[QuaternionToRotationMatrix] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`QuaternionToRotationMatrix

Symbol

QuaternionVectorToRotationMatrix[{b,c,d}] converts quaternion to rotation matrix R.

Documentation Local »

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`QuaternionVectorToRotationMatrix

Symbol

RandomString[n] generates a random string of length n with letters and numbers.

Documentation Local »
Default Definitions SyntaxInformation[RandomString] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RandomString

Symbol

RescaleData[data, dim] rescales the data to given dimensions.

RescaleData[data, {vox1, vox2}] rescales the data from size vox1 to size vox2.

Documentation Local »
Default Definitions SyntaxInformation[RescaleData] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options InterpolationOrder → 3
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RescaleData

Symbol

ReverseCrop[data, dim, crop] reverses the crop on the cropped data with crop values crop to the original size dim.

ReverseCrop[data, dim, crop, {voxOrig, voxNew}] reverses the crop on the cropped data with crop values crop to the original size dim.

Documentation Local »
Default Definitions SyntaxInformation[ReverseCrop] = {ArgumentsPattern → {_, _, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`ReverseCrop

Symbol

ReverseDimensions[data] reverses the dimensions of the data.

Documentation Local »
Default Definitions SyntaxInformation[ReverseDimensions] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`ReverseDimensions

Symbol

RMSNoZero[vec] return the RMS error of the vec which can be any dimension array. if vec={0...} the output is 0. Zeros are ignored.

Documentation Local »
Default Definitions SyntaxInformation[RMSNoZero] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RMSNoZero

Symbol

RotateDimensionsLeft[data] rotates the dimensions of the data one to the left.

RotateDimensionsLeft[data, i] rotates the dimensions of the data i to the left.

Documentation Local »
Default Definitions SyntaxInformation[RotateDimensionsLeft] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RotateDimensionsLeft

Symbol

RotateDimensionsRight[data] rotates the dimensions of the data one to the right.

RotateDimensionsRight[data, i] rotates the dimensions of the data i to the right.

Documentation Local »
Default Definitions SyntaxInformation[RotateDimensionsRight] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RotateDimensionsRight

Symbol

RotationMatrixToQuaternion[R] converts rotation matrix to quaternions {a, b,c,d}.

Documentation Local »
Default Definitions SyntaxInformation[RotationMatrixToQuaternion] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RotationMatrixToQuaternion

Symbol

RotationMatrixToQuaternionVector[R] converts rotation matrix to quaternions {b,c,d}.

Documentation Local »
Default Definitions SyntaxInformation[RotationMatrixToQuaternionVector] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`RotationMatrixToQuaternionVector

Symbol

SaveImage[image] exports graph to image, ImageSize, FileType and ImageResolution can be given as options.

SaveImage[image, "filename"] exports graph to image with "filename", ImageSize, FileType and ImageResolution can be given as options.

Documentation Local »
Default Definitions SyntaxInformation[SaveImage] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options {ImageSize → 6000, FileType → .jpg, ImageResolution → 300}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`SaveImage

Symbol

SetDemoDirectory[] Sets the directory to the demo data directory.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`SetDemoDirectory

Symbol

SignNoZero[val] gives the sign of the val, where the sign of val > 0 is 1 and val < 0 is −1.

Documentation Local »
Default Definitions SyntaxInformation[SignNoZero] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`SignNoZero

Symbol

Squeeze[data] Removes the singleton dimensions from data.

Documentation Local »
Default Definitions SyntaxInformation[Squeeze] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`Squeeze

Symbol

StandardDeviationNoZero[data] calculates the mean of the data ignoring the zeros.

Documentation Local »
Default Definitions SyntaxInformation[StandardDeviationNoZero] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`StandardDeviationNoZero

Symbol

StdFilter[data] StandardDeviation filter of data using gaussian kernel 2.

StdFilter[data, ker] StandardDeviation filter of data using kernel with size ker.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`StdFilter

Symbol

StichData[dataLeft, dataRight] joins left and right part of the data generated by CutData.

Documentation Local »
Default Definitions SyntaxInformation[StichData] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`StichData

Symbol

StringPadInteger[num] converts the integer num to a string and pads it with zeros to length 3.

StringPadInteger[{num, len}] converts the integer num to a string and pads it with zeros to length len.

StringPadInteger[pre, num] the same but with prefix pre.

StringPadInteger[pre, {num, len}] the same but with prefix pre.

StringPadInteger[num, post] the same but with postfix post.

StringPadInteger[{num, len}, post] the same but with postfix post.

StringPadInteger[pre, num, post] the same but with pre and postfix pre and post.

StringPadInteger[post, {num, len}, post] the same but with pre and postfix pre and post.

Documentation Local »

Default Definitions SyntaxInformation[StringPadInteger] = {ArgumentsPattern → {_, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`StringPadInteger

Symbol

SumOfSquares[{data1, data2, ... dataN}] calculates the sum of squares of the datasets.

Output is the SoS and the weights, or just the SoS.

Documentation Local »

Default Definitions SyntaxInformation[SumOfSquares] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options OutputWeights → True

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`SumOfSquares

Symbol

TensMat[tensor] transforms tensor form vector format {xx,yy,zz,xy,xz,yz} to matrix format {{xx,xy,xz},{xy,yy,yz},{xz,yz,zz}}.

Documentation Local »

Default Definitions SyntaxInformation[TensMat] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`TensMat

Symbol

TensVec[tensor] transforms tensor form matrix format {{xx,xy,xz},{xy,yy,yz},{xz,yz,zz}} to vector format {xx,yy,zz,xy,xz,yz}.

Documentation Local »
Default Definitions SyntaxInformation[TensVec] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`TensVec

Symbol

VectorToData[vec, {dim, pos}] converts the vectorized data from DataToVector back to its original Dimensions.

Documentation Local »
Default Definitions SyntaxInformation[VectorToData] = {ArgumentsPattern → {_, {_, _}, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`VectorToData

Options

Symbol

AllowSelfDependencies is an option for MakeFunctionGraph. Can be True or False. If True a function that calls itself is also shown.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`AllowSelfDependencies

Symbol

CenterRange is an option for MakeIntFunction. If set True the centers of the dataset is the origin else its the corner.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`CenterRange

Symbol

CenterVoxel is an option for MakeIntFunction. If set True the centers of the voxels are interpolated else its the corners.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`CenterVoxel

Symbol

CropAlways is an option for ApplyCrop. If set True is will always crop even if outside the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`CropAlways

Symbol

CropInit is an option for CropData. By default the crop is not initialized bu can be with {{xmin,xmax},{ymin,ymax},{zmin,zmax}}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`CropInit

Symbol

CropOutput is an option for CropData, can be "All" (data and crop values), "Data" (just the cropped data) or "Crop" (just the crop values).

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`CropOutput

Symbol

CropPadding is an option for AutoCropData or FindCrop. It specifies how much padding to use around the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`CropPadding

Symbol

FileType["*file*"] gives the type of a file, typically File, Directory or None.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`FileType

Symbol

FilterInternal is an option for MakeFunctionGraph. If set to True it will filter out all internal functions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`FilterInternal

Symbol

ImageResolution is an option for Export, Rasterize, and related functions that specifies at what resolution bitmap images should be rendered.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageResolution

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageSize

Symbol

InterpolationOrder is an option for Interpolation, as well as ListLinePlot, ListPlot3D, ListContourPlot, and related functions, that specifies what order of interpolation to use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`InterpolationOrder

Symbol

LabelPlacement is an option for MakeFunctionGraph. Defines where to place the label of the function graph. Accepts values that can be used in Placed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GeneralTools`LabelPlacement

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »
Attributes {HoldAll, Protected, ReadProtected}
Full Name System`Monitor

Symbol

OutputWeights is an option for SumOfSquares. If True it also output the SoS weights.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`OutputWeights

Symbol

Padding is an option to various array and image operations that specifies what padding to use when extending beyond the original data specified.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`Padding

Symbol

PadDirection is an option for PadToDimensions. It specifies the direction of padding, "Center", "Left" or "Right".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`PadDirection

Symbol

PadValue is an option for PadToDimensions. It specifies the value of the padding.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`PadValue

Symbol

SplineDegree is an option for spline functions and graphics primitives that specifies the degree of polynomial basis to use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`SplineDegree

Symbol

SplineKnotsNumber is an option for BSplineCurveFit and defines how many knots the b–spline has.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`SplineKnotsNumber

Symbol

SplineRegularization is an option for BSplineCurveFit and defines the amount of regularization for the linear fit.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GeneralTools`SplineRegularization

Symbol

System`WindowTitle

Documentation Local » | Web »
Attributes {Protected}
Full Name System`WindowTitle

GradientTools

Functions

Symbol

Bmatrix[bvec, grad] creates bmatrix form grad and bvec in form {–bxx, –byy, –bzz, –bxy, –bxz, –byz ,1}.

Bmatrix[{bvec, grad}] creates bmatrix form grad and bvec in form {bxx, byy, bzz, bxy, bxz, byz}.

Documentation Local »
Default Definitions SyntaxInformation[Bmatrix] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options Method → DTI
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`Bmatrix

Symbol

BmatrixCalc["folder", grads] calculates the true bmatrix from the exported sequence parameters from the philips scanner that are stored in "folder" for each of the gradient directions grads.

Documentation Local »
Default Definitions SyntaxInformation[BmatrixCalc] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › UseGrad → {1, 1, {1, 1}, 1, 1} … (8 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BmatrixCalc

Symbol

BmatrixConv[bm] converts the bmatrix form 7 to 6 or from 6 to 7.

Documentation Local »
Default Definitions SyntaxInformation[BmatrixConv] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BmatrixConv

Symbol

BmatrixInv[bm] generates a bvecotr and gradient directions form a given bmatrx.

BmatrixInv[bm, bvi] generates a bvecotr and gradient directions form a given bmatrx using the given bvalues bvi.

Documentation Local »
Default Definitions SyntaxInformation[BmatrixInv] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BmatrixInv

Symbol

BmatrixRot[bmat, rotmat] Rotates the B–matrix.

Documentation Local »
Default Definitions SyntaxInformation[BmatrixRot] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BmatrixRot

Symbol

BmatrixToggle[bmat, axes, flip], axes can be any order of {"x","y","z"}. flip should be {1,1,1},{1,1,–1},{1,–1,1} or {–1,1,1}.

Documentation Local »
Default Definitions SyntaxInformation[BmatrixToggle] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BmatrixToggle

Symbol

BVector[bvec, grad, gradField] Calculates the actual b–value based on the gradField. Ghis field is calculated by the GradientCoilTensor functions.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`BVector

Symbol

CalculateMoments[{gt, hw, te}, t] calculates the 0th to 3th order moments of the sequence created by GradSeq. Output is {{gt, M0, M1, M2, M3}, vals}.

Documentation Local »
Default Definitions SyntaxInformation[CalculateMoments] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`CalculateMoments

Symbol

ConditionNumberCalc[grads] calcualtes the condition number of the gradient set.

Documentation Local »
Default Definitions SyntaxInformation[ConditionNumberCalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`ConditionNumberCalc

Symbol

ConvertGrads[grad, bv] converts the gradients to txt format, which is needed for FinalGrads.

Documentation Local »
Default Definitions SyntaxInformation[ConvertGrads] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`ConvertGrads

Symbol

CorrectBmatrix[bmat, transformation] corrects the bmatrix bmat with the transformation parameters from RegisterData or RegisterDiffusionData.

Output is the corrected bmatrix.

Documentation Local »
Default Definitions SyntaxInformation[CorrectBmatrix] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options MethodReg → Full
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`CorrectBmatrix

Symbol

CorrectGradients[grad, transformation] corrects the gradient directions grad with the transformation parameters from RegisterData or RegisterDiffusionData.

Output is the corrected gradient vector.

Documentation Local »
Default Definitions SyntaxInformation[CorrectGradients] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options MethodReg → Rotation
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`CorrectGradients

Symbol

EnergyCalc[grads] calcualtes the total Energy of the gradient set.

Documentation Local »
Default Definitions SyntaxInformation[EnergyCalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`EnergyCalc

Symbol

FinalGrads[grtxt,{int,intn},{rand,order}] finalizes the gradient txt file.

grtxt is the output from the function ConvertGrads, which convert the grad to txt format.

int is True or False, if set to True it interleaves b=0 gradients every intn directions.

rand indicates if the gradients need to be randomized, for this it uses the order which is the output of FindOrder.

Documentation Local »
Default Definitions SyntaxInformation[FinalGrads] = {ArgumentsPattern → {_, {_, _, _}, {_, _}}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`FinalGrads

Symbol

FindOrder[grad,bv] finds the optimal order of the gradient directions which minimizes the duty cycle.

The output is needed for FinalGrads.

grad is a list of gradient sets and bv is a list of b–values with the same number as the list of gradient sets.

Documentation Local »

Default Definitions SyntaxInformation[FindOrder] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options OrderSpan → Auto

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`FindOrder

Symbol

FullGrad is an option for Grad. Default is True. When true the gradient directions wil be loaded with the first gradient {0,0,0}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`FullGrad

Symbol

GenerateGradients[numb] optimizes a set with numb gradients, numb mus be an integer.

GenerateGradients[{numb, fixed}] optimizes a set with numb gradients, numb must ba an integer and fixed a list of 3D coordinates e.g. {{0,0,1},{0,1,0}}. The fixed gradients will not be moved.

GenerateGradients[{numb1, numb2 ...}, alpha] optimizes a multi shel gradient set with numb gradients per shel. If alpha is set to 0.5 equal importance is

given tothe optimal distribution of each shell en the enitre set. if alpha is 0 only the sub shels will be optimized, if alpha is set to 1 only the global set wil be optimized.

GenerateGradients[] is based on DOI: 10.1002/mrm.26259 and 10.1002/(SICI)1522–2594(199909)42:3<515::AID–MRM14>3.0.CO;2–Q.

Documentation Local »

Default Definitions SyntaxInformation[GenerateGradients] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › Steps → 1000 … (6 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GenerateGradients

Symbol

GenerateGradientsGUI[] runs the GenerateGradients function in GUI with output for the philips system.

GenerateGradientsGUI[] is based on DOI: 10.1002/mrm.26259 and 10.1002/(SICI)1522–2594(199909)42:3<515::AID–MRM14>3.0.CO;2–Q.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GenerateGradientsGUI

Symbol

GetGradientScanOrder[grad, bval] determines the scanorder based on the txt file provided to the scanner as input.

GetGradientScanOrder[file, grad, bval] determines the scanorder based on the txt file provided to the scanner as input.

Documentation Local »

Default Definitions SyntaxInformation[GetGradientScanOrder] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GetGradientScanOrder

Symbol

GetSliceNormal[file] imports the slice normal from a dicom image.

Documentation Local »

Default Definitions SyntaxInformation[GetSliceNormal] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GetSliceNormal

Symbol

GetSliceNormalDir[file] imports the slice normal from a enhanced dicom image.

Documentation Local »

Default Definitions SyntaxInformation[GetSliceNormalDir] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GetSliceNormalDir

Symbol

GradBmatrix[gt, hw, te, t] Calculates the true bmatrix from the sequence created by GradSeq.

Documentation Local »
Default Definitions SyntaxInformation[GradBmatrix] = {ArgumentsPattern → {_, _, _, _, OptionsPattern[]}}

Options {OutputPlot → False, Method → Analytical, StepSizeI → 0.025}
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`GradBmatrix

Symbol

GradientCoilTensor[mask, vox, off, dint] makes the gradient coil tensor from the mask, with the resolution defined by vox.

The offset off is the left most vox coordinate in the scanner coordinate systeme and the gradient tensor dint comes from the function MakeGradientDerivatives.

Documentation Local »
Default Definitions SyntaxInformation[GradientCoilTensor] = {ArgumentsPattern → {_, _, _, _, OptionsPattern[]}}

Options Method → Vector
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`GradientCoilTensor

Symbol

GradSeq[pars, t, grad] Creates a sequence from the gradient pars imported by ImportGradObj.

Documentation Local »
Default Definitions SyntaxInformation[GradSeq] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options › UseGrad → {0, 1, {1, 0}, 1} … (6 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`GradSeq

Symbol

ImportGradObj[folder] Imports the gradient par files exported from the philips scanner.

Documentation Local »
Default Definitions SyntaxInformation[ImportGradObj] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`GradientTools`ImportGradObj

Symbol

MakeGradientDerivatives[vox, type] generates two interpolation functions using the resolution defined by vox.

One of the gradient field and one of the dx dy and dz of each of the gradient componets.

The type is the gradient of the system which now can be "WA1" or "WA2".

Documentation Local »

Default Definitions SyntaxInformation[MakeGradientDerivatives] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`MakeGradientDerivatives

Symbol

MakeGradientMaps[{int, dint}, fov, vox, start] will produce the gradient maps from the interpolation functions int and dint.

The field of view is defined by fov, the resolution by vox and the start is the left most coordinate in the scanner coordinate system.

The interpolation functions int and dint are generated by MakeGradientDerivatives.

Documentation Local »

Default Definitions SyntaxInformation[MakeGradientMaps] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`MakeGradientMaps

Symbol

OverPlusCalc[grads] determines the minimal overplus factor of of the gradient set.

Documentation Local »

Default Definitions SyntaxInformation[OverPlusCalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`OverPlusCalc

Symbol

SelectBvalueData[{data, vals}, sel] selects the volumes and bvalues based on sel. The value of sel can be a number, for which al higher bvalues are selected, or a list {min, max}.

SelectBvalueData[{data, vals, grads}, sel] the same but also including the gradients.

Output is {data, vals} or {data, vals, grads}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`SelectBvalueData

Symbol

UniqueBvalPosition[bval] generates a list of all the unique bvalues and their positions.

UniqueBvalPosition[bval, num] generates a list of all the unique bvalues and their positions that are present in the dataset equal or more than num times.

Documentation Local »

Default Definitions SyntaxInformation[UniqueBvalPosition] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`UniqueBvalPosition

Options

Symbol

ConditionCalc is an option for GenerateGradients if set to true GenerateGradients will also give the condition number evolution of the system.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`ConditionCalc

Symbol

FlipAxes is an option for GradSeq. Defaul value is {{1,1,1},{1,1,1}}. First three values are for diffusion gradients last three are for the acquisition gradients.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`FlipAxes

Symbol

FlipGrad is an option for GradSeq. When FlipGrad is true the gr180 is fliped.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`FlipGrad

Symbol

FullSphere is an option for GenerateGradients. If set True the gradients will be optimized on a full sphere rather than half a sphere.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`FullSphere

Symbol

GradType is an option GenerateGradients. It specifies what type of gradient set wil be produced, "Normal" or "OverPlus".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`GradType

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

MethodReg is an options for RegisterData, RegisterDiffusionData, RegisterCardiacData and RegisterDataTransform.

It specifies which registration method to use.

Methods can be be "translation", "rigid", "affine", "bspline", "rigidDTI", "affineDTI", "PCAtranslation", "PCArigid", "PCAaffine", or "PCAbspline".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ElastixTools`MethodReg

Symbol

OrderSpan is an options for FindOrder.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`OrderSpan

Symbol

OutputPlot is an option for GradBmatrix. It specifies if the plots of the gradients should also be exported.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`OutputPlot

Symbol

OutputType is an option for BmatrixCalc. Values can be "Matrix" of "Gradients".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`OutputType

Symbol

PhaseEncoding is an options of GradSeq. Values can be "A", "P", "R" and "L".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`PhaseEncoding

Symbol

Runs is an option for GenerateGradients. Set how often the minimization function is run. The best solution of all runs is the output. Default value is 1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`Runs

Symbol

Steps is an option GenerateGrads and is the number of step that is used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`Steps

Symbol

StepSizeI is an option for GradBmatrix. Specifies the integration stepsize is Method –> "Numerical" is used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`StepSizeI

Symbol

SwitchAxes is an option for GradSeq. Defaul value is {{1,2,3},{1,2,3}}. First three values are for diffusion gradients last three are for the acquisition gradients.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`SwitchAxes

Symbol

UnitMulti is an option for GradSeq. Defaul value is 10^–3. Defines the scaling of the gradient strength.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`UnitMulti

Symbol

UseGrad is an option for GradSeq. The default value is {0, 1, {1, 0}, 1} where {grex, gr180, {grepi1, grepi2}, grdiff, grflow}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`UseGrad

Symbol

VisualOpt is an option for GenerateGradients. Show the minimization proces of eacht calculation step. Default is False.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`GradientTools`VisualOpt

IVIMTools

Functions

Symbol

IVIMCalc[data, bInp, init] calculates the IVIM fit.

data should be 1D ,2D, 3D or 4D.

bInp should be full bmatrix which can be calculated from the bvecs en bvals using Bmatrix with the bvalues in s/mm^2.

init should are the initialization parameters for 2 comp this is {s0, f, D, Dp} for 3 comp this is {s0, f1, f2, D, Dp1, Dp2}.

The fraction is defined between 0 and 1, the D, Dp, Dp1 and Dp2 is in mm^2/s.

output is {s0, f1, D, pD1} or {s0, f1, f2, D, pD1, pD2}.

Documentation Local »

Default Definitions SyntaxInformation[IVIMCalc] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › Method → Automatic ... (6 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`IVIMCalc

Symbol

IVIMCorrectData[data, {s0, f, pdc}, bval] removes the ivim signal from the data.

data is the original data.

{s0, f, pdc} are the solution to a 2 compartment IVIM fit using IVIMCalc or BayesianIVIMFit2.

bval are the bvalues.

The fraction is defined between 0 and 1, the pdc is in mm^2/s.

output is the corrected data.

Documentation	Local »
Default Definitions	SyntaxInformation[IVIMCorrectData] = {ArgumentsPattern → {_, {_, _, _}, _, OptionsPattern[]}}
Options	{FilterMaps → True, FilterType → Median, FilterSize → 1}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`IVIMTools`IVIMCorrectData

Symbol

IVIMFunction[] gives the IVIM function with 2 comps.

IVIMFunction[comp] gives the IVIM function.

IVIMFunction[comp, type] gives the IVIM function.

type can be "Normal" or "Exp".

comp can be 2 or 3.

output is the function with b, s0, f1, f2, D, pD1, pD2 as parameters. The fraction is defined between 0 and 1, the D, Dp, Dp1 and Dp2 is in mm^2/s.

Documentation	Local »
Default Definitions	SyntaxInformation[IVIMFunction] = {ArgumentsPattern → {_, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`IVIMTools`IVIMFunction

Symbol

IVIMResiduals[data, bInp, pars] calculates the root mean square residuals of an IVIM fit using IVIMCalc, BayesianIVIMFit2 or BayesianIVIMFit3.

Documentation Local »
Default Definitions SyntaxInformation[IVIMResiduals] = {ArgumentsPattern → {_, _, _}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`IVIMTools`IVIMResiduals

Symbol

IVIMTensFit is an option for IVIMCalc. When set True the tissue diffusion component wil be calculated as a tensor.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`IVIMTools`IVIMTensFit

Symbol

MeanBvalueSignal[data, bval] calculates the geometric mean of the data for each unique bval.

output is the mean data and the unique bvalues.

Documentation Local »
Default Definitions SyntaxInformation[MeanBvalueSignal] = {ArgumentsPattern → {_, _}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`IVIMTools`MeanBvalueSignal

Options

Symbol

FilterMaps is an option for IVIMCorrectData. If True the IVIM parameter maps are filtered before signal correction.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`IVIMTools`FilterMaps

Symbol

FilterSize is an option for IVIMCorrectData. If FilterMaps is True it gives the kernel size.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`FilterSize

Symbol

FilterType is an option for IVIMCorrectData. If FilterMaps is True it tells which filter to use. can be "Median" of "Gaussian".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`FilterType

Symbol

IVIMComponents is an option for IVIMCalc. Default value is 2, the tissue and the blood component. can also be set to 3.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`IVIMComponents

Symbol

IVIMConstrained is an option for IVIMCalc. When set True the fit wil be constrained to the values given in IVIMConstrains.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`IVIMConstrained

Symbol

IVIMConstrains is an option for IVIMCalc.

Default values are: {{0.8, 1.2}, {0, 1}, {0.0005, 0.0035}, {0.005, 0.5}, {0.002, 0.015}}.

Where {{s0 in percentage},{fractions},{tissue diffusion},{blood compartment Dp},{third compartment}}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`IVIMConstrains

Symbol

IVIMFixed is an option for IVIMCalc and the default value is False.

When set True the pseudo diffusion wil be fixed to the parameter given as init.

When set to "One" only the fast component of a 3 compartment fit is fixed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`IVIMTools`IVIMFixed

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

Parallelize[*expr*] evaluates *expr* using automatic parallelization.

Definitions

```
Parallelize[Parallel`Kernels`Private`args$___] := (Parallel`Protected`doAutolaunch[TrueQ[Parallel`Static`$enableLaunchFeedback]];
Parallelize[Parallel`Kernels`Private`args$])
```

Documentation Local » | Web »

Options {DistributedContexts :→ $Context, Method → Automatic, ProgressReporting :→ $ProgressReporting}

Attributes {HoldFirst, Protected}

Full Name System`Parallelize

JcouplingTools

Functions

Symbol

GetSpinSystem[name] get a spin system that can be used in SimHamiltonian. Current implements systems are "glu", "lac", "gaba", "fatGly", "fatAll", "fatEnd", "fatDouble", "fatStart", and "fatMet".

Documentation Local »
Default Definitions SyntaxInformation[GetSpinSystem] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options CenterFrequency → 0
Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`GetSpinSystem

Symbol

MakeSpinSystem[name, freqs, jcoup] makes a spin system for jcoupling simulations. The with name is defined by the freqs of the nuclei and the jcoup values {{n1, nx}, j} between nuclei.

MakeSpinSystem[{name,labs}, freqs, jcoup] same but each nuclei has a specific name, e.g.{"ATP", {"γ","α","β"}.

MakeSpinSystem[name, freqs, jcoup, scales] same but each nuclei has a scale, default scales are 1.

MakeSpinSystem[{name,labs}, freqs, jcoup, scales] same as alle before.

Documentation Local »
Default Definitions SyntaxInformation[MakeSpinSystem] = {ArgumentsPattern → {_, _., _., _., OptionsPattern[]}}

Options CenterFrequency → 0
Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`MakeSpinSystem

Symbol

SequencePulseAcquire[din, ham] performs a pulse acquire experiment of the spin system din given the hamiltonian ham with a 90 Degree pulse.

SequencePulseAcquire[din, ham, b1] performs a pulse acquire experiment of the spin system din given the hamiltonian ham with a 90 Degree pulse and b1.

The output is a new spin system dOut.

Documentation Local »
Default Definitions SyntaxInformation[SequencePulseAcquire] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`SequencePulseAcquire

Symbol

SequenceSpaceEcho[din, ham, t1, t2, nEcho, b1] performs a multi echo spin echo

experiment with a 90 degree spin echo, with t1 the time between the 90 degree RF pulse and the first 180 degree RF pulse,

t2 the time between a 180 degree RF pulse and the following readout (and 2xt1 the time between two consecutive 180 degree RF pulses.

Further defines nEcho the number of 180 degree RF pulses, din the spin system given the hamiltonian ham using b1.

The t1 and t2 are defined in ms, and b1 of 100% is defines as 1.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SequenceSpaceEcho] = {ArgumentPattern → {_, _, _, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SequenceSpaceEcho

Symbol

SequenceSpinEcho[din, ham, te] performs a spin echo experiment with echo time te of the spin system din given the hamiltonian ham with a 90 and 180 Degree pulse.

SequenceSpinEcho[din, ham, te, b1] performs a spin echo experiment with echo time te of the spin system din given the hamiltonian ham with a 90 and 180 Degree pulse and b1.

The te is defined in ms and the b1 of 100% is defined as 1.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SequenceSpinEcho] = {ArgumentsPattern → {_, _, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SequenceSpinEcho

Symbol

SequenceSteam[din, ham, {te, tm}] performs a stimulated echo experiment with echo time te and mixing time tm of the spin system din given the hamiltonian ham with 3 90 Degree pulses.

The te and tm are defined in ms.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SequenceSteam] = {ArgumentsPattern → {_, _, {_, _}}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SequenceSteam

Symbol

SequenceTSE[din ,ham, {te, nEcho}, {ex, ref}] performs a multi echo spin echo experiment with

echo time te with nEcho echos of the spin system din given the hamiltonian ham using ex Degree excitation and ref Degree refocus pulses.

SequenceTSE[din ,ham, {te, nEcho}, {ex, ref}, b1] performs a multi echo spin echo experiment with echo

time te with nEcho echos of the spin system din given the hamiltonian ham using ex Degree excitation and ref Degree refocus pulses and b1.

The te is defined in ms, the ex and ref are defined in degree and b1 of 100% is defined as 1.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SequenceTSE] = {ArgumentsPattern → {_, _, {_, _, _}, {_, _}, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SequenceTSE

Symbol

SimAddPhase[din ,ham ,phase] adds phase to the spin system din given the hamiltonian ham.

din and ham are generated by SimHamiltonian.

The phase is defined in degree.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SimAddPhase] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SimAddPhase

Symbol

SimEvolve[din,ham,t] evolves the spin system din given the hamiltonian ham over a time t. din and ham are generated by SimHamiltonian.

The output is a new spin system dOut.

Documentation Local »

Default Definitions SyntaxInformation[SimEvolve] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SimEvolve

Symbol

SimHamiltonian[sys] simulates the hamiltonian for a given spin system. The spin system is generated by GetSpinSystem.

The output is the spin system and hamiltonian structure.

SimHamiltonian[] is based on DOI: 10.1016/j.jmr.2010.12.008 and 10.1002/mrm.24340.

Documentation	Local »
Default Definitions	SyntaxInformation[SimHamiltonian] = {ArgumentsPattern → {_, OptionsPattern[]}}
Options	{FieldStrength → 3, SimNucleus → 1H, CenterFrequency → 0}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`JcouplingTools`SimHamiltonian

Symbol

SimReadout[din, ham] performs a readout of a spin system din with hamiltonian ham.

Output is {time,fids,ppm,spec,dOut}, which are the free induction decay fids with its time, the spectrum spec with its ppm and the evolved spin system dOut.

Documentation	Local »
Default Definitions	SyntaxInformation[SimReadout] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options	ReadoutOutput → all … (8 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`JcouplingTools`SimReadout

Symbol

SimRotate[din, ham ,angle] rotates the spin system din given the hamiltonian ham over angle with phase 90 degrees.

SimRotate[din, ham ,angle, phase] rotates the spin system din given the hamiltonian ham over angle with phase.

din and ham are generated by SimHamiltonian.

The angle and phase are defined in degree.

The output is a new spin system dOut.

Documentation	Local »
Default Definitions	SyntaxInformation[SimRotate] = {ArgumentsPattern → {_, _, _, _.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`JcouplingTools`SimRotate

Symbol

SimSignal[din, ham] performs a readout of a spin system din with hamiltonian ham.

Output is the complex signal.

Documentation Local »
Default Definitions SyntaxInformation[SimSignal] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options ReadoutOutput → all
Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`SimSignal

Symbol

SimSpoil[din] spoils all the non zeroth order states of a spin system.

The output is a new spin system dOut.

Documentation Local »
Default Definitions SyntaxInformation[SimSpoil] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`SimSpoil

Symbol

SysTable[sys] shows the spin system as a table. The spin system is obtained form GetSpinSystem.

Documentation Local »
Default Definitions SyntaxInformation[SysTable] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`SysTable

Options

Symbol

CenterFrequency is an option for GetSpinSystem and defines the center frequency in ppm.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`JcouplingTools`CenterFrequency

Symbol

FieldStrength is an option for SimHamiltonian. It defines the field strength for which the hamiltonian is calculated defined in Tesla.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`FieldStrength

Symbol

Linewidth is an option for SimReadout and defines the spectral linewidth in Hz.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`Linewidth

Symbol

LinewidthShape is an option for SimReadout and defines the linewidth shape, values can be "Lorentzian", "Gaussian" or "Voigt".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`LinewidthShape

Symbol

ReadoutBandwidth is an option for SimReadout defines the spectral bandwidth in Hz.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`ReadoutBandwidth

Symbol

ReadoutMethod is an option of SimReadout and can be "Fid" or "Echo". With "Fid" it is also possible to define a delay time in ms {"Fid", delay}.

With "Echo" it is also possible to define a delay time in ms {"Echo", delay} and it than assumes te is half the readout, or a custom te can be defined {"Echo", delay, te}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`ReadoutMethod

Symbol

ReadoutOutput is an option for SimReadout and SimSignal and values can be "all" and

"each". When set to "all" the total signal and signal is given, when set to "each" the signal or spectrum for each peak is given separately.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`ReadoutOutput

Symbol

ReadoutPhase is an option for SimReadout and defines the readout phase in degrees.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`ReadoutPhase

Symbol

ReadoutSamples is an option for SimReadout and defines the number of readout samples for the spectrum.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`ReadoutSamples

Symbol

SimNucleus is an option for SimHamiltonian. It defines the nucleus for which to simulate the spectra.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`SimNucleus

LoggingTools

Functions

Symbol

AddToLog[list] add the list to the log at level 1. All elements of the list are converted to strings and joined with spaces.

AddToLog[list, level] add the list to the log at level.

AddToLog[list, True] add the list to the log at level 1 with a timestamp.

AddToLog[list, True, level] specifies both the level and the timestamp.

AddToLog[list, level, True] specifies both the level and the timestamp.

Documentation	Local »
Default Definitions	SyntaxInformation[AddToLog] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}
Options	SaveLogFile → True
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`LoggingTools`AddToLog

Symbol

CheckFile[{fol, name, tag}, stat, ver] checs a checkfile for status stat and software version.

Documentation	Local »
Default Definitions	SyntaxInformation[CheckFile] = {ArgumentsPattern → {{_, _, _}, _, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`LoggingTools`CheckFile

Symbol

DirectoryTree[dir] makes a directory tree from the dir.

Documentation	Local »
Default Definitions	SyntaxInformation[DirectoryTree] = {ArgumentsPattern → {_}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`LoggingTools`DirectoryTree

Symbol

ExportDirectoryTree[fol] make a DirectoryTree for the folder and exports it there as FileTree.txt.

ExportDirectoryTree[fol,name] make a DirectoryTree for the folder and exports it there as name .txt.

Documentation Local »
Default Definitions SyntaxInformation[ExportDirectoryTree] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`ExportDirectoryTree

Symbol

ExportLog[file] exports the log as a plain text to file.

Documentation Local »
Default Definitions SyntaxInformation[ExportLog] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`ExportLog

Symbol

ImportLog[file] imports the log as a list of string from a plain text file.

Documentation Local »
Default Definitions SyntaxInformation[ImportLog] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`ImportLog

Symbol

MakeCheckFile[{fol, name, tag}, stat] makes a check json file in the fol with

filename containing name and tag. Where stat is a list of rules that contain any information that needs to be put in the check file.

Documentation Local »
Default Definitions SyntaxInformation[MakeCheckFile] = {ArgumentsPattern → {{_, _, _}, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`MakeCheckFile

Symbol

PrintDirectoryTree[tree] prints the directory tree generated by DirectoryTree.

Documentation Local »
Default Definitions SyntaxInformation[PrintDirectoryTree] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`PrintDirectoryTree

Symbol

ResetLog[] resets the log to {}.

Documentation Local »
Default Definitions SyntaxInformation[ResetLog] = {ArgumentsPattern → {}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`ResetLog

Symbol

SetLogFile[file] sets the log file name to file.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`SetLogFile

Symbol

ShowLog[] shows the log in a popup window.

ShowLog[False] shows the log in the notebook.

Documentation Local »
Default Definitions SyntaxInformation[ShowLog] = {ArgumentsPattern → {_.}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`LoggingTools`ShowLog

Options

Symbol

SaveLogFile is an option for AddToLog. If True the log is saved to the log file each time something is added.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`LoggingTools`SaveLogFile

MaskingTools

Functions

Symbol

DilateMask[mask,size] if size > 0 the mask is dilated and if size < 0 the mask is eroded.

Documentation Local »

Default Definitions SyntaxInformation[DilateMask] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`DilateMask

Symbol

GetCommonSegmentation[dat, seg, vox] For a list of multiple datasets dat the common segmentations from the list seg are determined.

Output is a list of segmentations where for each region only the part present in all datasets is selected.

Documentation Local »

Default Definitions SyntaxInformation[GetCommonSegmentation] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`GetCommonSegmentation

Symbol

GetSegmentationLabels[segmentation] gives a list of all labels in the segmentation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`GetSegmentationLabels

Symbol

HomogenizeData[data, mask] tries to homogenize the data within the mask by removing intensity gradients.

Documentation Local »
Default Definitions SyntaxInformation[HomogenizeData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options FitOrder → 5
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`HomogenizeData

Symbol

JoinSegmentations[seg, joinRules] joins the segmentations in seg according to the rules in joinRules.

JoinRules is a list of rules {{join, new}..} where join is a list of labels to be joined and new is the new label number.

For example {{1, 2}, 3} joins the labels 1 and 2 to label 3. A single rule {join, new} can also be given.

Documentation Local »
Default Definitions SyntaxInformation[JoinSegmentations] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`JoinSegmentations

Symbol

Mask[data] creates a mask by automatically finding a threshold.

Mask[data, min]creates a mask which selects only data above the min value.

Mask[data,{min,max}] creates a mask which selects data between the min and max value.

Documentation Local »
Default Definitions SyntaxInformation[Mask] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › MaskSmoothing → False … (6 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`Mask

Symbol

MaskData[data, mask] applies a mask to data. mask can be 2D or 3D, data can be 2D, 3D or 4D.

Documentation Local »

Default Definitions SyntaxInformation[MaskData] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskData

Symbol

MaskSegmentation[seg, mask] applies a mask to a spited segmentation seg from SplitSegmentations.

The mask is 3D, seg is 4D.

Documentation Local »

Default Definitions SyntaxInformation[MaskSegmentation] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskSegmentation

Symbol

MaskVolume[mask] calculates the volume of the mask.

MaskVolume[{mask, ..}] calculates the Median volume of the List of masks.

MaskVolume[mask, vox] calculates the volume of the mask in cm^3 where vox is in mm.

MaskVolume[{mask, ..}, vox] calculates the Median volume of the List of masks in cm^3 where vox is in mm.

Documentation Local »

Default Definitions SyntaxInformation[MaskVolume] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskVolume

Symbol

MergeSegmentations[masks, labels] generates an ITKsnap or slices3D compatible segmentation from individual masks and label numbers.

Output is a labeled segmentation.

MergeSegmentations[masks] does the same but automatically numbers the segmentations.

The masks are {z, masks, y, x}. Voxels where masks overlap are set to 0.

Documentation Local »
Default Definitions SyntaxInformation[MergeSegmentations] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`MergeSegmentations

Symbol

NormalizeData[data] normalizes the data to the mean signal of the data. For 4D data it normalizes to the first volume of the 4th dimension.

NormalizeData[data, mask] normalizes the data based on the mean signal only within the mask.

Documentation Local »
Default Definitions SyntaxInformation[NormalizeData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options NormalizeMethod → Set
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`NormalizeData

Symbol

NormalizeMeanData[data] calculates the mean normalized data from a 4D dataset.

Documentation Local »
Default Definitions SyntaxInformation[NormalizeMeanData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options NormalizeMethod → Set
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`NormalizeMeanData

Symbol

RemoveMaskOverlaps[mask] removes the overlaps between multiple masks. Mask is a 4D dataset with {z, masks, x, y}.

Documentation Local »

Default Definitions SyntaxInformation[RemoveMaskOverlaps] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`RemoveMaskOverlaps

Symbol

ReplaceSegmentations[seg, labs, new] replaces the labels labs from the segmentation seg for labels new. Both labs and new should be lists of integers of the same size. If seg contains more labels then given in labs these will be replaced by 0.

ReplaceSegmentations[seg, labs] replaces the labels labs by a random permutation of Range[Length[labs]], using a fixed random seed.

ReplaceSegmentations[seg] does the same for all labels in seg.

When labs is given, the input seg can also be a split segmentation {masks, labels}, in which case the output is {masks, labels} sorted by label.

Documentation Local »

Default Definitions SyntaxInformation[ReplaceSegmentations] = {ArgumentsPattern → {_, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`ReplaceSegmentations

Symbol

RescaleSegmentation[data, dim] rescales segmentations to given dimensions.

RescaleSegmentation[data, {vox1, vox2}] rescales segmentations from voxel size vox1 to voxel size vox2.

Documentation Local »

Default Definitions SyntaxInformation[RescaleSegmentation] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`RescaleSegmentation

Symbol

SegmentationCrossSection[seg] calculates the maximum cross section of each label in the segmentation in voxels.

SegmentationCrossSection[seg, vox] calculates the maximum cross section of each label in the segmentation in cm^2 where vox is in mm.

Documentation Local »
Default Definitions SyntaxInformation[SegmentationCrossSection] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options Method → Max
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`SegmentationCrossSection

Symbol

SegmentationVolume[seg] calculates the volume of each label in the segmentation in voxels.

SegmentationVolume[seg, vox] calculates the volume of each label in the segmentation in cm^3 where vox is in mm.

Documentation Local »
Default Definitions SyntaxInformation[SegmentationVolume] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`SegmentationVolume

Symbol

SegmentMask[mask, n] divides a mask in n segments along the slice direction, n must be an integer. The mask is divided in n equal parts where each parts has the same number of slices.

Documentation Local »
Default Definitions SyntaxInformation[SegmentMask] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`SegmentMask

Symbol

SelectMaskComponents[mask] selects the largest connected component in the mask.

SelectMaskComponents[mask,n] selects the n largest connected components in the mask.

Documentation Local »
Default Definitions SyntaxInformation[SelectMaskComponents] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`SelectMaskComponents

Symbol

SelectSegmentations[seg, labs] selects only the segmentations from seg with label number labs and returns a labeled segmentation.

SelectSegmentations[seg, labs, join] does the same but if join is False the output is split as {masks, labels}, see SplitSegmentations.

The input seg can also be a split segmentation {masks, labels}, in which case the output is always {masks, labels} sorted by label.

Documentation Local »

Default Definitions SyntaxInformation[SelectSegmentations] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`SelectSegmentations

Symbol

SmoothMask[mask] generates one clean masked volume form a noisy mask.

Documentation Local »

Default Definitions SyntaxInformation[SmoothMask] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ❯ MaskComponents → 1 … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`SmoothMask

Symbol

SmoothSegmentation[segmentation] smooths segmentations and removes the overlaps between multiple segmentations.

SmoothSegmentation[segmentation, labs] only smooths the selected label number labs.

Documentation Local »

Default Definitions SyntaxInformation[SmoothSegmentation] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options ❯ MaskComponents → 1 … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`SmoothSegmentation

Symbol

SplitSegmentations[segmentation] splits a label mask from ITKSnap or slicer3D in separate masks and label numbers.

Output is masks and label numbers, {mask, labs}.

Documentation Local »

Default Definitions SyntaxInformation[SplitSegmentations] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`SplitSegmentations

Options

Symbol

FitOrder is an option for HomogenizeData. It specifies the order of harmonics to be used for the homogenization.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`FitOrder

Symbol

MaskClosing is an option for Mask, SmoothMask and SmoothSegmentation. The size of the holes in the mask that will be closed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskClosing

Symbol

MaskComponents is an option for Mask, SmoothMask and SmoothSegmentation. Determines the amount of largest clusters used as mask.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskComponents

Symbol

MaskDilation is an option for Mask, SmoothMask and SmoothSegmentation. If the value is greater than 0 it will dilate the mask, if the value is smaller than 0 it will erode the mask.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskDilation

Symbol

MaskFiltKernel is an option for Mask, SmoothMask and SmoothSegmentation. How much the contours are smoothed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskFiltKernel

Symbol

MaskSmoothing is an options for Mask, SmoothMask and SmoothSegmentation, if set to True it smooths the mask, by closing holes and smoothing the contours.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`MaskSmoothing

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

NormalizeMethod is an option for NormalizeData. Can be "Set" or "Volumes" which normalizes to the first volume or normalizes each volume individually, respectively.

If "Uniform" normalizes the histogram of the data to have a uniform distribution between 0 and 1 where 0 is treated as background of the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`NormalizeMethod

Symbol

SmoothIterations is an option for Mask, SmoothMask and SmoothSegmentation and defines how often the smoothing is repeated.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MaskingTools`SmoothIterations

MuscleBidsTools

Functions

Symbol

BidsDcmToNii[dir] converts the bids sourceFolder with dicom files to raw nii files based on the config file.

Documentation Local »
Default Definitions SyntaxInformation[BidsDcmToNii] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {BidsIncludeSession → True, ProcessSubjects → All}
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`BidsDcmToNii

Symbol

CheckBidsOutput[dir, step] checks for each subject if the output expected from the config file in dir is present after the given step.

Currently step can be "Convert", which checks for each Label if the json and nii files are present in the rawData folder.

Missing files are highlighted and can be shown only using the checkbox.

Documentation Local »
Default Definitions SyntaxInformation[CheckBidsOutput] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options ProcessSubjects → All
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`CheckBidsOutput

Symbol

CheckConfigLabels[dir] checks for each subject and session if the protocol names that are defined as Label in the config file in dir are present. Missing labels are highlighted and can be shown only using the checkbox.

Only subjects in the rawData folder are shown, so run it after BidsDcmToNii.

CheckConfigLabels[dir, subs] does the same for the subjects subs.

Documentation Local »
Default Definitions SyntaxInformation[CheckConfigLabels] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`CheckConfigLabels

Symbol

CheckDataDescription[description, method] checks the data description config file used in BidsDcmToNii, MuscleBidsConvert, MuscleBidsProcess, and MuscleBidsMerge.

Documentation Local »
Default Definitions SyntaxInformation[CheckDataDescription] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`CheckDataDescription

Symbol

GenerateBidsFileName[parts] generates a Bids file name from the Bids labels association which can be generated by PartitionBidsName.

GenerateBidsFileName[fol, parts] the same but with a custom root folder.

Documentation Local »
Default Definitions SyntaxInformation[GenerateBidsFileName] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GenerateBidsFileName

Symbol

GenerateBidsFolderName[parts] generates a Bids folder name from the Bids labels association which can be generated by PartitionBidsFolderName.

GenerateBidsFolderName[fol, parts] the same but with a custom root folder.

Documentation Local »
Default Definitions SyntaxInformation[GenerateBidsFolderName] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GenerateBidsFolderName

Symbol

GenerateBidsName[parts] generates a Bids file name from the Bids labels association which can be generated by PartitionBidsName. The labels are joined in the order {"sub","ses","vol","stk","rep","chunk","acq","part","type","suf"}.

Documentation Local »
Default Definitions SyntaxInformation[GenerateBidsName] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GenerateBidsName

Symbol

GetConfig[folder] imports a Muscle Bids config file from the given folder.

GetConfig[folder, name] same but for a config file with the given name.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GetConfig

Symbol

GetJSONPosition[{json..}, {{key, value}..}] gets the position from a list of JSON association lists where keys have the given value.

GetJSONPosition[{json..}, {{key, value}..}, sortKey] same but finally sorts the positions for the value of the sortKey.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GetJSONPosition

Symbol

GetProtocolNames[config, subs] gets the list of protocol names in the specified subject folder.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`GetProtocolNames

Symbol

ImportJSON[file] imports a JSON file as rawJSON.

Documentation Local »
Default Definitions SyntaxInformation[ImportJSON] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`ImportJSON

Symbol

ManageCheckFiles[dir] shows per processing step the check files expected from the config file in dir and their status.

Selected check files can be deleted such that the next run of that step processes the data again. Reload scans again.

The options are ProcessSubjects to limit the subjects and BidsOutputImages to define the expected Analysis images.

Documentation Local »
Default Definitions SyntaxInformation[ManageCheckFiles] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ProcessSubjects → All, BidsOutputImages → All}
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`ManageCheckFiles

Symbol

MergeConfig[assoc, replace] merges the replace association with the assoc association.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`MergeConfig

Symbol

MergeJSON[{json..}] merges a list of JSON association lists where duplicate keys with the same values are removed and duplicate keys with different values are merged.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`MergeJSON

Symbol

MuscleBidsAnalysis[dir] performs analysis on the Muscle–Bids named nii based on the config file in the bids sourceFolder dir. If a segmentation is present it is used to calculate the mean per segmentation.

Documentation Local »
Default Definitions SyntaxInformation[MuscleBidsAnalysis] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ProcessSubjects → All, VersionCheck → False, BidsOutputImages → All}
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`MuscleBidsAnalysis

Symbol

MuscleBidsCheck[dir] shows ViewProtocolNames, CheckConfigLabels, CheckBidsOutput and ManageCheckFiles for the config file in dir as tabs.

The Custom configs tab compares the custom config of each subject with its copies in the other stages.

The Report tab exports the problems of a step, or of all steps, as md or pdf report.

The data is scanned when the function is called, the Reload buttons scan again. The options ProcessSubjects and BidsOutputImages are passed on.

Documentation Local »
Default Definitions SyntaxInformation[MuscleBidsCheck] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ProcessSubjects → All, BidsOutputImages → All}
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`MuscleBidsCheck

Symbol

MuscleBidsConvert[dir] converts all raw nii data in the to Muscle–Bids named nii based on the config file in the bids sourceFolder dir.

Documentation Local »
Default Definitions SyntaxInformation[MuscleBidsConvert] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {DeleteAfterConversion → True, ProcessSubjects → All}
Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`MuscleBidsConvert

Symbol

MuscleBidsGUI[dir] shows a GUI to run the Muscle Bids processing steps for the config file in dir, with the options of each step and the subjects to process.

Each step runs in a separate kernel, nothing is printed in the notebook and the progress is shown from the log files.

The tabs Check and Config show MuscleBidsCheck and ViewConfig.

Documentation Local »

Default Definitions SyntaxInformation[MuscleBidsGUI] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsGUI

Symbol

MuscleBidsMerge[dir] merges multiple stack data for all Muscle–Bids named nii based on the config file in the bids sourceFolder dir.

Documentation Local »

Default Definitions SyntaxInformation[MuscleBidsMerge] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ProcessSubjects → All, VersionCheck → False}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsMerge

Symbol

MuscleBidsProcess[dir] processes all Muscle–Bids named nii based on the config file in the bids sourceFolder dir.

Documentation Local »

Default Definitions SyntaxInformation[MuscleBidsProcess] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ProcessSubjects → All, VersionCheck → False}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsProcess

Symbol

MuscleBidsReport[dir, step, file] writes the problem report of the config file in dir to file, for the step or for all steps when step is "All".

The steps are "Protocol names", "Config labels", "Convert", "Custom configs", "Process", "Merge", "Segment", "Tractography" and "Analysis".

Problems are grouped by class: error, unreadable, missing, duplicate and outdated. The extension of file selects the format, ".md" or ".pdf".

The options are ProcessSubjects to limit the subjects and BidsOutputImages to define the expected Analysis images. Returns the full path of file.

Default Definitions SyntaxInformation[MuscleBidsReport] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options {ProcessSubjects → All, BidsOutputImages → All}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsReport

Symbol

MuscleBidsSegment[dir] segments the data of Muscle–Bids named nii based on the config file in the bids sourceFolder dir. The segmentations are generated by the function SegmentData.

Documentation Local »

Default Definitions SyntaxInformation[MuscleBidsSegment] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ProcessSubjects → All, VersionCheck → False}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsSegment

Symbol

MuscleBidsTractography[dir] performs tractography on the Muscle–Bids named nii based on the config file in the bids sourceFolder dir. If a segmentation is present it is used as a mask for per muscle segmentation.

Documentation Local »

Default Definitions SyntaxInformation[MuscleBidsTractography] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {ProcessSubjects → All, VersionCheck → False, BidsTractographyMethod → Full}

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`MuscleBidsTractography

Symbol

PartitionBidsFolderName[fol] partitions the Bids folder and file name. It returns the bids root folder and the label parts using PartitionBidsName.

Documentation Local »
Default Definitions SyntaxInformation[PartitionBidsFolderName] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`PartitionBidsFolderName

Symbol

PartitionBidsName[name] converts a Bids name to the Bids labels as an association, i.e. {"sub","ses","vol","stk","rep","chunk","acq","part","type","suf"}.

Documentation Local »
Default Definitions SyntaxInformation[PartitionBidsName] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`PartitionBidsName

Symbol

SelectBidsFolders[fol, tag] selects all folders in the fol with the name tag.

Documentation Local »
Default Definitions SyntaxInformation[SelectBidsFolders] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`SelectBidsFolders

Symbol

SelectBidsSessions[fol] selects all sessions in the bids subject folder.

Documentation Local »
Default Definitions SyntaxInformation[SelectBidsSessions] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`MuscleBidsTools`SelectBidsSessions

Symbol

SelectBidsSubjects[fol] selects all subjects in the bids folder.

Documentation	Local »
Default Definitions	SyntaxInformation[SelectBidsSubjects] = {ArgumentsPattern → {_}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`MuscleBidsTools`SelectBidsSubjects

Symbol

SelectSubjects[dir] selects the subjects in the given data directory which has a config file.

SelectSubjects[dir, Dynamic[sel]] does the same but stores the selected subjects in the variable sel.

Documentation	Local »
Default Definitions	SyntaxInformation[SelectSubjects] = {ArgumentsPattern → {_, _.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`MuscleBidsTools`SelectSubjects

Symbol

ViewConfig[config] shows a config file for Muscle Bids processing.

ViewConfig[folder] same but imports the config file from the given folder.

ViewConfig["default"] shows the default config.

Documentation	Local »
Default Definitions	SyntaxInformation[ViewConfig] = {ArgumentsPattern → {_}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`MuscleBidsTools`ViewConfig

Symbol

ViewProtocolNames[folder] shows the protocol names in the radData folder specified by the config file with

the series number and protocol name from the json files in the raw folders for each subject and session.

Duplicate protocol names are highlighted and can be shown only using the checkbox. The config file only needs to define the folders.

Documentation Local »

Default Definitions SyntaxInformation[ViewProtocolNames] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ProcessSubjects → All

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`ViewProtocolNames

Symbol

$debugBids is a debug flag for Bids functionality within MuscleBidsTools.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`MuscleBidsTools`$debugBids

Options

Symbol

BidsIncludeSession is an option for BidsDcmToNii. If True session folders will be used in output even if

they are not specified.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`BidsIncludeSession

Symbol

BidsOutputImages is an option for MuscleBidsAnalysis and ManageCheckFiles. It sets which images are saved in the output folder,

one of "All", "Quantitative", "Segmentation", "Tractography" or "None".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`BidsOutputImages

Symbol

BidsTractographyMethod is an option for MuscleBidsTractography and can be "Full", "Tractography" or "Segmentation". With "Tractography" the tractography is performed without segmentation. With "Segmentation" only the segmentation is performed without tractography. With Full both are performed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`BidsTractographyMethod

Symbol

DeleteAfterConversion is an option for MuscleBidsConvert. If set True all files that have been converted will be deleted.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`DeleteAfterConversion

Symbol

ProcessSubjects is an option for Bids functions. Can be a list of bids subject names else it is All.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`ProcessSubjects

Symbol

VersionCheck is an option for all Bids functions. If set True data processed with an old version is reprocessed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`MuscleBidsTools`VersionCheck

NeuralNetworkTools

Functions

Symbol

ActivationLayer[type] is a wrapper around ElementwiseLayer needed in MakeNode.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`ActivationLayer

Symbol

AddLossLayer[net] adds all loss layers to a NetGraph. The DiceLossLayer, JaccardLossLayer, TverskyLossLayer, MSDLossLayer, TopK, and CELossLayer are added.

AddLossLayer[net, loss] only adds the loss layers given in loss, loss can be a string or a list of strings chosen from {"Dice", "Jaccard", "Tversky", "MSD", "MAE", "MSDM", "MAEM", "CE", "Focal", "TopK"}.

"MSDM" and "MAEM" are the masked versions of "MSD" and "MAE": the loss is only computed over the part of the target that is nonzero.

Documentation Local »

Default Definitions SyntaxInformation[AddLossLayer] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`AddLossLayer

Symbol

AnalyzeNetworkFeatures[net, data] gives overview of the information density of the network features by analyzing them with SVD.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`AnalyzeNetworkFeatures

Symbol

CELossFunction[] is a generalization for pixel based loss layers like CELossLayer, FocalLossLayer and TopKLossLayer.

Documentation Local »

Default Definitions SyntaxInformation[CELossFunction] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`CELossFunction

Symbol

CELossLayer[] represents a net layer that computes the Cross Entropy loss by comparing input class probability vectors with the target class vector. Identical to CrossEntropyLossLayer["Probability"].

Documentation Local »
Default Definitions SyntaxInformation[CELossLayer] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`CELossLayer

Symbol

ChangeNetDimensions[netIn] returns netIn unchanged; use the "Dimensions", "Channels", and "Classes" options to actually change the input patch dimension, input channels, or output classes of netIn, e.g. ChangeNetDimensions[netIn, "Dimensions" –> dim, "Channels" –> nChan, "Classes" –> nClass].

Documentation Local »
Default Definitions SyntaxInformation[ChangeNetDimensions] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {Dimensions → None, Channels → None, Classes → None}
Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`ChangeNetDimensions

Symbol

ClassConfidence[probability] gives the confidence of a probability vector as produced by a network's softmax output, i.e. the same input ClassDecoder takes, without collapsing it to a class label. Confidence is 1 minus the normalized entropy of the vector, so it is 1 for a fully certain (one–hot) vector and 0 for a maximally uncertain (uniform) vector.

Documentation Local »
Default Definitions SyntaxInformation[ClassConfidence] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`ClassConfidence

Symbol

ClassDecoder[probability] decodes a probability vector of 1 and 0 into Integers of 0 to the value of the last dimension of probability minus one.

ClassDecoder[probability, nClass] decodes a probability vector of 1 and 0 into Integers of 0 to nClass – 1.

Documentation Local »
Default Definitions SyntaxInformation[ClassDecoder] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`ClassDecoder

Symbol

ClassEncoder[label] encodes Integer label data of 0 to max value of label into a nClass + 1 vector of 1 and 0 as the last dimension.

ClassEncoder[label, nClass] encodes Integer label data of 0 to nClass into a nClass + 1 vector of 1 and 0 as the last dimension.

Documentation Local »
Default Definitions SyntaxInformation[ClassEncoder] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`ClassEncoder

Symbol

DiceLossLayer[] represents a net layer that computes the Dice loss by comparing input class probability vectors with the target class vector.

DiceLossLayer[n] does the same but n defines the power of the denominator, with n=2 the squared dice score, is calculated.

Documentation Local »
Default Definitions SyntaxInformation[DiceLossLayer] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`DiceLossLayer

Symbol

FocalLossLayer[] represents a net layer that computes the Focal loss by comparing input class probability vectors with the target class vector.

FocalLossLayer[g] does the same but uses g as the tunable focusing parameter gamma which needs to be larger than one.

Documentation Local »
Default Definitions SyntaxInformation[FocalLossLayer] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`FocalLossLayer

Symbol

GetNetNodes[net] returns a list of all the nodes in the network net.

Documentation Local »
Default Definitions SyntaxInformation[GetNetNodes] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`GetNetNodes

Symbol

JaccardLossLayer[] represents a net layer that computes the Jaccard loss by comparing input class probability vectors with the target class vector.

JaccardLossLayer[n] does the same but n defines the power of the denominator, with n=2 the squared Jaccard score is calculated.

Documentation Local »
Default Definitions SyntaxInformation[JaccardLossLayer] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`JaccardLossLayer

Symbol

MakeClassifyImage[data] makes a image of the input data. The data is automatically cropped to remove the background and normalized.

If the input data is 3D a list of images is returned.

Documentation Local »
Default Definitions SyntaxInformation[MakeClassifyImage] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ImageSize → {128, 128}
Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`MakeClassifyImage

Symbol

MakeClassifyNetwork[classes] makes a classify network with three convolution layers and 3 fully connected layers.

The input classes should be a list of strings. The input image dimensions should not be smaller than 64x64.

Documentation Local »
Default Definitions SyntaxInformation[MakeClassifyNetwork] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ImageSize → {128, 128}
Attributes {Protected, ReadProtected}
Full Name QMRITools`NeuralNetworkTools`MakeClassifyNetwork

Symbol

MakeNode[scale, conn, blockConfig] makes a node for a UNET.

The input scale defines the input and output scaling, is either an integer or a vector of length dim.

The input conn defines the connections, is a list of two integer values defining the number of input and output ports.

The blockConfig is defined as {{blockType, settings}, {features,..}, {act, dim}}.

Documentation Local »

Default Definitions SyntaxInformation[MakeNode] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options {DropoutRate → 0, RescaleMethod → Conv, CatenateMethod → Cat}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`MakeNode

Symbol

MakeUnet[nClasses, dimIn] Generates a UNET with one channel as input and nClasses as output.

MakeUnet[nChannels, nClasses, dimIn] Generates a UNET with nChannels as input and nClasses as output.

The number of features of the first convolution layer is set with the FeatureSchedule option. The data dimensions can be 2D or 3D and each of the dimensions should be 16, 32, 48, 64, 80, 96, 112, 128, 144, 160, 176, 192, 208, 224, 240 or 256. However dimensions can be different based on the network depth and the block type. The implemented block types are "Conv", "UNet", "ResNet", "DenseNet", "Inception", or "U2Net".

Documentation Local »

Default Definitions SyntaxInformation[MakeUnet] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › NetworkArchitecture → UNet … (12 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`MakeUnet

Symbol

NetDimensions[net] extracts the input channels, output classes, the input patch dimension, and the number of input filters.

NetDimensions[net, port] returns the dimensions at the given port instead. Port can be "Input", "Output", "FirstEncodingIn", "FirstEncodingOut", "LastEncodingIn", "LastEncodingOut", "MinEncodingOut", "AllEncodingOut", "AllMaxChannels", or "U2Encoding".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetDimensions

Symbol

NetSummary[net] gives a short summary of the convolution kernels and array elements in the network.

NetSummary[net, what] does the same but what can be "Full" which also includes net and node images or "Mem" which only reports the memory.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetSummary

Symbol

OverlapLossFunction[] is a generalization for overlap loss layers like TverskyLossLayer, JaccardLossLayer and DiceLossLayer.

Documentation Local »

Default Definitions SyntaxInformation[OverlapLossFunction] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`OverlapLossFunction

Symbol

TopKLossLayer[net] represents a net layer that computes the topK 10% loss.

TopKLossLayer[net, k] does the same but k defines the topK between 0 and 1.

Documentation Local »

Default Definitions SyntaxInformation[TopKLossLayer] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`TopKLossLayer

Symbol

TverskyLossLayer[] represents a net layer that computes the Tversky loss by comparing input class probability vectors with the target class vector.

TverskyLossLayer[b] does the same but b defines the Tversky beta factor. With beta = 0.5 its is the Dice coefficient. Here alpha + beta = 1.

Documentation Local »

Default Definitions SyntaxInformation[TverskyLossLayer] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`TverskyLossLayer

Symbol

$debugUnet is a debug flag for Unet functionality with in te NeuralNetworksTools.

Documentation Local »

Attributes {ReadProtected}

Full Name QMRITools`NeuralNetworkTools`$debugUnet

Options

`Missing[UnknownSymbol, Channels]`

`Missing[UnknownSymbol, Classes]`

Symbol

Dimensions[*expr*] gives a list of the dimensions of *expr*.

Dimensions[*expr*, *n*] gives a list of the dimensions of *expr* down to level *n*.

Documentation Local » | Web »

Options AllowedHeads → Automatic

Attributes {Protected}

Full Name System`Dimensions

Symbol

ActivationType is an option for MakeUnet. It specifies which activation layer is used in the network. It can be "LeakyRELU" or any type allowed by a "name" definition in ElementwiseLayer.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`ActivationType

Symbol

BlockType is an option for MakeUnet. It specifies the type of block used in the network. It can be "Conv", "UNet", "ResNet", "DenseNet", "Inception", or "U2Net".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`BlockType

Symbol

CatenateMethod is an option for MakeUnet. It specifies how the network catenates the skip and upscale channel. It can be "Tot", or "Cat".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`CatenateMethod

Symbol

DownsampleSchedule is an option for MakeUnet. It defines how the data is downsampled for each of the deeper layers of the Unet.

By default is is a factor two for each layer. A custom schedule for a 5 layer 3D Unet could be {{2,2,2},{1,2,2},{2,2,2},{1,2,2}, 1}.

The deepest layer is always downsampled by 1 and therefore not needed to be specified.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`DownsampleSchedule

Symbol

DropoutRate is an option for MakeUnet. It specifies how much dropout is used after each block. It is a value between 0 and 1, default is .2.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`DropoutRate

Symbol

FeatureSchedule is an option for MakeUnet. It defines how the number of features is up–sampled for each of the deeper layers of the Unet.

By default it increases the number of features by a factor 2 each layer, i.e. {1, 2, 4, 8, 16}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`FeatureSchedule

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageSize

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »

Attributes {HoldAll, Protected, ReadProtected}

Full Name System`Monitor

Symbol

NetworkArchitecture is an option for MakeUnet. It defines the architecture of the network. It can be "UNet", "UNet+", or "UNet++".

For "UNet+" or "UNet++" it can also be {arch, i} where i specifies how many of the top layers are connected to the mapping layer.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetworkArchitecture

Symbol

NetworkDepth is an option for MakeUnet. It specifies how deep the UNET will be.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetworkDepth

Symbol

NormalizationType is an option for MakeUnet. It specifies which normalization layer is used in the network. It can be "Batch" or "Instance" for either BatchNormalization or NormalizationLayer which is instance normalization.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NormalizationType

Symbol

RescaleMethod is an option for MakeUnet. It specifies how the network rescales. It can be "Conv" or "Pool".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`RescaleMethod

Symbol

SettingSchedule is an option for MakeUnet. It defines the settings for the Unet blocks. If one setting is given it applied to all layers.

If a list of settings is given the settings can be different per layer. The following settings are the default settings.

"Unet": convolution block repetitions, 2, "ResNet" –> convolution block repetitions, 2, "DenseNet" –> {dense depth, block repetitions}, {4,2},

"Inception" –> {inception width, block repetitions}, {4,2}, "U2Net"–> {Unet depth, downscale}, {5, True}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`SettingSchedule

NiftiTools

Functions

Symbol

CompressNiiFiles[] prompts for a folder. It then compresses all nii files to .nii.gz files in the selected folder.

CompressNiiFiles[folder] compresses all nii files to .nii.gz files in folder.

Documentation Local »

Default Definitions SyntaxInformation[CompressNiiFiles] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options MonitorCalc → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`CompressNiiFiles

Symbol

CorrectNiiOrientation[data,hdr] corrects the data orientation based on the nii header.

Documentation Local »

Default Definitions SyntaxInformation[CorrectNiiOrientation] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`CorrectNiiOrientation

Symbol

DcmToNii[] converts a dicom folder to nii, you will be prompted for the location of the folders.

DcmToNii[{"input","output"}] converts the "input" dicom folder to nii files which are place in the "output" folder.

For this function to work the dcm2niix.exe file should be present in the QMRITools application folder.

Documentation Local »

Default Definitions SyntaxInformation[DcmToNii] = {ArgumentsPattern → {_., _., OptionsPattern[]}}

Options › CompressNii → True ... (8 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`DcmToNii

Symbol

ExportBmat[bmat] exports the diffusion bmatrix to a bmat file in *.txt format.

ExportBmat[bmat, "file"] exports the diffusion bmatrix to "file" in *.txt format.

Documentation Local »

Default Definitions SyntaxInformation[ExportBmat] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ExportBmat

Symbol

ExportBval[bval] exports the diffusion bvalues to a *.bval format.

ExportBval[bval, "file"] exports the diffusion bvalues to "file" *.bval format.

Documentation Local »

Default Definitions SyntaxInformation[ExportBval] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ExportBval

Symbol

ExportBvalvec[{bval, grad}] exports the diffusion values and gradients to a *.bval and *.bvec format.

ExportBvalvec[{bval, grad}, "file"] exports the diffusion values and gradients to "file" to a *.bval and *.bvec format.

Documentation	Local »
Default Definitions	SyntaxInformation[ExportBvalvec] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options	{FlipBvec → False, PositiveZ → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ExportBvalvec

Symbol

ExportBvec[grad] exports the diffusion gradients to a *.bvec format.

ExportBvec[grad, "file"] exports the diffusion gradients to "file" *.bvec format.

Documentation	Local »
Default Definitions	SyntaxInformation[ExportBvec] = {ArgumentsPattern → {_, _., OptionsPattern[]}}
Options	{FlipBvec → False, PositiveZ → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ExportBvec

Symbol

ExportNii[data, vox] exports the nii file and will prompt for a file name.

ExportNii[data, vox, "file"] exports the nii file to the location "file".

Documentation	Local »
Default Definitions	SyntaxInformation[ExportNii] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}
Options ❯	NiiDataType → Automatic … (5 total)
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ExportNii

Symbol

ExtractNiiFiles[] prompts for a folder. It then extracts all nii.gz files to .nii files in the selected folder.

ExtractNiiFiles[folder] extracts all nii.gz files to .nii files in folder.

Documentation	Local »
Default Definitions	SyntaxInformation[ExtractNiiFiles] = {ArgumentsPattern → {_., _., OptionsPattern[]}}
Options	MonitorCalc → False
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ExtractNiiFiles

Symbol

GetNiiOrientation[hdr] get the s–form and q–form orientations from a nii header.

Documentation	Local »
Default Definitions	SyntaxInformation[GetNiiOrientation] = {ArgumentsPattern → {_}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`GetNiiOrientation

Symbol

ImportBmat[] will prompt to select the *.txt file containing the bmatrix.

ImportBmat[*.txt] imports the given *.txt file containing the bmatrix.

Documentation	Local »
Default Definitions	SyntaxInformation[ImportBmat] = {ArgumentsPattern → {_., _.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ImportBmat

Symbol

ImportBval[] will prompt to select the *.bval file.

ImportBval[*.bval] imports the given *.bval file.

Documentation	Local »
Default Definitions	SyntaxInformation[ImportBval] = {ArgumentsPattern → {_.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ImportBval

Symbol

ImportBvalvec[] will prompt to select the *.bval and *.bvec files.

ImportBvalvec[file] if file is either a *.bval or *.bvec it will automatically import the *.bval and *.bvec files.

ImportBvalvec[*.bvec,*.bval] imports the given *.bval and *.bvec files.

Documentation	Local »
Default Definitions	SyntaxInformation[ImportBvalvec] = {ArgumentsPattern → {_., _., OptionsPattern[]}}
Options	{FlipBvec → False, PositiveZ → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ImportBvalvec

Symbol

ImportBvec[] will prompt to select the *.bvec file.

ImportBvec[*.bvec] imports the given *.bvec file.

Documentation	Local »
Default Definitions	SyntaxInformation[ImportBvec] = {ArgumentsPattern → {_., OptionsPattern[]}}
Options	{FlipBvec → False, PositiveZ → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ImportBvec

Symbol

ImportExploreDTITens["file"] imports the *.nii export for the tensor from explore DTI.

Documentation	Local »
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`NiftiTools`ImportExploreDTITens

Symbol

ImportNii[] prompts to select the nii file to import.

ImportNii["file"] imports the nii file.

The default output is {data, vox}, however using NiiMethod various outputs can be given.

The Nii import is also supported using the native Import function from Mathematica.

Documentation Local »

Default Definitions SyntaxInformation[ImportNii] = {ArgumentsPattern → {_., OptionsPattern[]}}

Options {NiiMethod → default, NiiScaling → False}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ImportNii

Symbol

ImportNiiDiff[] will prompt for the *.nii, *.bvec and *.bval file to import.

ImportNiiDiff[*.nii] will import the *.nii file and automatically also imports the *.bvec and *.bval is they have the same name.

ImportNiiDiff[*.nii,*.bvec,*.bval] will import the given files.

The output will be {data,grad,bvec,vox}.

Documentation Local »

Default Definitions SyntaxInformation[ImportNiiDiff] = {ArgumentsPattern → {_., _., _., OptionsPattern[]}}

Options {RotateGradients → False, FlipBvec → True}

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ImportNiiDiff

Symbol

ImportNiiDix["file"] imports the dixon nii file which should contain all possible outputs given by the scanner and corrects them accordingly.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ImportNiiDix

Symbol

ImportNiiT1["file"] imports the t1 file which should contain the echos and the T1map calculated by the scanner and corrects them accordingly.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ImportNiiT1

Symbol

ImportNiiT2["file"] imports the t2 file which should contain the echos and the T2map calculated by the scanner and corrects them accordingly.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`ImportNiiT2

Symbol

MakeNiiOrientationQ[rot] makes the q vector from rotation matrix rot.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`MakeNiiOrientationQ

Symbol

MakeNiiOrientationS[off, vox] maxes the s–row values for nii header assuming not rot and Q.

MakeNiiOrientationS[off, vox, rot] maxes the s–row values for nii header using rotation rot.

MakeNiiOrientationS[off, vox, rot, Q] maxes the s–row values for nii header using rotation rot and skew Q.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`MakeNiiOrientationS

Options

Symbol

CompressNii is an option for DcmToNii and ExportNii. If set True .nii.gz files will be created.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`CompressNii

Symbol

DecompressDcm is an option for DcmToNii. If set True, all dcm files in the input folder are decompressed with dcmdjpeg.exe (DCMTK) before the conversion.

This is needed for JPEG2000 or JPEG–LS compressed dicom that dcm2niix cannot read. Only for Windows.

Note that the input files are overwritten, so use a copy of the data.

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`DecompressDcm

Symbol

DeleteOutputFolder is an option of DcmToNii. If the output folder already exists it will be deleted.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`DeleteOutputFolder

Symbol

FlipBvec is an option for ImportBvalvec.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`FlipBvec

Symbol

MergeEchos is an option for DcmToNii. If set True the echos are merged into one file.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`MergeEchos

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

NiiDataType is an option of ExportNii. The number type of Nii file can be "Integer", "Real", "Complex", or "Automatic".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiDataType

Symbol

NiiLegacy is an option for ExportNii, if set True default orientations are set instead of unknown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiLegacy

Symbol

NiiMethod is an option for ImportNIi. Values can be "data", "dataTR", "header", "scaling", "headerMat", "rotation", "all".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiMethod

Symbol

NiiOffset is an option of ExportNii. Is {xOff, yOff, zOff}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiOffset

Symbol

NiiScaling is an option for ImportNii. It scales the nii values with scale slope and offset for quantitative data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiScaling

Symbol

NiiSliceCode is an option for Export nii. With this you can set the slice code of the nii file.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`NiiSliceCode

Symbol

PositiveZ is an options for GradientPlot. If True all Gradients are displayed with a positive z direction.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`PositiveZ

Symbol

RotateGradients is an option for ImportNiiDiff.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`RotateGradients

Symbol

UseSubFolders is an option for DcmToNii. If set True the nii conversion is done for each folder in the selected input folder.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`UseSubFolders

Symbol

UseVersion is an option for DcmToNii. For windows it allows to switch between different versions of dcm2niix.exe.

Set to 1 for the default bundled version, an integer matching a dated "DcmToNii–<version>" asset (e.g. 25 for dcm2niix–20250506.exe),

or "Own" for a custom dcm2niix–own.exe build registered as the "DcmToNii–Own" asset in PacletInfo.wl.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NiftiTools`UseVersion

PlottingTools

Functions

Symbol

ColorFAPlot[tenor] create a color coded FA map from the tensor for l1, l2 and l3.

Documentation Local »

Default Definitions SyntaxInformation[ColorFAPlot] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ColorFAPlot

Symbol

GenerateRotationFrames[3Dgraphics, nFrames] generates nFrames of the 3D graphics rotating around the z–axis.

Documentation Local »

Options ImageSize → 500

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`GenerateRotationFrames

Symbol

GetSliceData[data, offsets] gets the slices from the data defined by offsets which are obtained by GetSlicePosisions.

GetSliceData[data, offsets, vox] gets the slices from the data defined by offsets which are obtained by GetSlicePosisions in mm.

The offsets can also be provided manually which is {{AX,..},{COR,..},{SAG,..}}.

Documentation Local »

Default Definitions SyntaxInformation[GetSliceData] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`GetSliceData

Symbol

GetSlicePositions[data] finds the position of slices with the maximal signal in voxel index.

GetSlicePositions[data, vox] find the position of slices with the maximal signal in mm.

Documentation Local »
Default Definitions SyntaxInformation[GetSlicePositions] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {MakeCheckPlot → False, DropSlices → {1, 1, 1}, PeakNumber → {1, 1, 2}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`GetSlicePositions

Symbol

GradientPlot[bvec, bval] plots the given bvec with position of the gradients scaled according to the bval.

Documentation Local »
Default Definitions SyntaxInformation[GradientPlot] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › PlotSpace → bspace … (4 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`GradientPlot

Symbol

LegendImage[colf, w, lab] generates a legend image for the color function colf with a width of w and labels lab.

Documentation Local »
Default Definitions SyntaxInformation[LegendImage] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`LegendImage

Symbol

Link3DGraphic[3Dgraphics] creates a 3D graphic with a linked view options between all 3D graphics that have been linked.

Documentation Local »
Default Definitions SyntaxInformation[Link3DGraphic] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`Link3DGraphic

Symbol

ListSpherePlot[points] plots 3D points as spheres.

Documentation Local »

Default Definitions SyntaxInformation[ListSpherePlot] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {SphereSize → 2, SphereColor → Automatic}

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ListSpherePlot

Symbol

LoessPlot[data] plots the data with a loess fit.

Documentation Local »

Options ❯ Bandwidth → Scaled[0.5] … (10 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`LoessPlot

Symbol

MakeSliceImages[imgData] generates images from the imgData which is obtained form GetSliceData.

MakeSliceImages[imgData, vox] generates images from the imgData which is obtained form GetSliceData, vox is used for the correct aspect ratio of the images.

MakeSliceImages[imgData, {labData, labels}] generates images from the imgData which is obtained

form GetSliceData with an overlay of the segmentations in labData, which can also be obtained using GetSliceData on the segmentations.

labels should be the label numbers used in the original segmentation (to allow correct scaling between slices).

MakeSliceImages[imgData, {labData, labels},vox] generates images from the imgData which is obtained form GetSliceData with an overlay

of the segmentations in labData, which can also be obtained using GetSliceData on the segmentations, vox is used for the correct aspect ratio of the images.

Documentation Local »

Default Definitions SyntaxInformation[MakeSliceImages] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options ❯ PlotRange → Automatic … (7 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`MakeSliceImages

Symbol

PlotContour[data, vox] creates a contour of the data.

The data can be 3D or 4D, when its 4D each contour is given a random color.

Documentation Local »

Default Definitions SyntaxInformation[PlotContour] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options ❯ ContourColor → ■ ... (8 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`PlotContour

Symbol

PlotCorrection[w] plots deformation vectors w {w1,w2..} generated by Registration2D and Registration3D for multiple datasets or registration steps.

Documentation Local »

Default Definitions SyntaxInformation[PlotCorrection] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`PlotCorrection

Symbol

PlotData[data] plots the data.

PlotData[data, vox] plots the data and for 3D and 4D data assumes the voxelsize vox (z,x,y).

PlotData[data1, data2] plots data1 and data2.

PlotData[data1, data2, vox] plots data1 and data2 and for 3D and 4D data assumes the voxelsize vox (z,x,y).

Documentation Local »

Default Definitions SyntaxInformation[PlotData] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {PlotRange → Auto, ColorFunction → BlackToWhite}

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`PlotData

Symbol

PlotData3D[data,vox] is a 3D dataviewer, data is the 3D dataset and voxsize the size of the voxels in mm (z,x,y).

Documentation Local »
Default Definitions SyntaxInformation[PlotData3D] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotData3D

Symbol

PlotDuty[{grad, bval, ord}, mode] plot the gradient dutycycle.

Documentation Local »
Default Definitions SyntaxInformation[PlotDuty] = {ArgumentsPattern → {{_, _, _}, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotDuty

Symbol

PlotIVIM[vals, data, bvals] plots the results of the IVIM fits from IVIMCalc.

Documentation Local »
Default Definitions SyntaxInformation[PlotIVIM] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options › Method → ... (5 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotIVIM

Symbol

PlotMoments[{G(t),..}, te, t] plots the moments generated by CalculateMoments.

Documentation Local »
Default Definitions SyntaxInformation[PlotMoments] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotMoments

Symbol

PlotSegmentations[seg, vox] crates a contour of each of the segmentations in seg. Uses PlotContour.

PlotSegmentations[seg, bone, vox] same but also generates an extra contour with the bones in gray.

Documentation Local »
Default Definitions SyntaxInformation[PlotSegmentations] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › ImageSize → 400 … (8 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotSegmentations

Symbol

PlotSequence[seq,var] where seq is the output from GradSeq.

Documentation Local »
Default Definitions SyntaxInformation[PlotSequence] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotSequence

Symbol

ResetLink3DGraphic[] resets the linked view options of all 3D graphics that have been linked with Link3DGraphic to their defaults.

Documentation Local »
Default Definitions SyntaxInformation[ResetLink3DGraphic] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ResetLink3DGraphic

Symbol

ShowLink3DGraphic[] shows the linked view options between all 3D graphics that have been linked.

Documentation Local »
Default Definitions SyntaxInformation[ShowLink3DGraphic] = {ArgumentsPattern → {}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ShowLink3DGraphic

Symbol

$debugPlot is a debug flag for the functions in PlottingTools, when True the progress and memory use are printed.

Documentation Local »
Attributes {ReadProtected}
Full Name QMRITools`PlottingTools`$debugPlot

Symbol

$plotOptions is a list of options for plotting.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`$plotOptions

Options

Symbol

Axes is an option for graphics functions that specifies whether axes should be drawn.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Axes

Symbol

Bandwidth is an option for LoessPlot. It specifies the bandwidth of the loess fit. Its a number between 0 and 1 and specifies which % of the data to include for the local regression.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`Bandwidth

Symbol

ClippingStyle is an option for plotting functions that specifies the style of what should be drawn when curves or surfaces would extend beyond the plot range.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ClippingStyle

Symbol

ColorFunction is an option for graphics functions that specifies a function to apply to determine colors of elements.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ColorFunction

Symbol

ColorLabels is an option for PlotSegmentations. If a list of labels is given only these labels are

plotted and each label gets a fixed color based on its position in the list. Default is Automatic, which uses the labels in the segmentation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ColorLabels

Symbol

ConfidenceLevel is an option for LinearModelFit and other fitting functions that specifies the level to use in various confidence and prediction intervals and bands.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ConfidenceLevel

Symbol

ContourColor is an option for PlotContour. It specifies the contour color and can be a single color or an array with the same dimensions as the data.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ContourColor

Symbol

ContourColorRange is an option for PlotContour. If the ContourColor is an array this values gives the plot range.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ContourColorRange

Symbol

ContourOpacity is an option for PlotContour and PlotSegmentations. It specifies the opacity of the contour.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ContourOpacity

Symbol

ContourResolution is an option for PlotContour. It defines the mesh resolution used, can be a single number or a list of 3 numbers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ContourResolution

Symbol

ContourScaling is an option for PlotCountour. The value can be "World" or "Voxel", if the value is "Wold" the segmentation is in mm else in voxel coordinates.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ContourScaling

Symbol

ContourSmoothRadius is and option for PlotContour. It defines the smoothing radius with an integer, None or 0 does not smooth.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ContourSmoothRadius

Symbol

ContourSpecularity is an option for PlotContour. It specifies the specularity of the contour.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`ContourSpecularity

Symbol

DropSlices is an option for GetSlicePositions and specifies how many slices from the beginning and and should be ignored.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`DropSlices

Symbol

FitKernel is an option for LoessPlot. It specifies the kernel used for the loess fit. Can be "Triangular" or "Gaussian".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`FitKernel

Symbol

FitOrder is an option for HomogenizeData. It specifies the order of harmonics to be used for the homogenization.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`MaskingTools`FitOrder

Symbol

Frame is an option for Graphics, Grid, and other constructs that specifies whether to include a frame.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Frame

Symbol

FrameStyle is an option for Graphics, Grid, and other constructs that specifies the style in which to draw frames.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`FrameStyle

Symbol

ImageLegend is an option for MakeSliceImages, if set true a barlegend is added to the image.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ImageLegend

Symbol

ImageOrientation is an option for MakeSliceImages. Can be Automatic, "Vertical" or "Horizontal".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ImageOrientation

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ImageSize

Symbol

LabelStyle is an option for formatting and related constructs that specifies the style to use in displaying their label–like elements.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`LabelStyle

Symbol

MakeCheckPlot is an option for GetSlicePositions and if set true gives a plot of the slices locations.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`MakeCheckPlot

Symbol

MaskOpacity is an option for MakeSliceImages. It specifies the opacity of the mask.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`MaskOpacity

Symbol

MaxPlotPoints is an option for plotting functions like ArrayPlot and ListPlot3D that specifies the maximum number of points that will explicitly be included in the output.

Documentation Local » | Web »

Attributes {Protected, ReadProtected}

Full Name System`MaxPlotPoints

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

NormalizeIVIM is an option for IVIMplot. If True the signal at b=0 is 1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`NormalizeIVIM

Symbol

PeakNumber is an option of GetSlicePositions and specifies how many slices per direction need to be found.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`PeakNumber

Symbol

PerformanceGoal is an option for various algorithmic and presentational functions that specifies what aspect of performance to try to optimize with Automatic settings for options.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PerformanceGoal

Symbol

PlotColor is an option for GradientPlot can be any color or gradient color name.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotColor

Symbol

PlotHighlighting is an option to Plot, ListPlot and related visualization functions that specifies how points and curves should be highlighted.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotHighlighting

Symbol

PlotRange is an option for graphics functions that specifies what range of coordinates to include in a plot.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotRange

Symbol

PlotSpace is an option for GradientPlot can be "bspace" or "qspace".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotSpace

Symbol

PlotStyle is an option for plotting and related functions that specifies styles in which objects are to be drawn.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PlotStyle

Symbol

PositiveZ is an options for GradientPlot. If True all Gradients are displayed with a positive z direction.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PositiveZ

Symbol

PredictionInterval is an option for LoessPlot. It specifies the prediction interval of the loess fit. If True it uses the prediction confidence bands if False it uses the standard error of the local mean.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PredictionInterval

Symbol

RandomizeColor is an option for PlotSegmentations. If True the colors are randomly assigned to each segmentation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`RandomizeColor

Symbol

SphereColor ListSpherePlor. Default value is Automatic, If a color is given this color will be used for all spheres.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`SphereColor

Symbol

SphereSize is an option for GradientPlot and ListSpherePlor. Sets the size of the spheres thar represent the gradients.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`PlottingTools`SphereSize

ProcessingTools

Functions

Symbol

B1MapCalc[data, TR, alpha] calculates the B1 map from a dual TR {tr1, tr2} acquisition (AFI) using magnitude data with reference angle alpha.

data has dimensions {z, {tr1,tr2}, x, y}.

B1MapCalc[dataTr1, dataTr2, TR, alpha] where dataTr1 and and dataTr2 can have any dimensions.

The Output can be "Map", "MagPhase", or "Complex"}

B1MapCalc[] is based on DOI: 10.1002/mrm.21120.

Documentation Local »

Default Definitions SyntaxInformation[B1MapCalc] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options › B1Output → Map … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1MapCalc

Symbol

B1Shimming[b10, b190, mask] finds the optimal shim values to shim to 100% b1. Assumes B1Scaling "Relative".

B1Shimming[b10, b190, mask, target] finds the optimal shim values to shim to target, which can be a number or a map.

Documentation Local »

Default Definitions SyntaxInformation[B1Shimming] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › B1ShimMethod → All … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1Shimming

Symbol

CombineB1[b10,b190,{f1,f2,ang}] combines the complex b1 maps with relative amplitudes f1 and f2 using phase angle ang.

Documentation Local »
Default Definitions SyntaxInformation[CombineB1] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options B1Scaling → Relative
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`CombineB1

Symbol

CorrectJoinSetMotion[[{dat1, dat2, ...}, vox, over] motion corrects multiple sets with overlap. Over is the number of slices overlap between stes. A Translation registration is performed.

Documentation Local »
Default Definitions SyntaxInformation[CorrectJoinSetMotion] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options {JoinSetSplit → True, PadOverlap → 2, MonitorCalc → True}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`CorrectJoinSetMotion

Symbol

DataTransformation[data, vox, w] transforms a 3D dataset according to the affine transformation vector w.

Documentation Local »
Default Definitions SyntaxInformation[DataTransformation] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options {InterpolationOrder → 1, PadOutputDimensions → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`DataTransformation

Symbol

DatTot[{data1, data2, ..}, name, vox] calculates the parameter table containing the volume, mean, std and 95 CI for each of the diffusion parameters.

Documentation Local »
Default Definitions SyntaxInformation[DatTot] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`DatTot

Symbol

DatTotXLS[{data1, data2, ..}, name, vox] is the same as DatTot, but gives the parameters as strings for easy export to excel.

Documentation Local »
Default Definitions SyntaxInformation[DatTotXLS] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`DatTotXLS

Symbol

ErrorPlot[data, xdata] plots a error plot of the data where the first dim of the data is the xrange which matches the xdata list.

ErrorPlot[data, xdata, range] similar with a given y range.

Documentation Local »
Default Definitions SyntaxInformation[ErrorPlot] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options ❯ ColorValue → {■, ■} … (5 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`ErrorPlot

Symbol

FindOutliers[data] finds the outliers of a list of data.

Documentation Local »
Default Definitions SyntaxInformation[FindOutliers] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options ❯ OutlierMethod → IQR … (6 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`FindOutliers

Symbol

FitData[data,range] converts the data into 100 bins within the +/– range around the mean. Function is used in ParameterFit.

Documentation Local »
Default Definitions SyntaxInformation[FitData] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`FitData

Symbol

GetMaskData[data, mask] returns the data selected by the mask.

Documentation Local »
Default Definitions SyntaxInformation[GetMaskData] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options {GetMaskOutput → All, GetMaskOnly → False}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`GetMaskData

Symbol

GetMaskMeans[dat, mask] calculates the mean, std, 5,50 and 95% CI form the given data for each of the given masks.

Mask can be generated by SegmentTracts.

GetMaskMeans[dat, mask, name] where name is a string that is added to the header.

Documentation Local »
Default Definitions SyntaxInformation[GetMaskMeans] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options MeanMethod → SkewNormalDist
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`GetMaskMeans

Symbol

GetTractMeans[dat, tracts, vox] calculates the mean, std, 5,50 and 95% CI form the given data for each of the given tracts.

Tracts can be by SplitSegmentations. name is a string that is added to the header.

GetTractMeans[dat, tracts, vox, name] where name is a string that is added to the header.

Documentation Local »
Default Definitions SyntaxInformation[GetTractMeans] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options {MeanMethod → SkewNormalDist, InterpolationOrder → 0}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`GetTractMeans

Symbol

Hist[data, range] plots a probability density histogram of the data from xmin to xmax with a fitted (skew)normal distribution. Uses ParameterFit.

Hist[data, range, label] plots a probability density histogram of the data from xmin to xmax with a fitted (skew)normal distribution and label as x–axis label.

Hist[{data1.., data2,..}, {range1, range2,..}] plots a probability density histogram of the data from xmin to xmax with a fitted (skew)normal distribution. Uses ParameterFit.

Hist[{data1, data2,..}, {range1, range2,..}, {label1, label2,..}] plots a probability density histogram of the data from xmin to xmax with a fitted (skew)normal distribution and label as x–axis label.

Documentation Local »

Default Definitions SyntaxInformation[Hist] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options ❯ ColorValue → {{, }, , , } … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`Hist

Symbol

Hist2[pars, range] plots a probability density histogram of the data over range with two fitted (skew)normal distribution. Uses ParameterFit2.

Hist2[pars, range, label] plots a probability density histogram of the data over range with two fitted (skew)normal distribution. Uses ParameterFit2.

Documentation Local »

Default Definitions SyntaxInformation[Hist2] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options Scaling → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`Hist2

Symbol

InvertDataset[data] inverts the data along the x y and z axes. In other words it is rotated around the origin such that (x,y,z)=(−x,−y,−z) and (0,0,0)=(0,0,0).

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`InvertDataset

Symbol

JoinSets[{dat1, dat2, ...}, over] joins dat1, dat2, ... with over slices overlap.

JoinSets[{dat1, dat2, ...},{over1, over2, ...}] joins dat1 and dat2 with over1 slices overlap, Joins dat2 and dat3 with over2 slices overlap and so on.

JoinSets[{dat1, dat2, ...},{{over, drop1, drop2}, ...}] joins dat1, dat2 with over slices overlap and drops drop1 slices for dat1 and drop2 from drop 2.

DOI: 10.1148/radiol.14140702.

Documentation Local »
Default Definitions SyntaxInformation[JoinSets] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options › ReverseSets → False ... (8 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`JoinSets

Symbol

MeanRange[Range] calculates the median (50%) and standard deviation (14% and 86%) range and reports it as a string.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`MeanRange

Symbol

MeanSignal[data] calculates the mean signal per volume of 4D data.

MeanSignal[data, pos] calculates the mean signal per volume of 4D data for the given list of positions.

Documentation Local »
Default Definitions SyntaxInformation[MeanSignal] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options UseMask → True
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`MeanSignal

Symbol

MeanStd[data] calculates the mean and standard deviation and reports it as a string.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`MeanStd

Symbol

MeanStdRange[Range] calculates the mean and standard deviation and the min max range and reports it as a string.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MeanStdRange

Symbol

MedCouple[data] calculates the medcouple of a list of data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MedCouple

Symbol

MinMaxRange[Range] calculates the min and max range and reports it as a string.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MinMaxRange

Symbol

NumberTableForm[data] makes a right aligned table of the numbers with 3 decimal precision.

NumberTableForm[data, n] makes a right aligned table of the numbers with n decimal precision.

Documentation Local »

Default Definitions SyntaxInformation[NumberTableForm] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › TableMethod → NumberForm … (6 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`NumberTableForm

Symbol

ParameterFit[data] fits a (skew)Normal probability density function to the data.

ParameterFit[{data1, data2,...}] fits a (skew)Normal probability density function to each of the datasets. Is used in Hist.

Documentation Local »
Default Definitions SyntaxInformation[ParameterFit] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {FitFunction → SkewNormal, FitOutput → Parameters, Method → Automatic}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`ParameterFit

Symbol

ParameterFit2[data] fits two skewNormal probability density functions to the data. Assuming two compartments, one for fat and one for muscle.

Documentation Local »
Default Definitions SyntaxInformation[ParameterFit2] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options FitOutput → BestFitParameters
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`ParameterFit2

Symbol

RotateData[data] rotates the data 180 degree, e.g. inversion of the z direction.

Documentation Local »
Default Definitions SyntaxInformation[RotateData] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`RotateData

Symbol

RotateTensor[tens] rotates the tensor 180 degree, e.g. inversion of the z direction with also inverting the tensor orientation.

Documentation Local »
Default Definitions SyntaxInformation[RotateTensor] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`RotateTensor

Symbol

SmartMask[input] crates a smart mask of input, which is either the tensor or the tensor parameters calculated using ParameterCalc.

SmartMask[input, mask] crates a smart mask of input and used the mask as a prior selection of the input.

Documentation Local »
Default Definitions SyntaxInformation[SmartMask] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › Strictness → 0.5 … (4 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SmartMask

Symbol

SNRCalc[data, sigma] calculates the ANR of the data using a noise sigma map for example generated by PCADeNoise.

Documentation Local »
Default Definitions SyntaxInformation[SNRCalc] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SNRCalc

Symbol

SNRMapCalc[data1, noisemap] calculates the signal to noise ratio of the data using MN[data]/(1/sqrt[pi/2] sigma),

where sigma is the local mean of the noise map assuming it is a Rician distribution.

SNRMapCalc[{data1,data2}] calculates the signal to noise ratio from two identical images using

MN[data1,data2] / (.5 SQRT[2] STDV[data2−data1]).

SNRMapCalc[{data1, .. dataN}] calculates the signal to noise ratio of the data using MN/sigma where the mean signal MN is the average voxel value over all dynamics N and the sigma is the standard deviation over all dynamics N.

Documentation Local »
Default Definitions SyntaxInformation[SNRMapCalc] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {OutputSNR → SNR, SmoothSNR → 2}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SNRMapCalc

Symbol

SplitSets[data, Nsets, Nover] splits the data in Nsets with Nover slices overlap.

Documentation Local »
Default Definitions SyntaxInformation[SplitSets] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options {ReverseSets → False, ReverseData → True, PadOverlap → 0}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SplitSets

Options

Symbol

AxesLabel is an option for graphics functions that specifies labels for axes.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`AxesLabel

Symbol

B1EqualPower is an option for B1shimming. If true equal power for both channels is used.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`B1EqualPower

Symbol

B1FilterData is an option for B1MapCalc. If True HammingFilterData is applied to the data before B1 calculation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`B1FilterData

Symbol

B1Masking is an option for B1MapCalc. If True then values where S2 is larger than S1 are masked.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`B1Masking

Symbol

B1MaxPower is an option for B1Shimming. Specifies how much power can be used per channel.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1MaxPower

Symbol

B1Output is an option for B1MapCalc. Values can be "Map", "MagPhase", or "Complex".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1Output

Symbol

B1Scaling is an option for B1Shimming and CombineB1. Values can be "Relative" or "Absolute". "Absolute" assumes b1 maps are given in uT, "Relative" assumes that maps are in %.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1Scaling

Symbol

B1ShimMethod is an option for B1Shimming. Values can be "All", "Phase" or "Magnitude".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`B1ShimMethod

Symbol

ColorValue is an option for Hist and ErrorPlot. Default {Black, Red}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`ColorValue

Symbol

FitFunction is an option for ParameterFit. Options are "Normal" or "SkewNormal". Indicates which function wil be fitted.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`FitFunction

Symbol

FitOutput is an option for ParameterFit and ParameterFit2. Option can be "Parameters", "Function" or "BestFitParameters".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`FitOutput

Symbol

GetMaskOnly is an option for GetMaskData. If set True all values in the mask are given, if set False only non zero values in the mask are give.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`GetMaskOnly

Symbol

GetMaskOutput is an option for GetMaskData. Default is "Slices" which gives the mask data per slices. Else the entire mask data is given as output.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`GetMaskOutput

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageSize

Symbol

InterpolationOrder is an option for Interpolation, as well as ListLinePlot, ListPlot3D, ListContourPlot, and related functions, that specifies what order of interpolation to use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`InterpolationOrder

Symbol

JoinSetSplit is an option ofr CorrectJoinSetMotion. If True RegisterDataTransformSplit is used else RegisterDataTransform is used.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`JoinSetSplit

Symbol

MaskCompartment is an option for SmartMask. Can be "Muscle" or "Fat".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`MaskCompartment

Symbol

MeanMethod is an option for GetMaskMeans. The option can be "NormalDist", "SkewNormalDist", or "Mean".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`MeanMethod

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

MotionCorrectSets is an option for JoinSets. True motion corrects the individual stacks before joining using CorrectJoinSetMotion.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MotionCorrectSets

Symbol

NormalizeOverlap is an option for JoinSets. True removes strong signal dropoff at the end of a stacks.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`NormalizeOverlap

Symbol

NormalizeSets is an option for JoinSets. True normalizes the individual stacks before joining.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`NormalizeSets

Symbol

OutlierCheckNormality is an option for FindOutliers. If set to True the data is checked for normality using the DistributionFitTest function.

If the data is not normal the data is log transformed before outlier detection.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierCheckNormality

Symbol

OutlierIncludeZero is an option for FindOutliers. If set to True all values that are zero are ignored and considered outliers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierIncludeZero

Symbol

OutlierIterations is an option for FindOutliers. Specifies how many iterations are used to find the outliers.

Each iteration the outliers are reevaluated on the data with the previously found outliers already rejected.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierIterations

Symbol

OutlierMethod is an option for FindOutliers. values can be "IQR", "SIQR" or "aIQR". "IRQ" is used for normally distributed data, "SIQR" or "aIQR" are better for skewed distributions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierMethod

Symbol

OutlierOutput is an option for FindOutliers. If value is "Mask" it gives a list of 1 for data and 0 for outliers. Else the output is {data, outliers}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierOutput

Symbol

OutlierRange is an option for FindOutliers. Specifies how many times the IQR is considered an outlier.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutlierRange

Symbol

OutputSNR is an option for SNRMapCalc.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`OutputSNR

Symbol

PadOutputDimensions is an options for DataTransformation. If False the data is the same dimensions as the input else the data is padded.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`PadOutputDimensions

Symbol

PadOverlap is an option of CorrectJoinSetMotion and JoinSets. it allows for extra motion in the z direction.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`PadOverlap

Symbol

PlotLabel is an option for graphics functions that specifies an overall label for a plot.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`PlotLabel

Symbol

ReferenceB1 is an option for B1MapCalc. Default value is None. Can be given a numeric value in uT.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`ReferenceB1

Symbol

ReverseData is an option for JoinSets. Reverses each individual dataset given as input for the JoinSets function. True by default.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`ReverseData

Symbol

ReverseSets is an option for JoinSets. Reverses the order of the datasets, False by default.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`ReverseSets

Symbol

Scaling is an option for Hist2. Scales the individual fits of the fat and muscle compartment.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`Scaling

Symbol

SmartMaskOutput is an option for Smartmask. Can be set to "mask" to output only the mask or "full" to also output the probability mask.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SmartMaskOutput

Symbol

SmartMethod is an option for SmartMask. This specifies how the mask is generated. Can be "Continuous" or "Catagorical".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ProcessingTools`SmartMethod

Symbol

SmoothSNR is an option for SNRMapCalc.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`SmoothSNR

Symbol

Strictness is an option for SmartMask value between 0 and 1. higher values removes more data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`Strictness

Symbol

System`TableAlignments

Documentation Local » | Web »

Attributes {Protected}

Full Name System`TableAlignments

Symbol

System`TableDepth

Documentation Local » | Web »

Attributes {Protected}

Full Name System`TableDepth

Symbol

System`TableDirections

Documentation Local » | Web »

Attributes {Protected}

Full Name System`TableDirections

Symbol

System`TableHeadings

Documentation Local » | Web »

Attributes {Protected}

Full Name System`TableHeadings

Symbol

TableMethod is an option for NumberTableForm. It specifies which number form to uses. Values can be NumberForm, ScientificForm or EngineeringForm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`TableMethod

Symbol

System`TableSpacing

Documentation Local » | Web »

Attributes {Protected}

Full Name System`TableSpacing

Symbol

UseMask is a function for MeanSignal and DriftCorrect.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`UseMask

ReconstructionTools

Functions

Symbol

AverageKspace[kspace, types, met] calculates the average of the kspace over the type "N_aver". The method that can be used are "Mean", "Total", "Hamming".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`AverageKspace

Symbol

CoilCombine[sig] combines the coil signals sig. Where sig is {nCoils, ...}.

CoilCombine[sig, cov] combines the coil signals sig. Where sig is {nCoils, ...} and cov the complex noise correlation matrix.

CoilCombine[sig, cov, sens] combines the coil signals sig. Where sig is {nCoils, ...} and cov the complex noise correlation matrix and sense the coils sensItivity.

Possible coil combination methods are "Sum", "RootSumSquares", "RoemerEqualNoise", "RoemerEqualSignal", "WSVD".

RootSumSquares needs the signal. Can be performed with and without the noise covariance matrix

RoemerEqualNoise needs the signal and the noise covariance matrix. Can be performed with

and without the sense data, without sense data the sensItivity is estimated using the signal and the RSS reconstruction of the signal.

RoemerEqualSignal needs the signal and the noise covariance matrix and the sense data.

WSVD needs the signal and the noise covariance matrix.

Documentation Local »

Default Definitions SyntaxInformation[CoilCombine] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options {Method → RoemerEqualNoise, SenseSmoothing → True, NoisePrewhiten → False}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`CoilCombine

Symbol

CoilWeightedRecon[kspace, noise, head] performs reconstruction of raw MS2D MRI data. The input kspace, noise and head are obtained using ReadListData.

The coil combination Methods can be "Roemer" or "RSS".

Documentation Local »

Default Definitions SyntaxInformation[CoilWeightedRecon] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › EchoShiftData → 0 ... (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`CoilWeightedRecon

Symbol

CoilWeightedReconCSI[kspace, noise, head] performs reconstruction of raw 3D CSI data. The input kspace, noise and head are obtained using ReadListData.

The coil combination Methods can be "Roemer" or "WSVD".

Documentation Local »
Default Definitions SyntaxInformation[CoilWeightedReconCSI] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › HammingFilter → False ... (7 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`CoilWeightedReconCSI

Symbol

DeconvolveCSIdata[spectra] deconvolves the CSI spectra after HammingFilterCSI to revert the blurring of the Hamming filtering.

DeconvolveCSIdata[spectra, ham] deconvolves the CSI spectra with the acquired weighting ham to revert the blurring of the kspace weighting.

Documentation Local »
Default Definitions SyntaxInformation[DeconvolveCSIdata] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {WienerRegularization → 0.007, DeconvolutionMethod → Wiener}
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`DeconvolveCSIdata

Symbol

FourierKspace2D[kspace,head] performs a 2D reconstruction of 2D kspace data. Where kspace and head are generated by ReadListData.

Documentation Local »
Default Definitions SyntaxInformation[FourierKspace2D] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierKspace2D

Symbol

FourierKspace3D[kspace,head] performs a 3D reconstruction of 3D kspace data. Where kspace and head are generated by ReadListData.

Documentation Local »
Default Definitions SyntaxInformation[FourierKspace3D] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierKspace3D

Symbol

FourierKspaceCSI[kspace,head] performs a 3D reconstruction of 3D CSI kspace data. Where kspace and head are generated by ReadListData.

Documentation Local »
Default Definitions SyntaxInformation[FourierKspaceCSI] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierKspaceCSI

Symbol

FourierRescaleData[data] rescales the data to double the dimensions using zero padding in fourier space.

FourierRescaleData[data, factor] rescales the data to factor times the dimensions using zero padding in fourier space.

Documentation Local »
Default Definitions SyntaxInformation[FourierRescaleData] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierRescaleData

Symbol

FourierShift[data] shift the data to the right by half the data dimensions.

Documentation Local »
Default Definitions SyntaxInformation[FourierShift] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierShift

Symbol

FourierShiftData[data, {dz, dy, dx}}] shifts the data with the specified voxels dx, dy and dz.

Output data is complex data.

Documentation Local »

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierShiftData

Symbol

FourierShifted[kspace] shifts the kspace half the kspace dimensions and then performs a FourierTransform on the kspace.

Documentation Local »
Default Definitions SyntaxInformation[FourierShifted] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`FourierShifted

Symbol

HammingFilterCSI[kspace] applies a Hamming filter to the k−space data. The data can be can be 1D, 2D or 3D, the spectral dimensions is the last dimensions (x,y,z, spectra).

Documentation Local »
Default Definitions SyntaxInformation[HammingFilterCSI] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`HammingFilterCSI

Symbol

HammingFilterData[kspace] applies a Hamming filter to the data. The data is in image space and can be 1D, 2D or 3D.

Documentation Local »
Default Definitions SyntaxInformation[HammingFilterData] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`HammingFilterData

Symbol

InverseFourierShift[data] shift the data to the left by half the data dimensions.

Documentation Local »
Default Definitions SyntaxInformation[InverseFourierShift] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`InverseFourierShift

Symbol

InverseFourierShifted[data] performs a InverseFourierTransform on the data and then shifts the kspace half the kspace dimensions.

Documentation Local »

Default Definitions SyntaxInformation[InverseFourierShifted] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`InverseFourierShifted

Symbol

MakeHammingFilter[xdim] makes a 1D HammingKernel for filtering k–space.

MakeHammingFilter[{xdim}] makes a 1D HammingKernel for filtering k–space.

MakeHammingFilter[{xdim, ydim}] makes a 2D HammingKernel for filtering k–space in 2D CSI data of size {xdim, ydim}.

MakeHammingFilter[{xdim, ydim, zdim}] makes a 3D HammingKernel for filtering k–space in 3D CSI data of size {xdim, ydim, zdim}.

Documentation Local »

Default Definitions SyntaxInformation[MakeHammingFilter] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`MakeHammingFilter

Symbol

MakeSense[coils, cov] makes a sense map for coils. Each coil signal is divided by the RSS reconstruction of the coils.

Documentation Local »

Default Definitions SyntaxInformation[MakeSense] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options {SenseSmoothing → Hamming, SenseWeight → 1}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`MakeSense

Symbol

MeanType[kspace, types, type] calculates the Mean of the kspace data on type, where type is part of types. The kspace and types are generated by ReadListData.

MeanType[{kspace, types}, type] calculates the Mean of the kspace data on type, where type is part of types.

MeanType[kspace, types, {type,..}] calculates the Mean of the kspace data on each of the list type, where type is part of types.

Output is {kspace, types}.

Documentation Local »
Default Definitions SyntaxInformation[MeanType] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`MeanType

Symbol

NoiseCorrelation[noise] calculates the noise correlation matrix, noise is {nrCoils, noise Samples}.

Documentation Local »
Default Definitions SyntaxInformation[NoiseCorrelation] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NoiseCorrelation

Symbol

NoiseCovariance[noise] calculates the noise covariance matrix, noise is {nrCoils, noise Samples}.

Documentation Local »
Default Definitions SyntaxInformation[NoiseCovariance] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NoiseCovariance

Symbol

NoisePrewhitening[sig, cov] prewhiten the signal sig using the noise covariance matrix cov. The signal and covariance matrix can be any dimension.

Documentation Local »
Default Definitions SyntaxInformation[NoisePrewhitening] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options Method → PCA
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NoisePrewhitening

Symbol

NormalizeSpectra[spec] normalizes spectra to be scaled to the max value of the absolute signal = 1000. Can be any dimension.

Documentation Local »
Default Definitions SyntaxInformation[NormalizeSpectra] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NormalizeSpectra

Symbol

OrderKspace[kspace, types, order] reorders the kspace data to order, where order is a list and each value is a part of types. The kspace and types are generated by ReadListData.

Documentation Local »
Default Definitions SyntaxInformation[OrderKspace] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`OrderKspace

Symbol

ReadListData[file] reads a list/data raw data file from the philips MR platform. The input file can either be .list or .data file.

Output is {{rawData, noise}, {head, types}}.

ReadListData[file, print] does the same but if print is set False no reporting is done.

Documentation Local »
Default Definitions SyntaxInformation[ReadListData] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`ReadListData

Symbol

SagittalTranspose[data] makes a transpose of the data of the second level and reverses the slices.

Documentation Local »
Default Definitions SyntaxInformation[SagittalTranspose] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`SagittalTranspose

Symbol

ShiftedFourier[kspace] performs a FourierTransform on the kspace and then shifts the data half the data dimensions.

Documentation Local »
Default Definitions SyntaxInformation[ShiftedFourier] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`ShiftedFourier

Symbol

ShiftedInverseFourier[data] shifts the data half the data dimensions and then performs a InverseFourierTransform on the data.

Documentation Local »
Default Definitions SyntaxInformation[ShiftedInverseFourier] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`ShiftedInverseFourier

Symbol

TotalType[kspace, types, type] calculates the Total of the kspace data on type, where type is part of types. The kspace and types are generated by ReadListData.

TotalType[{kspace, types}, type] calculates the Total of the kspace data on type, where type is part of types.

TotalType[kspace, types, {type,..}] calculates the Total of the kspace data on each of the list type, where type is part of types.

Output is {kspace, types}.

Documentation Local »
Default Definitions SyntaxInformation[TotalType] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`TotalType

Symbol

ZeroPad[data] up scales the data resolution by a factor 2 by zero padding.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`ZeroPad

Options

Symbol

AcquisitionMethod is an option for CoilWeightedReconCSI. Values can be "Fid" or "Echo".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`AcquisitionMethod

Symbol

CoilSamples is an option for CoilWeightedReconCSI and specifies how many fud samples are used to calculate the coil sensItivity for Roemer reconstruction.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`CoilSamples

Symbol

DeconvolutionMethod is an option for DeconvolveCSIdata. It specifies which deconvolution method to used.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`DeconvolutionMethod

Symbol

DenoiseCSI is an option for CoilWeightedReconCSI. Specifies if denoiseing needs to be applied. Can be "Coils", "Spectra" or False.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`DenoiseCSI

Symbol

EchoShiftData is an option for CoilWeightedRecon.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`EchoShiftData

Symbol

HammingFilter is an option for CoilWeightedReconCSI. If True it applies a spatial hamming filter to the data.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`HammingFilter

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

NoisePrewhiten is an option for CoilWeightedReconCSI. Specifies if noise pre whitening is done for Roemer.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NoisePrewhiten

Symbol

NormalizeOutputSpectra is an option for CoilWeightedReconCSI.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`NormalizeOutputSpectra

Symbol

OutputSense is an option for CoilWeightedRecon. If set true the function will also output the used Sense map.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ReconstructionTools`OutputSense

Symbol

ReconFilter is an option for CoilWeighted recon. If true the reconstruction gets a hamming filter.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`ReconFilter

Symbol

RescaleRecon is an option for CoilWeightedRecon. If set true the data will be scaled to the range 0–1000.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`RescaleRecon

Symbol

SenseSmoothing is an option for MakeSense. If set True the data and sense maps are smoothed using hamming filters.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`SenseSmoothing

Symbol

SenseWeight is an option for MakeSense. Is a integer between 0 and Infinity and define the amount of SOS weighting in the SENSE map.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`SenseWeight

Symbol

WienerRegularization is an option for DeconvolveCSIdata. It defines te amount of regularization used in the wiener deconvolution.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ReconstructionTools`WienerRegularization

RelaxometryTools

Functions

Symbol

CalibrateEPGT2Fit[datan, times, angle] calculates the Fat t2 ralaxation that will be used in the EPGT2fit.

Outputs the fat t2 value.

Documentation Local »

Default Definitions SyntaxInformation[CalibrateEPGT2Fit] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options ❯ EPGRelaxPars → {{0, 100}, {20, 300}, {1400., 365.}} … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`CalibrateEPGT2Fit

Symbol

CreateT2Dictionary[{T1m, T1f}, {nEcho, detlaTE}, angle] Creates a EPG signal dictionary used for EPGT2fit.

Every dictionary that is defined is cached.

The output is in units as defined by the detlaTE, e.g. if detlaTE is in ms the output is in ms.

The TR and TE should be in the same units as Dela.

Output is {dictionary, vals}.

Documentation Local »

Default Definitions SyntaxInformation[CreateT2Dictionary] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options ❯ DictB1Range → {0.5, 1.4, 0.01} … (7 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`CreateT2Dictionary

Symbol

DictionaryMinSearch[dictionary, y] performs dictionary minimization of data y. dictionary is generated with CreateT2Dictionary.

Output is {{t2, b1}, fwfraction, residualError}.

Documentation Local »

Default Definitions SyntaxInformation[DictionaryMinSearch] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`DictionaryMinSearch

Symbol

EPGSignal[{nEcho, echoSpace}, {t1, t2}, {ex_angle,ref_angle}, b1] generates a EPG t2 curve with stimulated echos.

t1, t2 and echoSpace are in ms, angel is in degree, b1 is between 0 and 1.

Output is the EPG Signal vector.

EPGSignal[] is based on DOI: 10.1002/jmri.24619.

Documentation Local »

Default Definitions SyntaxInformation[EPGSignal] = {ArgumentsPattern → {_, _, _, _, _., OptionsPattern[]}}

Options EPGOutput → Magnitude

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGSignal

Symbol

EPGT2Fit[data, {nEcho, detlaTE}, {exitation, refoucs}] fits the t2 based on Marty B

et.al. Simultaneous muscle water t2 and fat fraction mapping using transverse relaxometry with stimulated echo compensation.

Exitation and refocus are the RF pulse angles e.g. 90,180. They can also be a range of angeles over the slice profile as defined by GetSliceProfile.

The output is in units as defined by the detlaTE, e.g. if detlaTE is in ms the output is in ms.

The exitation and refocus are defined in Degrees.

Output is {{{T2map, T2fatMap, B1Map}, {wat, fat, fatMap}, residual}, callibration} or {{T2map, T2fatMap, B1Map}, {wat, fat, fatMap}, residual}.

The residual is the root mean square error of the fit over the echoes, in signal units.

EPGT2Fit[] is based on DOI: 10.1002/nbm.3459.

Documentation Local »

Default Definitions SyntaxInformation[EPGT2Fit] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options › EPGRelaxPars → {1400., 365.} … (18 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGT2Fit

Symbol

NonLinearEPGFit[{vals, T2cons}, y] performs dictionary minimization of data y. vals = {{T1muscle, T1fat, T2fat}, {nEcho, echoSpace, angle}}.

Output is {{t2, b1}, fwfraction, residualError}.

Documentation Local »

Default Definitions SyntaxInformation[NonLinearEPGFit] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`NonLinearEPGFit

Symbol

ShiftPulseProfile[angs, shift] shifts the reference pulse profile by shift and makes the.

ans = {exitation, refocus} as generated by GetPulseProfile. Shift is the shift in sample points.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`ShiftPulseProfile

Symbol

T1Fit[data, TR] fits the t1 value to the data using a nonlinear method. The output is in units as defined by the TR, e.g. if TR is in ms the TR is in ms. Output is {t1, apar, bpar}.

Documentation Local »

Default Definitions SyntaxInformation[T1Fit] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`T1Fit

Symbol

T1rhoFit[data, EchoTimes] fits the T1rho value to the data using linear or nonlinear methods.

The output is in units as defined by the EchoTimes, e.g. if EchoTimes is in ms the output is in ms.

Output is {S(0), T1rhomap}.

Documentation Local »

Default Definitions SyntaxInformation[T1rhoFit] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options Method → Linear

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`T1rhoFit

Symbol

T2Fit[data, EchoTimes] fits the t2 value to the data using linear or nonlinear

methods.vThe output is in units as defined by the EchoTimes, e.g. if EchoTimes is in ms the output is in ms.vOutput is {S(0), t2}.

Documentation Local »
Default Definitions SyntaxInformation[T2Fit] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options Method → Linear
Attributes {Protected, ReadProtected}
Full Name QMRITools`RelaxometryTools`T2Fit

Symbol

TriExponentialT2Fit[data, EchoTimes] fits the t2 based on Azzabou N et.al.

Validation of a generic approach to muscle water t2 determination at 3T in fat–infiltrated skeletal muscle. J. Magn. Reson. 2015.

The fat t2 parameters are automatically estimated from the high signal voxels from the last echo.

The output is in units as defined by the EchoTimes, e.g. if EchoTimes is in ms the output is in ms.

The output fraction is between 0 an 1.

Output is {{S(0), fatFraction, muscleFraction, T2map},callibration} or {S(0), fatFraction, muscleFranction, T2map}.

Documentation Local »
Default Definitions SyntaxInformation[TriExponentialT2Fit] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options OutputCalibration → False
Attributes {Protected, ReadProtected}
Full Name QMRITools`RelaxometryTools`TriExponentialT2Fit

Options

Symbol

DictB1Range is an option for CreateT2Dictionary and EPGT2Fit. It specifies the range and step of the b1 values in the dictionary {min, max, step}.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`RelaxometryTools`DictB1Range

Symbol

DictT2fRange is an option for CreateT2Dictionary and EPGT2Fit. is specifies the range and step of the t2 fat values in the dictionary {min, max, step} in ms.

If a single value is given this fixed value is used a long as EPGCalibrate is False.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`DictT2fRange

Symbol

DictT2fValue is an option for EPGFit.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`DictT2fValue

Symbol

DictT2IncludeWater is an options for EPGT2Fit.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`DictT2IncludeWater

Symbol

DictT2Range is an option for CreateT2Dictionary and EPGT2Fit. is specifies the range and step of the t2 values in the dictionary {min, max, step} in ms.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`DictT2Range

Symbol

EPGCalibrate is an option for EPGT2Fit. If set to True it does autmatic callibration of the t2 fat relaxation time.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGCalibrate

Symbol

EPGFatShift is an options for EPGT2Fit. Specifies the amount of shift of the fat refocusing pulse relative to the fat exitation pulse.

Can be obtained form GetPulseProfile.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGFatShift

Symbol

EPGFitFat is an option for EPGT2Fit.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGFitFat

Symbol

EPGFitPoints is a option for CalibrateEPGT2Fit and EPGT2Fit. Number of points is 200 by default.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGFitPoints

Symbol

EPGMethod is an optionf for EPGT2Fit. Values can be "NLLS", "dictionary" or "dictionaryM".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGMethod

Symbol

EPGMethodCal is an option for CalibrateEPGT2Fit and EPGT2Fit. The calibration can be done using "1comp", "2comp", "2compF".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGMethodCal

Symbol

EPGOutput is an option for EPGSignal. It specifies how to format the output, as magnitude or complex.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGOutput

Symbol

EPGRefocussingPhase is an option for EPGT2Fit and CreateT2Dictionary. It specifies the phase of the refocussing pulse.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGRefocussingPhase

Symbol

EPGRelaxPars is and option for EPGT2Fit. Needs to be {T1muscl, T1Fat, T2Fat} in ms, defaul is {1400,365,137} in ms.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGRelaxPars

Symbol

EPGSmoothB1 is an options for EPGT2Fit. If set to True the b1 map of the fit will be smoothed after which the minimization if perfomed again but with a fixed b1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`EPGSmoothB1

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

OutputCalibration is an option for EPGT2Fit and TriExponentialT2Fit. If true it outputs the calibartion values.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`OutputCalibration

Symbol

WaterFatShift is an options for EPGT2Fit. It specifies the amount of water fat shift in voxels.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`WaterFatShift

Symbol

WaterFatShiftDirection is an options for EPGT2Fit. It specifies the water fat shift direction: "left", "right", "up" and "down".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`RelaxometryTools`WaterFatShiftDirection

ScientificColorData

Functions

Symbol

AddScientificColors[dir] adds the scientific color data (https://zenodo.org/records/8409685) from the specified folder.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ScientificColorData`AddScientificColors

Symbol

ExtractColorData[] Extracts the scientific color data archive (https://zenodo.org/records/8409685).

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ScientificColorData`ExtractColorData

Options

SegmentationTools

Functions

Symbol

ApplySegmentationNetwork[data, net] segments data using net. Data can be an array or a nii file path. Net can be a network name, file, or NetGraph.

ApplySegmentationNetwork[data, net, node] returns the network output at the specified intermediate node rather than the final segmentation.

ApplySegmentationNetwork[{datFol, outFol}, net] processes all nii files in datFol and saves results to outFol.

ApplySegmentationNetwork[{{datFol, outFol}, {inTag, outTag}}, net] uses custom file tags to find input and name output files.

Documentation Local »

Default Definitions SyntaxInformation[ApplySegmentationNetwork] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › TargetDevice → CPU … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`ApplySegmentationNetwork

Symbol

AugmentImageData[image, {rotate, flip}] augments the input image by rotating between −180 and 180 degrees and flipping.

The inputs rotate and flip can be set to True or False.

AugmentImageData[{image, ..}, {rotate, flip}] same but for a list of images.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`AugmentImageData

Symbol

AugmentTrainingData[{data, seg}, vox] augments data and segmentation consistently.

AugmentTrainingData[{data, seg}, vox, aug] controls augmentation. aug can be True/False for all, or an association with any of the keys "Flip", "Rotate", "Scale", "Noise", "Blur", "Bias" (unspecified keys default to True).

AugmentTrainingData[data, vox] augments data only, returning the data without segmentation.

Documentation	Local »
Default Definitions	SyntaxInformation[AugmentTrainingData] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}
Options	{Augment2D → False, MaskedPretraining → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`SegmentationTools`AugmentTrainingData

Symbol

CheckSegmentation[seg] checks the segmentation for errors and returns a vector of two numbers, the first indicates if the segmentation has more than one region, the second indicates if it has holes.

Documentation	Local »
Default Definitions	SyntaxInformation[CheckSegmentation] = {ArgumentsPattern → {_, _.}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`SegmentationTools`CheckSegmentation

Symbol

ClassifyData[data, method] classifies the input data using the Body classification network.

Method "Body" returns {side, positions}. Any method containing "Side" returns the detected side only.

Any other method returns the raw network output.

Documentation	Local »
Default Definitions	SyntaxInformation[ClassifyData] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options	{TargetDevice → CPU, Monitor → False}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`SegmentationTools`ClassifyData

Symbol

CommonLabels[{seg1, seg2, ..}] gives the labels, without background 0, that are present in all of the segmentations.

Documentation Local »
Default Definitions SyntaxInformation[CommonLabels] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`CommonLabels

Symbol

CompareSegmentations[ref, pred, vox] compares the segmentations ref and pred for all labels that are present in both, using the voxel size vox.

CompareSegmentations[ref, pred, labs, vox] compares only the labels labs.

CompareSegmentations[segs, vox] with segs an association of named segmentations compares every pair of segmentations for the labels present in all of them.

The output is a Dataset with a row per label and a column per measure, for the association input it is an association with an entry "name1 vs name2" per pair.

The measures are set with the Method option, which can be "Dice", "Jaccard" or any method of SurfaceDistance ("Mean", "Median", "RMS", "HD", "HD95", "Std"), or a list of these.

Documentation Local »
Default Definitions SyntaxInformation[CompareSegmentations] = {ArgumentsPattern → {_, _, _., _., OptionsPattern[]}}

Options Method → {Dice, HD95}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`CompareSegmentations

Symbol

CopyTrainedNetwork[fileIn, loc] copy a trained network to final location.

CopyTrainedNetwork[fileIn, loc, dim] does the same but appends dim to the network name, used for the "2D" or "3D" segmentation network variants.

loc can be "HeadNeck", "Shoulder", "Torso", "Hip", "UpperLeg", "LowerLeg", "Arm", or "PosSide" to copy the body

position/side classification network.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`CopyTrainedNetwork

Symbol

DataToPatches[data, patchSize] creates non–overlapping patches covering data with minimal overlap.

DataToPatches[data, patchSize, n] creates patches with n additional overlap steps per dimension.

DataToPatches[data, patchSize, pts] extracts patches at the given pre–computed ranges pts.

DataToPatches[data, pts] extracts patches at ranges pts without resizing.

Output is {patches, ranges} where ranges can be passed to PatchesToData.

Documentation Local »
Default Definitions SyntaxInformation[DataToPatches] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options {PatchNumber → 0, PatchPadding → 0}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`DataToPatches

Symbol

DiceSimilarity[ref, pred] gives the Dice Similarity between 1 and 0 of segmentations ref and pred for class equals 1.

DiceSimilarity[x, y, class] gives the Dice Similarity of segmentations ref and pred for class.

DiceSimilarity[x, y, {class, ..}] gives the Dice Similarity of segmentations ref and pred for the list of gives classes.

Documentation Local »
Default Definitions SyntaxInformation[DiceSimilarity] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`DiceSimilarity

Symbol

FindPatchDim[net, dim] finds the largest patch size for the network net that fits within 8 GB for data of dimensions dim {z, y, x}.

FindPatchDim[net, dim, lim] uses a maximum memory of lim GB.

Output is {memory in GB, patch size}.

Documentation Local »
Default Definitions SyntaxInformation[FindPatchDim] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`FindPatchDim

Symbol

FreezeEncoderLayers[net] freezes the "start" stem node and the first 3 encoder blocks of net ("enc_1" .. "enc_3") for transfer learning, returning a LearningRateMultipliers spec to pass to NetTrain.

FreezeEncoderLayers[net, n] freezes "start" plus the first n encoder blocks instead.

FreezeEncoderLayers[net, n, includeStart] controls whether "start" is included, default True – set to False if the input channel count differs from the pretrained net, since "start" is then freshly initialized and must stay trainable.

Documentation Local »

Default Definitions SyntaxInformation[FreezeEncoderLayers] = {ArgumentsPattern → {_, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`FreezeEncoderLayers

Symbol

GetNeuralNet[name] loads a pre–trained neural net by asset name. The net is cached within a session.

GetNeuralNet[file] loads a net from a file path directly.

GetNeuralNet[net] passes through an existing NetGraph or NetChain unchanged.

GetNeuralNet["Clear"] clears the session cache.

Documentation Local »

Default Definitions SyntaxInformation[GetNeuralNet] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`GetNeuralNet

Symbol

GetTrainData[data, batch size, patch] creates a training batch of size batch size with patch size patch.

The input data can be out of memory in the form of a list of "*wxf" files that contain the data, segmentation and voxel size or a list of "*.nii" files in the form {{"data.nii", "segmentation.nii"}..}. The input data can be in memory in a list in the form {{data, segmentation, vox}..}

GetTrainData[data, batch size, patch, nClass] If nClass is set to an value n > 0 the segmentations are decoded in n classes.

If MaskedPretraining –> True the target becomes the clean data itself and the input is masked with noise filled blocks, for self supervised pretraining.

Documentation Local »

Default Definitions SyntaxInformation[GetTrainData] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › PatchesPerSet → 1 … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`GetTrainData

Symbol

ImportITKLabels[] loads the default MusclesLegLabels asset.

ImportITKLabels[file] imports an ITKSnap label file by path or asset name.

ImportITKLabels[file, "Labels"] returns label integer –> name replacement rules.

ImportITKLabels[file, "Names"] returns name –> label integer replacement rules.

ImportITKLabels[file, "List"] returns {names, labels} as separate lists.

Documentation Local »

Default Definitions SyntaxInformation[ImportITKLabels] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`ImportITKLabels

Symbol

JaccardSimilarity[ref, pred] gives the Jaccard Similarity between 1 and 0 of segmentations ref and pred for class equals 1.

JaccardSimilarity[x, y, class] gives the Jaccard Similarity of segmentations ref and pred for class.

JaccardSimilarity[x, y, {class, ..}] gives the Jaccard Similarity of segmentations ref and pred for the list of gives classes.

Documentation Local »

Default Definitions SyntaxInformation[JaccardSimilarity] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`JaccardSimilarity

Symbol

MakeChannelClassGrid[data, label] makes a 3 x 3 grid of cross–sectional images of the channels data overlaid with a cross–sectional image of the classes label of a training dataset generated.

MakeChannelClassGrid[data, label, n] makes a n x n.

MakeChannelClassGrid[data, label, {n, m}] makes a n x m.

Documentation Local »

Default Definitions SyntaxInformation[MakeChannelClassGrid] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeChannelClassGrid

Symbol

MakeChannelClassImage[data, label] makes a cross–sectional image of the channels data overlaid with a cross–sectional image of the classes label of a training dataset generated.

MakeChannelClassImage[data, label, {off,max}] same but with explicit definition of background value b and number of classes n.

MakeChannelClassImage[data, label, vox] same but with the aspect ratio determined by vox.

MakeChannelClassImage[data, label, {off,max}, vox] same with explicit definition and aspect ratio definition.

Documentation Local »

Default Definitions SyntaxInformation[MakeChannelClassImage] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeChannelClassImage

Symbol

MakeChannelGrid[data] makes a 3 x 3 grid of cross–sectional images of the channels data of a training dataset generated by GetTrainData.

MakeChannelGrid[data, n] makes a n x n.

MakeChannelGrid[data, {n, m}] makes a n x m.

Documentation Local »

Default Definitions SyntaxInformation[MakeChannelGrid] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeChannelGrid

Symbol

MakeChannelImage[data] makes a cross–sectional image of the channels data of a training dataset generated by GetTrainData.

MakeChannelImage[data, vox] same but with the aspect ratio determined by vox.

Documentation Local »

Default Definitions SyntaxInformation[MakeChannelImage] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeChannelImage

Symbol

MakeClassImage[label] makes a cross–sectional image of the classes label of a training dataset generated by GetTrainData

MakeClassImage[label, {b, n}] same but with explicit definition of background value b and number of classes n.

MakeClassImage[data, vox] same but with the aspect ratio determined by vox.

MakeClassImage[label, {b, n}, vox] same with explicit definition and aspect ratio definition.

Documentation Local »

Default Definitions SyntaxInformation[MakeClassImage] = {ArgumentsPattern → {_, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeClassImage

Symbol

MakeDistanceMap[mask] makes a distance map of the given mask in voxels. The distance map is positive inside the mask and negative outside the mask.

MakeDistanceMap[mask, vox] makes a distance map of the given mask in the same unit as vox. The distance map is positive

inside the mask and negative outside the mask.

Documentation Local »

Default Definitions SyntaxInformation[MakeDistanceMap] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options DistanceRange → Automatic

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeDistanceMap

Symbol

MakeTrainData[{data1, data2, n}] makes multi contrast training data by blending the co–registered 3D datasets data1 and data2 in n linear steps from data1 to data2. Each blend is normalized and the result is masked. The output is a 4D dataset where each blend is a channel.

MakeTrainData[{{data1, data2, n}, ..}] does the same for multiple pairs and joins all blends as channels, e.g. {{outph, inph, 6}, {wat, fat, 5}}.

The output can be used as data for PrepareTrainingData, during training a random channel is selected for each sample which acts as contrast augmentation.

Documentation Local »

Default Definitions SyntaxInformation[MakeTrainData] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MakeTrainData

Symbol

MuscleLabelToName[{lab, ..}] converts list of lab, which need to be integers to names using the file GetAssetLocation["MusclesLegLabels"].

MuscleLabelToName[{lab, ..}, file] does the same but uses a user defined ITKSnap label definition file.

Documentation Local »

Default Definitions SyntaxInformation[MuscleLabelToName] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MuscleLabelToName

Symbol

MuscleNameToLabel[{name, ..}] converts list of muscle names to integer labels using the file GetAssetLocation["MusclesLegLabels"]

MuscleNameToLabel[{name, ..}, file] does the same but uses a user defined ITKSnap label definition file.

Documentation Local »

Default Definitions SyntaxInformation[MuscleNameToLabel] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MuscleNameToLabel

Symbol

PatchesToData[patches, ranges] reconstructs data from patches at the given ranges.

PatchesToData[patches, ranges, dim] reconstructs into an array of dimensions dim. Overlapping patches are averaged.

PatchesToData[patches, ranges, dim, labels] reconstructs segmentation patches. For each label only the largest connected component is kept and overlapping labels are resolved. Returns a merged integer segmentation.

Documentation Local »
Default Definitions SyntaxInformation[PatchesToData] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`PatchesToData

Symbol

PrepareTrainingData[inFolder, outFolder] prepares the data in de inFolder for training a neural network for segmentation and outputs in outFolder.

PrepareTrainingData[{labFolder, datFolder}, outFolder] does the same but the labels are stored in labFolder and data is stored in datFolder.

Documentation Local »
Default Definitions SyntaxInformation[PrepareTrainingData] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options ❯ LabelTag → label … (9 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`PrepareTrainingData

Symbol

RunMuscleMap[file] will run MuscleMap segmentation on the selected nii file.

RunMuscleMap[{data, vox}] will run MuscleMap segmentation of the data. Using TargetDevice "GPU" (default) or "CPU" the device is selected. Using Monitor –> True reports the steps while running.

Documentation Local »
Default Definitions SyntaxInformation[RunMuscleMap] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ❯ MuscleMapPythonEnvironment → Automatic … (5 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`RunMuscleMap

Symbol

SegmentData[data] segments the data using the default "Body" method.

SegmentData[data, what] segments using the specified anatomical region. With full body output labels what can be "Body", "LegsBody", "LegsHipBody", "HipBody", "UpperLegsBody", "LowerLegsBody", or "ShoulderBody". With legacy output labels what can be "Legs", "LegsHip", "UpperLegs", "LowerLegs", "Hip", "Shoulder", or "Arm".

SegmentData[{data, vox}] and SegmentData[{data, vox}, what] additionally give the voxel size of data. Unless vox is {1, 1, 1}, data is rescaled before segmentation to {vox[[1]], 1.5, 1.5} for SegmentationDimension –> "2D" or to {6, 1.5, 1.5} for "3D".

If NetworkOutput –> "Both" returns {segmentation, confidence}, any other value always returns just the segmentation.

Documentation Local »

Default Definitions SyntaxInformation[SegmentData] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › TargetDevice → CPU … (5 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`SegmentData

Symbol

SegmentDataGUI[] is a function that creates a graphical user interface (GUI) for segmenting data.

It prompts the user to enter the paths for the input and output files, and allows them to select the segmentation type.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`SegmentDataGUI

Symbol

ShowTrainLog[folder] shows an interactive plot of the training log stored in folder as JSON files.

ShowTrainLog[folder, min] filters rounds with fewer than min progress entries. Default is 5.

The interface allows selecting metrics, toggling log scale, Gaussian filtering, and live folder reloading.

Documentation Local »

Default Definitions SyntaxInformation[ShowTrainLog] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`ShowTrainLog

Symbol

SplitDataForSegmentation[data] splits data for "Legs" segmentation, detecting left/right side and upper/lower leg position automatically using a classification network.

SplitDataForSegmentation[data, what] splits for the specified region. What can be any region accepted by SegmentData, i.e. "Body", "LegsBody", "LegsHipBody", "HipBody", "UpperLegsBody", "LowerLegsBody", "ShoulderBody", "Legs", "LegsHip", "UpperLegs", "LowerLegs", "Hip", "Shoulder", or "Arm".

SplitDataForSegmentation[data, seg] splits both data and segmentation identically for "Legs".

SplitDataForSegmentation[data, seg, what] does the same for the specified region.

Output is {{patches, ranges, dim}, locations}.

Documentation	Local »
Default Definitions	SyntaxInformation[SplitDataForSegmentation] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}
Options	{Monitor → False, TargetDevice → CPU, SplitOverlap → 0.05}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`SegmentationTools`SplitDataForSegmentation

Symbol

SurfaceDistance[ref, pred] gives the surface distance of segmentations ref and pred for class 1, assuming isotropic unit voxels. Use the vox argument for correct physical distances.

SurfaceDistance[ref, pred, class] gives the surface distance for the given integer class.

SurfaceDistance[ref, pred, {class, ..}] gives the surface distance for each class in the list.

SurfaceDistance[ref, pred, class, vox] gives the surface distance in millimeters using voxel size vox.

SurfaceDistance[ref, pred, {class, ..}, vox] does the same for a list of classes.

The Method option controls the distance metric: "Mean", "Median", "RMS", "HD", "HD95" (default), or "Std". A list of methods can be given to return multiple metrics at once.

Documentation	Local »
Default Definitions	SyntaxInformation[SurfaceDistance] = {ArgumentsPattern → {_, _, _., _., OptionsPattern[]}}
Options	Method → HD95
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`SegmentationTools`SurfaceDistance

Symbol

TrainSegmentationNetwork[{inFol, outFol}] trains a segmentation network from scratch. Training data should be wxf files in inFol.
inFol can also be a list of folders, in which case wxf files from all folders are used. Progress is saved each round to outFol.
TrainSegmentationNetwork[{inFol, outFol}, netCont] continues training. netCont can be "Start" to restart, a NetGraph, a wlnet file,
or a previous outFol to continue from the last saved network.
Loss functions can be All or a subset of {"Dice", "Jaccard", "Tversky", "MSD", "CE", "Focal", "TopK"}.
If MaskedPretraining –> True self supervised pretraining is done instead: the network gets a single output channel, only the MSD and MAE losses are used, and the training data is masked and noise filled as generated by GetTrainData with MaskedPretraining –> True.

Documentation Local »
Default Definitions SyntaxInformation[TrainSegmentationNetwork] = {ArgumentsPattern → {{_, _}, _., OptionsPattern[]}}

Options › LoadTrainingData → True ... (30 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`TrainSegmentationNetwork

Symbol

$SegmentationLabels is an association of all valid values for what in SegmentData to the name of the labels asset of its output, see GetAssetLocation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`$SegmentationLabels

Options

`Missing[UnknownSymbol, Augment2D]`

Symbol

MaskedPretraining is an option for GetTrainData and TrainSegmentationNetwork. If set to True self supervised masked–pretraining data/mode is used instead of segmentation data: the target becomes the clean data itself and the input is masked with noise filled blocks.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MaskedPretraining

Symbol

ActivationType is an option for MakeUnet. It specifies which activation layer is used in the network. It can be "LeakyRELU" or any type allowed by a "name" definition in ElementwiseLayer.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`ActivationType

Symbol

AugmentData is an option for GetTrainData and TrainSegmentationNetwork. If set True the training data is augmented.

It can also be set to "2D" or "3D" to control if augmentation is done through plane or only in-plane.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`AugmentData

Symbol

BatchSize is an option for NetTrain and related functions that specifies the size of a batch of examples to process together.

Documentation Local » | Web »

Attributes {Protected, ReadProtected}

Full Name System`BatchSize

Symbol

BlockType is an option for MakeUnet. It specifies the type of block used in the network. It can be "Conv", "UNet", "ResNet", "DenseNet", "Inception", or "U2Net".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`BlockType

Symbol

CatenateMethod is an option for MakeUnet. It specifies how the network catenates the skip and upscale channel. It can be "Tot", or "Cat".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`CatenateMethod

Symbol

CleanUpSegmentations is an option for PrepareTrainingData. If set to True the segmentations are cleaned up by removing holes reducing to one volume and smoothing.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`CleanUpSegmentations

Symbol

DataPadding is an option for ApplySegmentationNetwork. Defines how much to pad the data patches in all directions.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`DataPadding

Symbol

DataTag is an option for PrepareTrainingData. It defines the tag used in the filenames of the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`DataTag

Symbol

DistanceRange is an option for MakeDistanceMap. It defines the range of the distance map outside the segmentation in voxels. Values can be Automatic, All, or a integer value. If All the distance map is calculated for the whole image. If 0 the distance map is only calculated inside the segmentation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`DistanceRange

Symbol

DownsampleSchedule is an option for MakeUnet. It defines how the data is downsampled for each of the deeper layers of the Unet.

By default is is a factor two for each layer. A custom schedule for a 5 layer 3D Unet could be {{2,2,2},{1,2,2},{2,2,2},{1,2,2}, 1}.

The deepest layer is always downsampled by 1 and therefore not needed to be specified.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`DownsampleSchedule

Symbol

DropoutRate is an option for MakeUnet. It specifies how much dropout is used after each block. It is a value between 0 and 1, default is .2.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`DropoutRate

Symbol

FeatureSchedule is an option for MakeUnet. It defines how the number of features is up–sampled for each of the deeper layers of the Unet.

By default it increases the number of features by a factor 2 each layer, i.e. {1, 2, 4, 8, 16}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`FeatureSchedule

Symbol

FreezeEncoderDepth is an option for TrainSegmentationNetwork. If set to an integer n, freezes the first n

encoder blocks of a pretrained input network so their weights are not updated during training, useful for transfer learning

from a pretrained encoder. Only applies when netCont is not "Start".

Default is None, which trains all layers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`FreezeEncoderDepth

Symbol

InputLabels is an option for PrepareTrainingData. Can be set to a list of integers corresponding to the labels to be used from the given segmentation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`InputLabels

Symbol

L2Regularization is an option for TrainSegmentationNetwork. It defines the L2 regularization factor.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`L2Regularization

Symbol

LabelTag is an option for PrepareTrainingData. It defines the tag used in the filenames of the label data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`LabelTag

Symbol

LearningRate is an option for NetTrain that specifies the rate at which to adjust neural net weights in order to minimize the training loss.

Documentation Local » | Web »

Attributes {Protected, ReadProtected}

Full Name System`LearningRate

Symbol

LoadTrainingData is an option for TrainSegmentationNetwork. If set to True the training data is loaded from the disk.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`LoadTrainingData

Symbol

LossFunction is an option for NetTrain that specifies how to compare actual and requested outputs from a neural net.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`LossFunction

Symbol

MaskedPretraining is an option for GetTrainData and TrainSegmentationNetwork. If set to True self supervised masked−pretraining data/mode is used instead of segmentation data: the target becomes the clean data itself and the input is masked with noise filled blocks.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MaskedPretraining

Symbol

MaxMemorySize is an option for SegmentData and ApplySegmentationNetwork. Defines the maximum memory a patch can use in Gb. Higher is better.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MaxMemorySize

Symbol

MaxTrainingRounds is an option for NetTrain and related functions that specifies the maximum number of rounds of training to do.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`MaxTrainingRounds

Symbol

Method is an option for various algorithm−intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »
Attributes {HoldAll, Protected, ReadProtected}
Full Name System`Monitor

Symbol

MonitorInterval is an option for TrainSegmentationNetwork. It defines how often the training is monitored.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MonitorInterval

Symbol

MultiChannel is an option for TrainSegmentationNetwork. Multi channel training is not yet implemented and the option currently has no effect.

Networks always have a single input channel, for 4D training data a random channel is selected for each sample.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MultiChannel

Symbol

MuscleMapLabels is an option for RunMuscleMap. If False the MuscleMap labels are converted to the QMRITools "MuscleLabels" numbering. If True the original MuscleMap labels are returned.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MuscleMapLabels

Symbol

MuscleMapPath is the path to the MuscleMap code that is cloned from github and installed according to the muscle map readme.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SegmentationTools`MuscleMapPath

Symbol

MuscleMapPythonEnvironment is the path to the conda environment of the MuscleMap installation. Fhe folder should contain the python.exe file.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`MuscleMapPythonEnvironment

Symbol

NetworkArchitecture is an option for MakeUnet. It defines the architecture of the network. It can be "UNet", "UNet+", or "UNet++".

For "UNet+" or "UNet++" it can also be {arch, i} where i specifies how many of the top layers are connected to the mapping layer.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetworkArchitecture

Symbol

NetworkDepth is an option for MakeUnet. It specifies how deep the UNET will be.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NetworkDepth

Symbol

NetworkOutput is an option for ApplySegmentationNetwork. Can be "Segmentation" (default) to return the segmentation, "Confidence" to return the per–voxel class confidence (see ClassConfidence), "Both" to return {segmentation, confidence}, or "Volume" for a network that outputs a reconstructed volume rather than class probabilities (e.g. a self–supervised pretrained network), which merges patches without ClassDecoder.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`NetworkOutput

Symbol

NormalizationType is an option for MakeUnet. It specifies which normalization layer is used in the network. It can be "Batch" or "Instance" for either BatchNormalization or NormalizationLayer which is instance normalization.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`NormalizationType

Symbol

OutputLabels is an option for PrepareTrainingData. Can be set to a list of integers. The used label number will be replaced by these numbers.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`OutputLabels

Symbol

OutputTag is an option for PrepareTrainingData. It defines the tag that will be added to the output training data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`OutputTag

Symbol

PadData is an option for GetTrainData and TrainSegmentationNetwork.. If set to an integers the that number of slices on the top and bottom of the data can be made 0. This is done to learn cut of datasets.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`PadData

Symbol

PatchesPerSet is an option for GetTrainData. Defines how many random patches per dataset are created within the batch.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`PatchesPerSet

Symbol

PatchNumber is an option for DataToPatches. Can be an integer value >= 0. The larger the number the more overlap the patches have.

The minimal number of patches in each direction is calculated, and then for each dimension the given number is added.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`PatchNumber

Symbol

PatchPadding is an option for DataToPatches. Can be an integer value >= 0. It pads the chosen patch size with the given number.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`PatchPadding

Symbol

PatchSize is an option for TrainSegmentationNetwork. Defines the patch size used in the network training.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`PatchSize

Symbol

RescaleMethod is an option for MakeUnet. It specifies how the network rescales. It can be "Conv" or "Pool".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`RescaleMethod

Symbol

RestartLearningCycle is an option for TrainSegmentationNetwork. If set to True and continuing training from a previous run, the one-cycle learning rate schedule (including its warmup) is restarted from 0 and fit to the remaining rounds, instead of resuming mid-cycle where the previous run left off.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`RestartLearningCycle

Symbol

RoundLength is an option for TrainSegmentationNetwork. Defines how many batches will be seen during each training round.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`RoundLength

Symbol

SegmentationDimension is an option for SegmentData. The default value is "3D" but can also be set to "2D" to used 2D slice by slice segmentation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`SegmentationDimension

Symbol

SegmentationsPerSlice is an option for PrepareTrainingData. It defines how many segmentations should be present in a slice for it to be used in the training data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`SegmentationsPerSlice

Symbol

SettingSchedule is an option for MakeUnet. It defines the settings for the Unet blocks. If one setting is given it applied to all layers.

If a list of settings is given the settings can be different per layer. The following settings are the default settings.

"Unet": convolution block repetitions, 2, "ResNet" –> convolution block repetitions, 2, "DenseNet" –> {dense depth, block repetitions}, {4,2}, "Inception" –> {inception width, block repetitions}, {4,2}, "U2Net"–> {Unet depth, downscale}, {5, True}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`NeuralNetworkTools`SettingSchedule

Symbol

SplitOverlap is an option of SplitDataForSegmentation. Has to be a value between 0 and 0.2 and defines how much the left and right side overlap.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`SplitOverlap

Symbol

System`TargetDevice

Documentation Local » | Web »

Attributes {Protected, ReadProtected}

Full Name System`TargetDevice

Symbol

TestRun is an option for PrepareTrainingData. If set to True the data is not saved only analyzed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`TestRun

Symbol

TrainVoxelSize is an option for PrepareTrainingData. It defines the voxel size of the training data. When set to Automatic the voxel size is that of the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`TrainVoxelSize

Symbol

UseParallelKernels is an option for TrainSegmentationNetwork. If set to True (default), producer kernels are launched as raw WSTP links, data loading and batch augmentation run on these separate kernels while NetTrain runs concurrently on its own kernel, which can substantially speed up training when augmentation is the bottleneck. If set to "Parallel", the older shared–queue Parallel–framework kernel pool is used instead. If set to {True, n} or {"Parallel", n} at most n producer kernels are used, useful for debugging on a small scale. If set to False batches are generated in–process.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SegmentationTools`UseParallelKernels

ShapeTools

Functions

Symbol

ApplyShapeModel[model, std] applies the std to the shape model to create the points. The model is made by MakeShapeModel, the std is obtained by FitShapeModel. The std can be a number, vector or matrix.

ApplyShapeModel[model, std, n] does the same but std is a number and n the integer indicating which vector to be used.

Documentation Local »

Default Definitions SyntaxInformation[ApplyShapeModel] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`ApplyShapeModel

Symbol

EvaluateModel[points, mesh] return a dynamic plot showing an evaluation of the shape model.

The points and mesh are obtained from TemplateToVolume.

EvaluateModel[{mean, mat}, points, cells] does the same but for a explicit shape model given by mean and mat.

the shape model can be obtained from MakeShapeModel.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`EvaluateModel

Symbol

FitShapeModel[{meanPoints, model}, points] fits the shape model to the points, the shape model is defined by meanPoints and model which are obtained form MakeShapeModel.

FitShapeModel[{meanPoints, model}, points, n] fits the model using the first n shape vectors from the model.

Documentation Local »

Default Definitions SyntaxInformation[FitShapeModel] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`FitShapeModel

Symbol

MakeMuscleTemplate[masks, vox] makes a muscle template form a series of masks. The output is the mean template muscle mask as a binary volume.

Documentation Local »
Default Definitions SyntaxInformation[MakeMuscleTemplate] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MakeMuscleTemplate

Symbol

MakeRegionMesh[data] makes a MeshRegion from a binary mask assuming an 1x1x1mm isotropic voxel size and 1000 mesh coordinates.

MakeRegionMesh[data, n] makes a MeshRegion with exactly n mesh coordinates.

MakeRegionMesh[data, vox] makes a MeshRegion taking in account the given voxel size vox with 1000 mesh coordinates.

MakeRegionMesh[data, vox, n] makes a MeshRegion taking in account the given voxel size vox with exactly n mesh coordinates.

Documentation Local »
Default Definitions SyntaxInformation[MakeRegionMesh] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options MeshOutput → Mesh
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MakeRegionMesh

Symbol

MakeShapeModel[points] makes the statistical shape model for the points given. The points per muscle must have the same number of coordinates.

The points can be obtained form TemplateToVolume. The output is {meanPoints, model} which are needed in FitShapeModel.

Documentation Local »
Default Definitions SyntaxInformation[MakeShapeModel] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MakeShapeModel

Symbol

MakeTemplatePlot[meshes, points] visualizes the template alignment. The input meshes and points are made by TemplateToVolume with the TemplateOutput set to "Meshes".

Documentation Local »
Default Definitions SyntaxInformation[MakeTemplatePlot] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MakeTemplatePlot

Symbol

MaskFromDistanceMap[dist] converts a mask distance map created with MaskToDistanceMap back to a binary mask.

MaskFromDistanceMap[dist, method] converts a mask distance map created with MaskToDistanceMap back to a binary mask. The method can be "both", "out" or "in" which calculates the mask distance inside and outside, outside, or inside the mask.

Documentation Local »
Default Definitions SyntaxInformation[MaskFromDistanceMap] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MaskFromDistanceMap

Symbol

MaskToDistanceMap[mask] converts a binary mask into a mask distance map.

MaskToDistanceMap[mask, method] converts a binary mask into a distance mask. The method can be "both", "out" or "in" which calculates the mask distance inside and outside, outside, or inside the mask.

Documentation Local »
Default Definitions SyntaxInformation[MaskToDistanceMap] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MaskToDistanceMap

Symbol

MeshGridPlot[meshes, points] gives a grid of shape meshes defined by meshes and points. These can be obtained from with the TemplateOutput set to "Meshes".

Documentation Local »
Options MeshesPerRow → 5
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MeshGridPlot

Symbol

PlotMesh[mesh] Plots a MeshRegion as a GraphicsComplex.

PlotMesh[points, cells] plots a MeshRegion which has been split into the coordinates and cells using SplitRegionMesh.

Documentation Local »
Default Definitions SyntaxInformation[PlotMesh] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › MeshOpacity → 1 … (5 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`PlotMesh

Symbol

PlotShapeVariation[points, mesh] plots the template mesh which shows where the model has variation for each principal component. The needed points and mesh are obtained from TemplateToVolume.

PlotShapeVariation[points, mesh, n] does the same but only shows the first n principal components.

Documentation Local »
Options MeshesPerRow → 5
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`PlotShapeVariation

Symbol

ScaleToVolume[mask, vox] rescales a series of mask all to the same volume, which will be the median volume of all the masks.

ScaleToVolume[data, vox, vol] does the same but then rescales to the target volume vol.

Documentation Local »
Default Definitions SyntaxInformation[ScaleToVolume] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`ScaleToVolume

Symbol

SplitRegionMesh[mesh] splits the mesh in the MeshCoordinates and the mesh cells.

Documentation Local »
Default Definitions SyntaxInformation[SplitRegionMesh] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`SplitRegionMesh

Symbol

TemplateToVolume[{masks, voxM}, {template, voxT}] registers a mask or a series of masks to the given muscle template.

The default output are the mesh points per mask volume and the cell needed to generate the mesh as {points, mesh}.

Documentation Local »
Default Definitions SyntaxInformation[TemplateToVolume] = {ArgumentsPattern → {{_, _}, {_, _}, OptionsPattern[]}}

Options › MeshPoints → {500, 2000}… (7 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`TemplateToVolume

Options

Symbol

MeshColor is an option for PlotMesh. It specifies the color of the mesh. The color can be a single color or a list of colors with the same length as the mesh coordinates.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`MeshColor

Symbol

MeshEdgeColor is an option for PlotMesh. It specifies the color for the border of the mesh vertices.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshEdgeColor

Symbol

MeshesPerRow is and option for MeshGridPlot and PlotShapeVariation. It specifies how many volumes are shown per row.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshesPerRow

Symbol

MeshOpacity is an option for PlotMesh. It specifies how much opacity the mesh has.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshOpacity

Symbol

MeshOutput is an option for MakeRegionMesh. Values can be "Mesh", "Points", "Cells", "PointsCells", "All".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshOutput

Symbol

MeshPointColor is an option for PlotMesh. It specifies the color of the mesh coordinates. The points need to be one and the points are shown as spheres.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshPointColor

Symbol

MeshPoints is an option for TemplateToVolume. It specifies how many mesh coordinates will be used in the template.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshPoints

Symbol

MeshPointSize is an option for PlotMesh. It specifies how large the points will be, its defined in mm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`MeshPointSize

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »

Attributes {HoldAll, Protected, ReadProtected}

Full Name System`Monitor

Symbol

SplineSpacing is an option for TemplateToVolume. It defines how big the b–splines are during the registration process. Its value is multiplied with the voxel size which then becomes the spline spacing.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`SplineSpacing

Symbol

TemplateDilation is an option for TemplateToVolume. It specifies how much the template is dilated during registration.

The value is specified in mm and it helps to properly align the points.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ShapeTools`TemplateDilation

Symbol

TemplateOutput is an option for TemplateToVolume. The value can be "Meshes" "PointsCells" or "Points". The "Meshes" "PointsCells" outputs are needed for template and model visualization. The "Points" is the only thing needed for model fitting.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`TemplateOutput

Symbol

TemplatePadding is an option for TemplateToVolume. It specifies how much the template will be padded to prevent cut off of larger muscles.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`TemplatePadding

Symbol

VolumeRescale is an option for TemplateToVolume. If this option is set to True the volume of each muscle will be normalized to the mean volume of the template volumes or the template volume.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`ShapeTools`VolumeRescale

SimulationTools

Functions

Symbol

AddNoise[data, noise] ads rician noise to the data with a given sigma or SNR value.

Documentation Local »
Default Definitions SyntaxInformation[AddNoise] = {ArgumentsPattern → {_, _, OptionsPattern[]}}
Options {NoiseSize → Sigma, NoiseType → Absolute}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`AddNoise

Symbol

BlochSeries[vectorIn, delta_t, freqRange, B1] performs a Bloch simulation of an RF pulse.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`BlochSeries

Symbol

CalculateGFactor[factors, sensitivity, wMat] calculates a gFactor for given sensitivity maps and noise corelation w. given the sense factors which is a list of three integers.

Documentation Local »

Default Definitions SyntaxInformation[CalculateGFactor] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options GRegularization → 0.

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`CalculateGFactor

Symbol

CreateDiffData[sig, eig, bvec, gradients, dim] creates a DTI datasets of dimensions dim with sig as unweighted signal s0 and bvec and gradients.

eig can be {l1, l2, l3}, {{l1, l2, l3}, {e1, e2, e3}}, {{l1, l2, l3}, "Random"}, {{l1, l2, l3}, "RandomZ"} or {{l1, l2, l3}, "OrtRandom"}.

Uses Tensor internally.

CreateDiffData[] is based on DOI: 10.1002/nbm.2959.

Documentation Local »

Default Definitions SyntaxInformation[CreateDiffData] = {ArgumentsPattern → {_, _, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`CreateDiffData

Symbol

ErnstAngle[] shows Ernst angle plot for t1 = 1400ms and tr = 15 ms.

ErnstAngle[t1] shows Ernst angle plot for t1 and tr = 15 ms.

ErnstAngle[t1, tr] shows Ernst angle plot for t1 and tr.

Documentation Local »

Default Definitions SyntaxInformation[ErnstAngle] = {ArgumentsPattern → {_., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`ErnstAngle

Symbol

GESignal[ang, {tr, t1}] calculates the gradient echo signal for flip angles ang using tr and t1.

GESignal[ang, {{tr1, tr2}, t1}] calculates the dual tr gradient echo signal for flip angles ang using tr1, tr2 and t1.

GESignal[ang, {{tr1, tr2}, t1}, f] does the same but the flip angle of the second tr is f ang.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`GESignal

Symbol

GetPulseProfile[excitation, refocus] gives the pulse angle profiles for the excitation and refocusing pulses.

a pulse is defined as {"name", flip angle, {G_strength, Dur, BW}}.

GetPulseProfile[{"name", flip angle, {G_strength, Dur, BW}}] gives detailed slice profile information of one pulse.

output is {ex_angle_profile, ref_angel_profile, {plots}}.

output for single pulse is {{distance, Mt, Mz, Mx, My, ang, phase}, plots}.

Documentation Local »

Default Definitions SyntaxInformation[GetPulseProfile] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options › MagnetizationVector → {0, 0, 1} … (4 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`GetPulseProfile

Symbol

GFactorSimulation[sensitivity, cov, {dir,sense}] calculates the g–factor maps for given sensitivity maps and noise corelation cov in one direction.

The sense factors are a list of integers in a given direction: "LR", "FH", or "AP".

GFactorSimulation[sensitivity, cov, {dir1,sense1}, {dir2,sense2}] calculates the g–factor maps for given sensitivity maps and noise corelation w in two directions.

Documentation Local »

Default Definitions SyntaxInformation[GFactorSimulation] = {ArgumentsPattern → {_, _, _, _., _., OptionsPattern[]}}

Options {GRegularization → 0., GOutput → Grid}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`GFactorSimulation

Symbol

PlotSimulation[pars, xval, true, label, color] plots the pars (output form Parameters). Using label as PlotLabel and xval as x axis thicks.

tr are the true parameter values. color are the color used for the plot.

Documentation Local »

Default Definitions SyntaxInformation[PlotSimulation] = {ArgumentsPattern → {_, _, _, _, _, OptionsPattern[]}}

Options PlotRange → {{0, 3}, {0, 3}, {0, 3}, {0, 3}, {0, 1}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSimulation

Symbol

PlotSimulationAngle[par, xData, label, col] plots pars (output from AngleParameters).

Documentation Local »

Options PlotRange → {0, 90}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSimulationAngle

Symbol

PlotSimulationAngleHist[pars, label, xData] plots pars (output from AngleParameters).

Documentation Local »

Default Definitions SyntaxInformation[PlotSimulationAngleHist] = {ArgumentsPattern → {_, _, _, _, _., OptionsPattern[]}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSimulationAngleHist

Symbol

PlotSimulationHist[pars, label, xData, tr] plots the pars (output form Parameters).

Using label as plot label and xData as x axis label. tr are the true parameter values.

Documentation Local »

Default Definitions SyntaxInformation[PlotSimulationHist] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSimulationHist

Symbol

PlotSimulationVec[tens, xData, label] plots the eigenvectors from simulated tensors.

Documentation Local »

Default Definitions SyntaxInformation[PlotSimulationVec] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options SortVecs → True

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSimulationVec

Symbol

PlotSincProfile[] shows the SincProfile for a block pulse for f0 = GetGyro[7, "31P"].

PlotSincProfile[f0] does the same but for f0.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`PlotSincProfile

Symbol

Pulses[name] gives the pulse shape of some predefined Philips pulse shapes.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`Pulses

Symbol

Signal[par, tr, te] calculates the MRI signal at a given tr and te. Par is defined as {pd, t1, t2}.

Documentation Local »
Default Definitions SyntaxInformation[Signal] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`Signal

Symbol

SimAngleParameters[tens,vec] calculates the diffusion eigenvectors for tens compared to the true values vec.

The output can be used in PlotSimulationAngleHist and PlotSimulationAngle.

Documentation Local »
Default Definitions SyntaxInformation[SimAngleParameters] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`SimAngleParameters

Symbol

SimParameters[tens] calculates the diffusion parameters for tens. The output can be used in PlotSimulationHist and PlotSimulation.

Documentation Local »
Default Definitions SyntaxInformation[SimParameters] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options Reject → False
Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`SimParameters

Symbol

SimulateDualTR[] simulates the signal of a Dual tr t1 map, with an adjustable flip angle ratio between the second and first tr.

Documentation Local »
Default Definitions SyntaxInformation[SimulateDualTR] = {ArgumentsPattern → {_.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`SimulateDualTR

Symbol

SimulateSliceEPG[excitation, refocus, {{t1, t2}, {nEcho, echoSp}, b1}] gives a simulated slice profile and EPG signal plot.

excitation and refocus are generated by GetPulseProfile.

Documentation Local »

Default Definitions SyntaxInformation[SimulateSliceEPG] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options ReportFits → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`SimulateSliceEPG

Symbol

Tensor[{l1, l2, l3}] creates a diffusion tensor with vectors {{0,0,1},{0,1,0},{1,0,0}} and eigenvalues {l1, l2, l3}.

Tensor[{l1, l2, l3}, {e1, e2, e3}] creates a diffusion tensor with vectors {e1, e2, e3} and eigenvalues {l1, l2, l3}.

Tensor[{l1, l2, l3}, "Random"] creates a diffusion tensor with random orthogonal eigenvectors {e1, e2, e2} and eigenvalues {l1, l2, l3}.

Tensor[{l1, l2, l3}, "RandomZ"] creates a diffusion tensor with random orthogonal eigenvectors {{1,0,0}, e2, e3} with random eigenvectors and eigenvalues {l1, l2, l3}.

Tensor[{l1, l2, l3}, "OrtRandom"] creates a diffusion tensor with random orthogonal eigenvectors {{1,0,0},{0,1,0},{0,0,1}} and eigenvalues {l1, l2, l3}.

Tensor[] is based on DOI: 10.1002/nbm.2959.

Documentation Local »

Default Definitions SyntaxInformation[Tensor] = {ArgumentsPattern → {_, _., OptionsPattern[]}}

Options TensOutput → Vector

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`Tensor

Options

Symbol

FatFieldStrength is an option for GetPulseProfile. If the value >0 it will calculate the shift of the fat refocusing pulse compared to the fat excitation pulse.

The shift is in SliceRangeSamples steps.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`FatFieldStrength

Symbol

GOutput is an option for GFactorSimulation. can be "Grid" or "List".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`GOutput

Symbol

GRegularization is an option for CalculateGFactor and GFactorSimulation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`GRegularization

Symbol

MagnetizationVector is an option for GetPulseProfile. It defines the start magnetization vector for the bloch simulation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`MagnetizationVector

Symbol

NoiseSize is an option for AddNoise. Values can be "Sigma", then the noise sigma is given or "SNR", then the SNR is given.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`NoiseSize

Symbol

NoiseType is an option for AddNoise. Values can be "Absolute" or "Complex", and will add either Rician absolute noise or complex noise to the data.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`NoiseType

Symbol

PlotRange is an option for graphics functions that specifies what range of coordinates to include in a plot.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotRange

Symbol

Reject is an option for EigenvalCalc. It True then voxels with negative eigenvalues are rejected and set to 0.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`Reject

Symbol

ReportFits is an option for SimulateSliceEPG. If True it also reports the fit values.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`ReportFits

Symbol

SliceRange is an option for GetPulseProfile. It specifies over which range the slice profile is generated (in mm). the total profile is 2xSliceRange.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`SliceRange

Symbol

SliceRangeSamples is an option for GetPulseProfile. defines how many samples are used to generate half a puls profile.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SimulationTools`SliceRangeSamples

Symbol

SortVecs is an option for PlotSimulationVec.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`SortVecs

Symbol

TensOutput is an option for Tensor. Values can be "Vector" or "Matrix".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SimulationTools`TensOutput

SpectroTools

Functions

Symbol

ApodizeEcho[echo] performs apodization on the echo. The apodization function is set with the option ApodizationFunction.

Documentation Local »

Default Definitions SyntaxInformation[ApodizeEcho] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ApodizationFunction → Hanning

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`ApodizeEcho

Symbol

ApodizeFid[fid] performs apodization on the fid. The apodization function is set with the option ApodizationFunction.

Documentation Local »

Default Definitions SyntaxInformation[ApodizeFid] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options ApodizationFunction → Hanning

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`ApodizeFid

Symbol

ApodizePadEcho[echo] performs apodization on the echo and pads the echo with zeros to increase its length.

Documentation Local »
Default Definitions SyntaxInformation[ApodizePadEcho] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ApodizationFunction → Hanning, PaddingFactor → 2}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ApodizePadEcho

Symbol

ApodizePadFid[fid] performs apodization on the fid and pads the fid with zeros to increase its length.

Documentation Local »
Default Definitions SyntaxInformation[ApodizePadFid] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ApodizationFunction → Hanning, PaddingFactor → 2}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ApodizePadFid

Symbol

ApodizePadSpectra[spec] and doubles the number of spectral points while maintaining the dwell time.

Documentation Local »
Default Definitions SyntaxInformation[ApodizePadSpectra] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ApodizationFunction → Hanning, PaddingFactor → 2, ReadoutType → Fid}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ApodizePadSpectra

Symbol

ApodizeSpectra[spec] performs apodization of the spectra. The apodization function is set with the option ApodizationFunction.

Documentation Local »
Default Definitions SyntaxInformation[ApodizeSpectra] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {ApodizationFunction → Hanning, ReadoutType → Fid}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ApodizeSpectra

Symbol

ChangeDwellTimeFid[fid, dt, dtNew] changes the sampling time of an fid from dwell time dt to dwell time dtNew.

Documentation Local »
Default Definitions SyntaxInformation[ChangeDwellTimeFid] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ChangeDwellTimeFid

Symbol

CorrectTEFid[fid, dw, te] corrects the fid for 1st order phase by extrapolating the missing FID samples in the TE using Hankel matrix SVD analysis.

CorrectTEFid[fid, dw, te, gyro, ppmRan] corrects the fid for 1st order phase by extrapolating

the missing FID samples in the TE using Hankel matrix SVD analysis. Only the part of the spectra in the ppmRan is used for optimization.

Documentation Local »
Default Definitions SyntaxInformation[CorrectTEFid] = {ArgumentsPattern → {_, _, _, _., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`CorrectTEFid

Symbol

CorrectTESpec[spectra, dw, te] corrects the spectra for 1st order phase by extrapolating the missing FID samples in the TE using Hankel matrix SVD analysis.

CorrectTESpec[spectra, dw, te, gyro, ppmRan] corrects the spectra for 1st order phase by extrapolating

the missing FID samples in the TE using Hankel matrix SVD analysis. Only the part of the spectra in the ppmRan is used for optimization.

Documentation Local »
Default Definitions SyntaxInformation[CorrectTESpec] = {ArgumentsPattern → {_, _, _, _., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`CorrectTESpec

Symbol

ExportSparSdat[file, specs, {bw ,te}, {gyro ,nuc}] exports specs to file. Function not done.

Documentation Local »
Default Definitions SyntaxInformation[ExportSparSdat] = {ArgumentsPattern → {_, _, {_, _}, {_, _}, _., OptionsPattern[]}}

Options {SparName → QMRITools, SparOrientation → {0, 0}, SparID → }
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ExportSparSdat

Symbol

GetGyro[nuc, field] geth the gyromagnetic ratio with field the field strength in Tesla and nuc the nucleus available in GyromagneticRatio.

Documentation Local »
Default Definitions SyntaxInformation[GetGyro] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`GetGyro

Symbol

GetPpmRange[spec, {dt, field, nuc}] get the ppm values of the spec where dt is the well time in ms, field the field strength in Tesla and nuc the nucleus available in GyromagneticRatio.

GetPpmRange[spec, dt, field, nuc] get the ppm values of the spec.

GetPpmRange[spec, dt, gyro] get the ppm values of the spec.

Documentation Local »
Default Definitions SyntaxInformation[GetPpmRange] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`GetPpmRange

Symbol

GetSpectraBasisFunctions[{met1, ..., metN}] generates a list of spectra basis functions with names met1 to metN. The names are strings and are the metabolites available in GetSpinSystem.

GetSpectraBasisFunctions[{{prop1}, ..., {propN}}] generates a list of spectra basis functions with properties prop1 to propN. The properties are those specified in MakeSpinSystem.

GetSpectraBasisFunctions[inp, split] generates a list of spectra basis functions. Each metabolite name present in the list split wil be split in individual spectra per peak.

Documentation Local »
Default Definitions SyntaxInformation[GetSpectraBasisFunctions] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options › BasisSequence → {PulseAcquire, 0} … (6 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`GetSpectraBasisFunctions

Symbol

GetTimePpmRange[spec, {dt, field, nuc}] get the timing of the fid and the ppm values

of the spec where dt is the well time in ms, field the field strength in Tesla and nuc the nucleus available in GyromagneticRatio.

GetTimePpmRange[spec, dt, field, nuc] get the timing of the fid and the ppm values of the spec.

GetTimePpmRange[spec, dt, gyro] get the timing of the fid and the ppm values of the spec.

Documentation Local »
Default Definitions SyntaxInformation[GetTimePpmRange] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`GetTimePpmRange

Symbol

GetTimeRange[fid, dt] get the timing of the fid where dt is the well time in ms.

Documentation Local »
Default Definitions SyntaxInformation[GetTimeRange] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`GetTimeRange

Symbol

ImportSparSdat[fileSpar, fileSdat] imports fileSpar and fileSdat files. Function not done.

Documentation Local »
Default Definitions SyntaxInformation[ImportSparSdat] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ImportSparSdat

Symbol

PadEcho[echo] pads the echo with zeros to increase its length.

Documentation Local »
Default Definitions SyntaxInformation[PadEcho] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options PaddingFactor → 2
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`PadEcho

Symbol

PadFid[fid] pads the fid with zeros to increase its length.

Documentation Local »
Default Definitions SyntaxInformation[PadFid] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options PaddingFactor → 2
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`PadFid

Symbol

PadSpectra[spec] doubles the number of spectral points while maintaining the dwell time.

Documentation Local »
Default Definitions SyntaxInformation[PadSpectra] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {PaddingFactor → 2, ReadoutType → Fid}
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`PadSpectra

Symbol

PhaseCorrectSpectra[spec] performs 0th order phase correction of the spectra by minimizing the difference between the real and absolute spectra value.

PhaseCorrectSpectra[spec, dw] performs 0th order phase correction of the spectra using Hankel matrix SVD fitting.

PhaseCorrectSpectra[spec, dw, te] := performs 0th and 1st order phase correction of the spectra

using Hankel matrix SVD fitting. The first order phase is corrected by padding the fid with the missing values in the time before the TE.

PhaseCorrectSpectra[spec, dw, te, gyro, ppmRan] performs 0th and 1st order phase

correction of the spectra using Hankel matrix SVD fitting. Only the part of the spectra in the ppmRan is used for optimization.

Documentation Local »
Default Definitions SyntaxInformation[PhaseCorrectSpectra] = {ArgumentsPattern → {_, _., _., _., _., _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`PhaseCorrectSpectra

Symbol

PhaseShiftFid[spectra, phi0] applies the 0th order phase phi0 to the FID.

Documentation Local »

Default Definitions SyntaxInformation[PhaseShiftFid] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`PhaseShiftFid

Symbol

PhaseShiftSpectra[spectra, phi0] applies the 0th order phase phi0 to the spectra.

PhaseShiftSpectra[spectra, ppm, gyro, phi1] applies the 1st order phase phi1 to the spectra. The ppm can be obtained using GetPpmRange and gyro with GetGyro.

PhaseShiftSpectra[spec, ppm, gyro, {phi0, phi1}] applies the 0th and 1st order phases {phi0, phi1} to the spectra. The ppm can be obtained using GetPpmRange and gyro with GetGyro.

The 0th order phase phi0 is in radians and the 1st order phase phi1 is in ms.

Documentation Local »

Default Definitions SyntaxInformation[PhaseShiftSpectra] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`PhaseShiftSpectra

Symbol

PlotCSIData[spectra, {dwell, gyro}] plots the CSI spectra which has dimensions {z,y,x,nSamp}. The ppm axes is determined by dwell and gyro. Gyro can be obtained with GetGyro.

PlotCSIData[spectra, {dwell, field, nuc}] plots the CSI spectra which has dimensions {z,y,x,nSamp}. The ppm axes is determined by dwell and field and nuc.

Documentation Local »

Default Definitions SyntaxInformation[PlotCSIData] = {ArgumentsPattern → {_, _, _., _., OptionsPattern[]}}

Options PlotRange → Full

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`PlotCSIData

Symbol

PlotFid[fid, dwell] plots the fid assuming dwell as the sampling time.

PlotFid[time, fid] plot the fid where time is the timing of the fid which can be obtained with GetTimeRange.

Documentation Local »

Default Definitions SyntaxInformation[PlotFid] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options › PlotRange → Full ... (8 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`PlotFid

Symbol

PlotSpectra[spectra, {dwell, gyro}] plots the spectra, the ppm axes is determined by dwell and gyro. Gyro can be obtained with GetGyro.

PlotSpectra[spectra {dwell, field, nuc}] plots the spectra, the ppm axes is determined by dwell field and nuc.

PlotSpectra[ppm, spectra] plots the spectra where ppm is the pmm range of the spectra which can be obtained with GetPpmRange.

Documentation Local »

Default Definitions SyntaxInformation[PlotSpectra] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options › PlotRange → Full ... (12 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`PlotSpectra

Symbol

ReadjMRUI[file] read a jMRUI spectrum file.

Output is the {time, spec, {beginTime, samplingInterval}}.

Documentation Local »

Default Definitions SyntaxInformation[ReadjMRUI] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`ReadjMRUI

Symbol

ShiftSpectra[spectra, {dw, gyro}, shift] shifts the spectra by shift. The shift is in ppm.

Documentation Local »

Default Definitions SyntaxInformation[ShiftSpectra] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Options ReadoutType → Fid

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`ShiftSpectra

Symbol

SpectraSimulator[] opens a interactive spectra simulator.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraSimulator

Symbol

TimeShiftEcho[fid, time, gam] applies a line broadening with linewidth gam and a Voigt line shape to the fid. The time can be obtained using GetTimeRange.

TTimeShiftEcho[fid, time, {gam, f}] applies a line broadening with linewidth gam and a custom line shape f to the fid (f=0, "Gaussian", f=1 "Lorentzian").

TTimeShiftEcho[fid, time, gyro, {gam, eps}] applies a line broadening with linewidth gam to the fid and a phase eps that results in eps ppm shift of the spectra. The gyro can be obtained with GetGyro.

TTimeShiftEcho[fid, time, gyro, {gam, eps, f}] applies a line broadening with linewidth gam using a custom line shape f to the fid and a phase eps that results in eps ppm shift of the spectra.

The linewidth gam is given in ms and the spectra shift eps is given in ppm.

Documentation Local »

Default Definitions SyntaxInformation[TimeShiftEcho] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`TimeShiftEcho

Symbol

TimeShiftFid[fid, time, gam] applies a line broadening with linewidth gam and a Voigt line shape to the fid. The time can be obtained using GetTimeRange.

TimeShiftFid[fid, time, {gamL, gamG}] applies a line broadening with linewidth gamG "Gaussian" and gamL "Lorentzian".

TimeShiftFid[fid, time, gyro, {gam, eps}] applies a line broadening with linewidth gam to the fid and a phase eps that results in eps ppm shift of the spectra. The gyro can be obtained with GetGyro.

TimeShiftFid[fid, time, gyro, {{gamL, gamG}, eps}] applies a line broadening with

linewidth linewidth gamG "Gaussian" and gamL "Lorentzian" to the fid and a phase eps that results in eps ppm shift of the spectra.

The linewidth gam is given in ms and the spectra shift eps is given in ppm.

Documentation Local »

Default Definitions SyntaxInformation[TimeShiftFid] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`TimeShiftFid

Options

Symbol

ApodizationFunction is an options for ApodizeFid, ApodizeSpectra, ApodizePadFid, and ApodizePadSpectra. Values can be "Hanning", "Hamming", "Gaussian", "Lorentzian", and "Voigt".

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`ApodizationFunction

Symbol

AspectRatio is an option for Graphics and related functions that specifies the ratio of height to width for a plot.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`AspectRatio

Symbol

BasisSequence is an option for GetSpectraBasisFunctions and specifies which sequence to use.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`BasisSequence

Symbol

CenterFrequency is an option for GetSpinSystem and defines the center frequency in ppm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`JcouplingTools`CenterFrequency

Symbol

Filling is an option for ListPlot, Plot, Plot3D, and related functions that specifies what filling to add under points, curves, and surfaces.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Filling

Symbol

GridLines is an option for two–dimensional graphics functions that specifies grid lines.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`GridLines

Symbol

GridLineSpacing is an option of TransmuralPlot. It defines the spacing of the grid lines.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`CardiacTools`GridLineSpacing

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`ImageSize

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

PaddingFactor is an option for PadFid, PadSpectra, ApodizePadFid, ApodizePadSpectra. It Specifies with which factor to lengthen the fid.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`PaddingFactor

Symbol

PlotColor is an option for GradientPlot can be any color or gradient color name.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`PlotColor

Symbol

PlotLabel is an option for graphics functions that specifies an overall label for a plot.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PlotLabel

Symbol

PlotLabels is an option for visualization functions that specifies what labels to use for each data source.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotLabels

Symbol

PlotRange is an option for graphics functions that specifies what range of coordinates to include in a plot.

Documentation Local » | Web »
Attributes {Protected, ReadProtected}
Full Name System`PlotRange

Symbol

ReadoutType is an option for PadSpectra, ApodizeSpectra, ApodizePadSpectra, ShiftSpectra and FitSpectra. Value can be "Fid" or "Echo".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`ReadoutType

Symbol

SparID is an option for ExportSparSdat.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`SparID

Symbol

SparName is an option for ExportSparSdat.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`SparName

Symbol

SparOrientation is an option for ExportSparSdat.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`SpectroTools`SparOrientation

Symbol

SpectraBandwidth is an option for GetSpectraBasisFunctions and sets the bandwidth of the spectra.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraBandwidth

Symbol

SpectraFieldStrength is an option for GetSpectraBasisFunctions and sets the field strength at which the simulations and fitting is performed.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraFieldStrength

Symbol

SpectraNucleus is an option for GetSpectraBasisFunctions and specifies which nucleus to Simulate or fit, see GyromagneticRatio.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraNucleus

Symbol

SpectraPpmShift is an option for GetSpectraBasisFunctions and defines how much the center frequency is shifted, default is water at 4.65 ppm.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraPpmShift

Symbol

SpectraSamples is an option for GetSpectraBasisFunctions and sets the number of samples in the spectra.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraSamples

Symbol

SpectraSpacing is an option for PlotSpectra and defines the amount of spacing between spectra when multiple spectra are plotted.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`SpectroTools`SpectraSpacing

TaggingTools

Functions

Symbol

AnnalyzeTagging[gridC] tracks the motion in tagged cine data gridC, which is {slices, frames, y, x}.

The wave vectors of the tag grid are estimated from the power spectrum of the first frames. Per slice the displacement between consecutive frames is estimated

from the phase of the band pass filtered tag images for each tag direction and accumulated over the frames.

Output is {{ux, uy}, {im0, im, w0, w}}, all {slices, frames, y, x}, where ux and uy are the accumulated displacements in pixels along the first and second image dimension,

im0 are the reference images, im the motion corrected frames and w0 and w the corresponding tag grid weighting maps.

Documentation Local »

Options {HistoryWeighting → 0.7, MonitorTagging → True}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TaggingTools`AnnalyzeTagging

Symbol

CalculateDisplacementParameters[{motx, moty}, mask] calculates strain and rotation from the displacements motx and moty as given by AnnalyzeTagging, both {slices, frames, y, x}.

The mask is {slices, y, x} and a circle fitted to the mask of each slice defines the center for the circumferential and radial directions.

Output is {Exx, Eyy, Exy, Ecc, Ecr, rot}, all {slices, frames, y, x}, where Exx, Eyy and Exy are the Green–Lagrange strain components along the image axes,

Ecc is the circumferential strain, Ecr the radial strain and rot the rotation around the center in degrees.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TaggingTools`CalculateDisplacementParameters

Symbol

TaggingParPlot[data, label] plots a tagging parameter over time with one line per slice, data is {time, slices}.

TaggingParPlot[data, label, {min, max}] uses the given plot range, by default it is symmetric around zero with 1.2 times the maximal absolute value.

Documentation Local »
Default Definitions SyntaxInformation[TaggingParPlot] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TaggingTools`TaggingParPlot

Options

Symbol

HistoryWeighting is an options for AnnalyzeTagging. It is the weight of the new frame when updating the reference image, reference = (1 – w) reference + w frame.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TaggingTools`HistoryWeighting

Symbol

MonitorTagging is an options for AnnalyzeTagging. If True the intermediate images and displacements are shown during the calculation.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TaggingTools`MonitorTagging

TensorTools

Functions

Symbol

ADCCalc[eigenvalues] calculates the ADC from the given eigenvalues.

Documentation Local »
Default Definitions SyntaxInformation[ADCCalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`ADCCalc

Symbol

AngleCalc[data, vector] calculates the angle between the vector and the data. Data should be an array of dimensions {xxx,3}.

AngleCalc[data, vectors] does the same but calculates the angle between each element of data and the corresponding element of vectors, data and vectors should have the same dimensions.

Documentation Local »
Default Definitions SyntaxInformation[AngleCalc] = {ArgumentsPattern → {_, _, OptionsPattern[]}}

Options Distribution → 0–180
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`AngleCalc

Symbol

AngleMap[data] calculates the zenith and azimuth angles of a 3D dataset (z,x,y,3) containing vectors relative to the slice direction.

Documentation Local »
Default Definitions SyntaxInformation[AngleMap] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`AngleMap

Symbol

ConcatenateDiffusionData[{{data1, .., dataN}, {grad1, .., gradN}, {bval, .., bvalN}, {vox, .., voxN}}] concatenates the diffusion data sets.

ConcatenateDiffusionData[{data1, .., dataN}, {grad1, .., gradN}, {bval, .., bvalN}, {vox, .., voxN}] concatenates the diffusion data sets.

Documentation Local »
Default Definitions SyntaxInformation[ConcatenateDiffusionData] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`ConcatenateDiffusionData

Symbol

Correct[data, phase, shiftPar] corrects the dataset data using the phase map and the shiftPar and interpolation order 1.

Correct[data, phase, shiftPar, int] corrects the dataset data using the phase map and the shiftPar and interpolation order int.

Documentation Local »
Default Definitions SyntaxInformation[Correct] = {ArgumentsPattern → {_, _, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`Correct

Symbol

CreateRPBMDictionary[tm] creates a dictionary of RPBM functions for the given mixing times tm.

Options are the used parameters in the dictionary and the simulated RD values, {pars, sim}.

Documentation	Local »
Default Definitions	SyntaxInformation[CreateRPBMDictionary] = {ArgumentsPattern → {_, OptionsPattern[]}}
Options	{RPBMRangeTau → {100, 2000}, RPBMRangeZeta → {0.1, 3}, RPBMDictionarySize → {100, 1000}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`TensorTools`CreateRPBMDictionary

Symbol

Deriv[disp, vox] calculates the derivative of the displacement along the three main axes. disp is the displacement field, vox is the voxel size.

Deriv[disp, vox, mask] calculates the derivative of the displacement along the three main

axes. Sharp edges between the background and disp are solved by the mask. mask is a mask defining the edge of the displacement field.

Documentation	Local »
Default Definitions	SyntaxInformation[Deriv] = {ArgumentsPattern → {_, _, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`TensorTools`Deriv

Symbol

DriftCorrect[data, bval] drift corrects the data using the signals of the lowest bvalue that has 5 or more unique volumes.

For the function to work optimal it is best to have these volumes evenly spread throughout the data and for the first and last volume to have this low bvalue.

DriftCorrect[] is based on DOI: 10.1002/mrm.26124.

Documentation	Local »
Default Definitions	SyntaxInformation[DriftCorrect] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}
Options	{NormalizeSignal → True, UseMask → True}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`TensorTools`DriftCorrect

Symbol

ECalc[eigenvalues] calculates the E from the given eigenvalues.

Documentation Local »
Default Definitions SyntaxInformation[ECalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`ECalc

Symbol

EigensysCalc[tensor] calculates the eigensystem for the given tensor.

Documentation Local »
Default Definitions SyntaxInformation[EigensysCalc] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {RejectMap → False, Reject → True, PerformanceGoal → Speed}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`EigensysCalc

Symbol

EigenvalCalc[tensor] calculates the eigenvalues for the given tensor.

Documentation Local »
Default Definitions SyntaxInformation[EigenvalCalc] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {RejectMap → False, Reject → True, PerformanceGoal → Speed}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`EigenvalCalc

Symbol

EigenvecCalc[tensor] calculates the eigenvectors for the given tensor.

Documentation Local »
Default Definitions SyntaxInformation[EigenvecCalc] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {RejectMap → False, Reject → True, PerformanceGoal → Speed}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`EigenvecCalc

Symbol

ExpTensor[tensor] transforms the tensor from LogEuclidean space.

ExpTensor[] is based on DOI: 10.1109/42.963816.

Documentation Local »
Default Definitions SyntaxInformation[ExpTensor] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`ExpTensor

Symbol

FACalc[eigenvalues] calculates the FA from the given eigenvalues.

Documentation Local »
Default Definitions SyntaxInformation[FACalc] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`FACalc

Symbol

FitRPBMDictionary[sig, {pars, sim}, snr] fits the RPBM function to the simulated data sig using the parameters pars and the simulated data sim.

The needed dictionary {pars, sim} is generated using CreateRPBMDictionary. The snr is the estimated snr values of the data.

FitRPBMDictionary[sig, {pars, sim}, snr, d0] does the same but fixes the initial diffusion to the given d0.

Documentation Local »
Default Definitions SyntaxInformation[FitRPBMDictionary] = {ArgumentsPattern → {_, _, _, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`FitRPBMDictionary

Symbol

FitRPBMFunction[tms, dat] fits the RPBM function to the data dat using the mixing times tms using FindMinimum.

The output is a rule of the fitted parameters {d0–>val, tau–> tau, zeta –>zeta}.

FitRPBMFunction[tms, dat, init] does the same but uses init as initial guess values.

FitRPBMFunction[tms, dat, init, fix] does the same but uses init as initial guess values and fixes the parameters in fix.

Documentation Local »

Default Definitions SyntaxInformation[FitRPBMFunction] = {ArgumentsPattern → {_, _, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`FitRPBMFunction

Symbol

FlipGradientOrientation[grad, perm] permutes the internal orientation of the gradients, perm can be any permutation of {"x","y","z"}.

FlipGradientOrientation[grad, flip] flips the internal orientation of the gradients, flip can be {1,1,1}, {–1,1,1}, {1,–1,1} or {1,1,–1}.

FlipGradientOrientation[grad, flip, perm] flips and permutes the internal orientation of the gradients.

FlipGradientOrientation[grad, perm, flip]flips and permutes the internal orientation of the gradients.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`FlipGradientOrientation

Symbol

FlipTensorOrientation[tens, perm] permutes the internal orientation of the tensor, perm can be any permutation of {"x","y","z"}.

FlipTensorOrientation[tens, flip] flips the internal orientation of the tensor, flip can be {1,1,1}, {–1,1,1}, {1,–1,1} or {1,1,–1}.

FlipTensorOrientation[tens, flip, perm] flips and permutes the internal orientation of the tensor.

FlipTensorOrientation[tens, perm, flip]flips and permutes the internal orientation of the tensor.

Documentation Local »

Default Definitions SyntaxInformation[FlipTensorOrientation] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`FlipTensorOrientation

Symbol

GetRPBMValues[{d0, tau, zeta}] derives parameters from RPBM function. d0, tau and zeta are the parameters of the RPBM function.

output is a list containing {d0, tau, zeta, dInf, td, tr, l, sv, a, kappa}.

Documentation Local »

Default Definitions SyntaxInformation[GetRPBMValues] = {ArgumentsPattern → {_, _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`GetRPBMValues

Symbol

LogTensor[tensor] transforms the tensor to LogEuclidean space.

LogTensor[] is based on DOI: 10.1109/42.963816.

Documentation Local »

Default Definitions SyntaxInformation[LogTensor] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`LogTensor

Symbol

ParameterCalc[tensor] calculates the eigenvalues and MD and FA from the given tensor. The parameters are l1, l2, l3, MD and FA. l1, l2, l3, MD are in (10^-3 mm^2/s).

Documentation Local »

Default Definitions SyntaxInformation[ParameterCalc] = {ArgumentsPattern → {_, OptionsPattern[]}}

Options {Reject → False, PerformanceGoal → Speed}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`ParameterCalc

Symbol

RemoveIsoImages[data, grad, bval] Removes the ISO images from the philips scanner from the data. ISO images have g={0,0,0} and b>0.

Documentation Local »

Default Definitions SyntaxInformation[RemoveIsoImages] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RemoveIsoImages

Symbol

ResidualCalc[dti,{tensor,s0},gradients,bvector] calculates the tensor residuals for the given dataset.

ResidualCalc[dti,{tensor,s0},outlier,gradients,bvector] calculates the tensor residuals for the given dataset taking in account the outliers.

ResidualCalc[dti,{tensor,s0},bmat] calculates the tensor residuals for the given dataset.

ResidualCalc[dti,{tensor,s0},outlier,bmat] calculates the tensor residuals for the given dataset taking in account the outliers.

ResidualCalc[dti,tensor,gradients,bvector] calculates the tensor residuals for the given dataset. Tensor must contain Log[s0].

ResidualCalc[dti,tensor,outlier,gradients,bvector] calculates the tensor residuals for the given dataset taking in account the outliers. Tensor must contain Log[s0].

ResidualCalc[dti,tensor,bmat] calculates the tensor residuals for the given dataset. Tensor must contain Log[s0].

ResidualCalc[dti,tensor,outlier,bmat] calculates the tensor residuals for the given dataset taking in account the outliers. Tensor must contain Log[s0].

Documentation Local »

Default Definitions SyntaxInformation[ResidualCalc] = {ArgumentsPattern → {_, _, _, _, OptionsPattern[]}}

Options MeanRes → All

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`ResidualCalc

Symbol

RPBMFunction[tm, {d0, tau, zeta}] generates the RPBM function for the given mixing times tm, with d0, tau and zeta.

RPBMFunction[tm, {tau, zeta}] generates the RPBM function for the given mixing times tm, assuming a d0 of 1.

RPBMFunction[tm, {{d0, tau, zeta}, ..}] generates the RPBM function for the given mixing times tm, for a list of d0, tau and zeta.

Documentation Local »

Default Definitions SyntaxInformation[RPBMFunction] = {ArgumentsPattern → {_, _..}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RPBMFunction

Symbol

SigmaCalc[dti,grad,bvec] calculates the noise sigma based on the tensor residual, using a blur factor of 2.

SigmaCalc[dti,tens,grad,bvec] calculates the noise sigma based on the tensor residual, using a blur factor of 2.

SigmaCalc[dti,grad,bvec,blur] calculates the noise sigma based on the tensor residual, If blur is 1 there is no blurring.

SigmaCalc[dti,tens,grad,bvec,blur] calculates the noise sigma based on the tensor residual. If blur is 1 there is no blurring.

Documentation	Local »
Default Definitions	SyntaxInformation[SigmaCalc] = {ArgumentsPattern → {_, _, _, _., _., OptionsPattern[]}}
Options	FilterShape → Median
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`TensorTools`SigmaCalc

Symbol

SortDiffusionData[data, grad, bval] sorts the diffusion datasets grad and bval for magnitude of bvalue.

Documentation	Local »
Default Definitions	SyntaxInformation[SortDiffusionData] = {ArgumentsPattern → {_, _, _}}
Attributes	{Protected, ReadProtected}
Full Name	QMRITools`TensorTools`SortDiffusionData

Symbol

TensorCalc[data, gradients, bvec] calculates the diffusion tensor for the given dataset. allows for multiple unweighted images and multiple bvalues.

allows for different tensor fitting methods. gradient directions must be in the form {{x1,y1,z1}, ..., {xn,yn,zn}} with the unweighted direction as {0,0,0}.

bvec the b–vector, with a bvalue defined for each gradient direction. b value for unweighted images is 0.

TensorCalc[data, bmatrix] calculates the diffusion tensor for the given dataset. allows for multiple unweighted images and multiple bvalues.

bmat is the bmatrix which can be generated using Bmatrix.

The bvalue assumed to be is in s/mm^2 and therefore the output is in mm^2/2.

TensorCalc[] is based on DOI: 10.1016/j.neuroimage.2013.05.028 and 10.1002/mrm.25165.

Documentation Local »

Default Definitions SyntaxInformation[TensorCalc] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options › MonitorCalc → True ... (6 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`TensorCalc

Symbol

TensorCorrect[tensor, phase, shift, vox] corrects the tensor based on B0 field map. Can perform both translation and rotation of tensor.

Documentation Local »

Default Definitions SyntaxInformation[TensorCorrect] = {ArgumentsPattern → {_, _, _, _, _., OptionsPattern[]}}

Options RotationCorrect → False

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`TensorCorrect

Symbol

TransformTensor[tensor, disp, vox] corrects the tensor with voxel size vox based on the displacement field disp. The displacement field is te displacement in mm for each voxel location in x, y and z.

TransformTensor[] is based on DOI: 10.1109/42.963816.

Documentation Local »
Default Definitions SyntaxInformation[TransformTensor] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`TransformTensor

Symbol

WestinMeasures[eigenvalues] calculates the westin measures.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`WestinMeasures

Options

Symbol

Distribution is an option for AngleCalc. values can be "0–180", "0–90" and "–90–90".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`Distribution

Symbol

FilterShape is an option for SigmaCalc. Can be "Gaussian" or "Median".

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`FilterShape

Symbol

FullOutput is an option for TensorCalc when using b–vector. When True also the s0 is given as output.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`FullOutput

Symbol

MeanRes is an option for ResidualCalc. When True the root mean square of the residual is calculated.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`MeanRes

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »

Attributes {Protected}

Full Name System`Method

Symbol

MonitorCalc is an option for many processing functions. When true the process of the calculation is shown.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`MonitorCalc

Symbol

NormalizeSignal is an option for DriftCorrect.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`NormalizeSignal

Symbol

Parallelize[*expr*] evaluates *expr* using automatic parallelization.

Definitions

```
Parallelize[Parallel`Kernels`Private`args$___] := (Parallel`Protected`doAutolaunch[TrueQ[Parallel`Static`$enableLaunchFeedback]];
Parallelize[Parallel`Kernels`Private`args$])
```

Documentation Local » | Web »
Options {DistributedContexts :→ $Context, Method → Automatic, ProgressReporting :→ $ProgressReporting}
Attributes {HoldFirst, Protected}
Full Name System`Parallelize

Symbol

PerformanceGoal is an option for various algorithmic and presentational functions that specifies what aspect of performance to try to optimize with Automatic settings for options.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PerformanceGoal

Symbol

Reject is an option for EigenvalCalc. It True then voxels with negative eigenvalues are rejected and set to 0.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`Reject

Symbol

RejectMap is an option for EigenvalCalc. If Reject is True and RejectMap is True both the eigenvalues as well as a map showing the rejected values is returned.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TensorTools`RejectMap

Symbol

RobustFit is an option for TensorCalc. If true outliers will be rejected in the fit, only works with WLLS.

If FullOutput is given the outlier map is given.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RobustFit

Symbol

RobustFitParameters is an option for TensorCalc. gives the threshold for stopping the iterations and the kappa for the outlier margin, {tr,kappa}.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RobustFitParameters

Symbol

RotationCorrect is an option for TensorCorrect. Default is False. Is a tensor is deformed setting to True also the shear is accounted for by local rotation of the tensor.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RotationCorrect

Symbol

RPBMDictionarySize is an option for CreateRPBMDictionary. Default is {100, 1000}.

The number of dictionaries and number of points in the dictionary.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RPBMDictionarySize

Symbol

RPBMRangeTau is an option for CreateRPBMDictionary. Default is {100, 2000}. The range of the tau values used in the RPBM function.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RPBMRangeTau

Symbol

RPBMRangeZeta is an option for CreateRPBMDictionary. Default is {0.1, 3}. The range of the zeta values used in the RPBM function.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TensorTools`RPBMRangeZeta

Symbol

UseMask is a function for MeanSignal and DriftCorrect.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`ProcessingTools`UseMask

TractographyTools

Functions

Symbol

EndpointDensityMap[tracts, vox, dim] makes an endpoint density map based on the tracts endpoints.

Documentation Local »

Default Definitions SyntaxInformation[EndpointDensityMap] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`EndpointDensityMap

Symbol

ExportTCK[file, tracts, vox, dim] exports the tracts to the given *.tck file and automatically generates the header.

ExportTCK[file, tracts, header] exports the tracts to the given *.tck file. The header can be the header string or the header association as given by ImportTCK.

Documentation Local »

Default Definitions SyntaxInformation[ExportTCK] = {ArgumentsPattern → {_, _, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`ExportTCK

Symbol

ExportTracts[file, tracts, vox, dim, seeds] exports the tracts, vox, dim and seeds to *.trk file.

Documentation Local »

Default Definitions SyntaxInformation[ExportTracts] = {ArgumentsPattern → {_, _., _., _., _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`ExportTracts

Symbol

FiberTractography[tensor, vox] performs fiber tractography on the tensor with voxel dimensions vox.

FiberTractography[tensor, vox, {par, {min, max}}] performs fiber tractography on the tensor with voxel dimensions vox with additional stopping criteria par, where tracts are only generated between values of par min and max.

FiberTractography[tensor, vox, {{par, {min, max}}, ..}] performs fiber tractography on the tensor with voxel dimensions vox with multiple additional stopping criteria.

Documentation Local »

Default Definitions SyntaxInformation[FiberTractography] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › FiberLengthRange → {20, 500} ... (11 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`FiberTractography

Symbol

FilterTractLength[tracts, {min, max}] filters the tracts based on the length of the tracts.

Documentation Local »

Default Definitions SyntaxInformation[FilterTractLength] = {ArgumentsPattern → {_, _, _.}}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`FilterTractLength

Symbol

FilterTracts[tracts, vox, {select.. }] filters the tracts based on the list of select criteria.

Select criteria are defined as {"logic",{"how", criteria}}.

The "logic" parameter can be "and", "or" and "not".

The "how" parameter can be:

- "x", "y", or "z" for slice selection, here criteria is a slice number
- "through" for selecting tract that go through a roi, here criteria is a 3D mask.
- "within" for selecting tract that fit fully within the roi, here criteria is a 3D mask.
- "partWithin" for selecting the part of the tracts that fall within the roi, here criteria is a 3D mask.

Any number of select criteria can be listed.

Documentation Local »

Default Definitions SyntaxInformation[FilterTracts] = {ArgumentsPattern → {_, _, _, _., OptionsPattern[]}}

Options FiberLengthRange → {20, 500}

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`FilterTracts

Symbol

FindTensorPermutation[tensor, vox] performs tractography for all tensor permutations and gives back the one that has the longest tracts.

FindTensorPermutation[tensor, vox, {par, {min, max}}] same but with additional stopping criteria par, where tracts are only generated between values of par min and max.

FindTensorPermutation[tensor, vox, {{par, {min, max}}, ..}] same but with multiple additional stopping criteria.

Output = {permutations, flips, plot}

FindTensorPermutation[] is based on DOI: 10.1016/j.media.2014.05.012.

Documentation Local »

Default Definitions SyntaxInformation[FindTensorPermutation] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options › FiberLengthRange → {20, 500} … (7 total)

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`FindTensorPermutation

Symbol

FitTracts[tract] fits a tract or a list of tracts, each defined as a list of {x, y, z} coordinates with a polynomial function.

FitTracts[tract, vox, dim] does the same but constrains all the tract coordinates to the volume defined by dim.

Documentation Local »
Default Definitions SyntaxInformation[FitTracts] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options FittingOrder → 5
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`FitTracts

Symbol

GetTractValues[tracts, parameter, vox] gets the value of the parameter map at each tract coordinate.

Documentation Local »
Default Definitions SyntaxInformation[GetTractValues] = {ArgumentsPattern → {_, _, _., OptionsPattern[]}}

Options InterpolationOrder → 1
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`GetTractValues

Symbol

ImportTCK[file] imports a *.tck file. the output is {tracts, vox, header}.

Documentation Local »
Default Definitions SyntaxInformation[ImportTCK] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`ImportTCK

Symbol

ImportTracts[file] imports a *.trk file. It can contain {tracts, vox, dim, seeds}.

Documentation Local »
Default Definitions SyntaxInformation[ImportTracts] = {ArgumentsPattern → {_}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`ImportTracts

Symbol

MoveTracts[tracts, off] moves the tract coordinates by off, which is {x, y, z}.

Documentation Local »
Default Definitions SyntaxInformation[MoveTracts] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`MoveTracts

Symbol

PlotSegmentedTracts[tracts, segments, dim, vox] plots the tracts after segmenting each segment.

PlotSegmentedTracts[tracts, segments, bones, dim, vox] plots the tracts after segmenting each segment also rendering a bone volume.

Documentation Local »
Default Definitions SyntaxInformation[PlotSegmentedTracts] = {ArgumentsPattern → {_, _, _, _, _., OptionsPattern[]}}

Options › MaxTracts → 5000 … (11 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`PlotSegmentedTracts

Symbol

PlotTracts[tracts, vox] plots the tracts assuming a BoxRatio based on vox.

PlotTracts[tracts, vox, dim] plots the tracts assuming a BoxRatio based on vox with a PlotRange spanning the full dim.

Documentation Local »
Default Definitions SyntaxInformation[PlotTracts] = {ArgumentsPattern → {_, _., _., OptionsPattern[]}}

Options › MaxTracts → 2000 … (10 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`PlotTracts

Symbol

ResampleTracts[tracts, n] resample each Tract to exactly n vertices.

Documentation Local »
Default Definitions SyntaxInformation[ResampleTracts] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`ResampleTracts

Symbol

RescaleTracts[tracts, sc] scales the tract coordinates by 1/sc, which is {x, y, z} or single number.

Documentation Local »
Default Definitions SyntaxInformation[RescaleTracts] = {ArgumentsPattern → {_, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`RescaleTracts

Symbol

SeedDensityMap[seeds, vox, dim] makes a seed density map based on the seed locations.

Documentation Local »
Default Definitions SyntaxInformation[SeedDensityMap] = {ArgumentsPattern → {_, _, _}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`SeedDensityMap

Symbol

SegmentTracts[tracts, segments, vox, dim] segments the tracts based on segments.

Documentation Local »
Default Definitions SyntaxInformation[SegmentTracts] = {ArgumentsPattern → {_, _, _, _, OptionsPattern[]}}

Options › FiberLengthRange → {15, 500} … (5 total)
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`SegmentTracts

Symbol

TractAngle[tracts] calculates the angle of each tract segment with the z–plane.

TractAngle[tracts, v1] calculates the angle of each tract segment with the plane normal to v1.

TractAngle[tracts, {v1, v2}] calculates the elevation and azimuth angle of each tract segment with the plane normal to v1 and the azimuth angle in that plane relative to v2.

Documentation Local »
Default Definitions SyntaxInformation[TractAngle] = {ArgumentsPattern → {_, _.}}

Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractAngle

Symbol

TractAngleMap[tracts, vox, dim] makes a tract angle map based on the tracts angles with the z–plane.

TractAngleMap[tracts, v1, vox, dim] makes a tract angle map based on the tracts angles with the plane normal to v1.

TractAngleMap[tracts, {v1, v2}, vox, dim] makes a tract angle map based on the tracts elevation angles with the plane normal to v1 and the azimuth angle in that plane relative to v2.

Documentation Local »
Default Definitions SyntaxInformation[TractAngleMap] = {ArgumentsPattern → {_, _, _, _.}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractAngleMap

Symbol

TractCurvature[tracts] calculates the curvature of each tract segment.

Documentation Local »
Default Definitions SyntaxInformation[TractCurvature] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractCurvature

Symbol

TractCurvatureMap[tracts, vox, dim] makes a tract curvature map based on the tracts curvature.

Documentation Local »
Default Definitions SyntaxInformation[TractCurvatureMap] = {ArgumentsPattern → {_, _, _}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractCurvatureMap

Symbol

TractDensityMap[tracts, vox, dim] makes a tract density map based on the tracts vertices.

Documentation Local »
Default Definitions SyntaxInformation[TractDensityMap] = {ArgumentsPattern → {_, _, _, OptionsPattern[]}}
Options NormalizeDensity → True
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractDensityMap

Symbol

TractLength[tracts] calculates the length of each tract.

Documentation Local »
Default Definitions SyntaxInformation[TractLength] = {ArgumentsPattern → {_}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractLength

Symbol

TractLengthMap[tracts, vox, dim] makes a tract length map based on the tracts lengths.

Documentation Local »
Default Definitions SyntaxInformation[TractLengthMap] = {ArgumentsPattern → {_, _, _}}
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`TractLengthMap

Options

Symbol

Boxed is an option for Graphics3D that specifies whether to draw the edges of the bounding box in a three-dimensional picture.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Boxed

Symbol

ColorFunction is an option for graphics functions that specifies a function to apply to determine colors of elements.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ColorFunction

Symbol

ContourOpacity is an option for PlotContour and PlotSegmentations. It specifies the opacity of the contour.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`PlottingTools`ContourOpacity

Symbol

FiberAngle is an option for FiberTractography and specifies the allowed angle change per tract step.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`FiberAngle

Symbol

FiberLengthRange is an option for FiberTractography and specifies the allowed tract range.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`FiberLengthRange

Symbol

FittingOrder is an option for FitTracts. It specifies the polynomial order of the function to fit the tract.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`FittingOrder

Symbol

FitTractSegments is an option for SegmentTracts. If set True the segmented tracts are fitted with FitTracts.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`FitTractSegments

Symbol

ImageSize is an option that specifies the overall size of an image to display for an object.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`ImageSize

Symbol

InterpolationOrder is an option for Interpolation, as well as ListLinePlot, ListPlot3D, ListContourPlot, and related functions, that specifies what order of interpolation to use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`InterpolationOrder

Symbol

MaxSeedPoints is an option for FiberTractography and defines the maximum number of seed points to be used.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`MaxSeedPoints

Symbol

MaxTracts is an option for PlotTracts. It specifies how many tracts are plotted.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`MaxTracts

Symbol

Method is an option for various algorithm–intensive functions that specifies what internal methods they should use.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Method

Symbol

Monitor[*expr*, *mon*] generates a temporary monitor cell in which the continually updated current value of *mon* is displayed during the course of evaluation of *expr*.

Documentation Local » | Web »
Attributes {HoldAll, Protected, ReadProtected}
Full Name System`Monitor

Symbol

NormalizeDensity is an option for TractDensityMap. If set True the tract density is normalized, if False then it is the true tract count.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`NormalizeDensity

Symbol

OutputForm[*expr*] prints as a two-dimensional representation of *expr* using only keyboard characters.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`OutputForm

Symbol

Parallelization is an option for Compile that specifies whether it should create a compiled function that could run in parallel.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`Parallelization

Symbol

PerformanceGoal is an option for various algorithmic and presentational functions that specifies what aspect of performance to try to optimize with Automatic settings for options.

Documentation Local » | Web »
Attributes {Protected}
Full Name System`PerformanceGoal

Symbol

StepSize is an option for FiberTractography and defines the tractography step size.

Documentation Local »
Attributes {Protected, ReadProtected}
Full Name QMRITools`TractographyTools`StepSize

Symbol

StopThreshold is an option for FiberTractography and defines the stop threshold which is a value between 0 and 1.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`StopThreshold

Symbol

TensorFlips is an option for FiberTractography and specifies if the tensor orientation is flipped, see FlipTensorOrientation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TensorFlips

Symbol

TensorPermutations is an option for FiberTractography and specifies if the tensor orientation is permuted, see FlipTensorOrientation.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TensorPermutations

Symbol

TractColoring is an option for PlotTracts and sets how the tracts are colored. Values can be "Direction", "Length", "Angle", {par}, or RGBColor[].

For "Length", "Angle", {par} it can be defined in the form {..., {min, max}} where the {min, max} specifies the range of the color function.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractColoring

Symbol

TractMethod is an option for FiberTractography and FindTensorPermutation and defines the integration method of each tract step.

Values can be "Euler", "RK2" (midpoint, also "RK" or "RungeKutta"), "Heun" (also "RK2T"), "RK3" (strong stability preserving,

also "SSPRK3" or "RungeKutta3"), "Heun3" (also "RK3Heun") and "RK4" (also "RungeKutta4").

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractMethod

Symbol

TractMonitor is an option for FiberTractography. When set True it prints the progress.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractMonitor

Symbol

TractReduction is an option for PlotTracts. Value can be an Integer > 0, which determines with which factor the tract coordinates are sub sampled.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractReduction

Symbol

TractScaling is an option for PlotTracts. The value can be "World" or "Voxel", if the value is "World" the tracts are in mm else in voxel coordinates.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractScaling

Symbol

TractSize is an option for PlotTracts. When tubes are used it specifies the tube width.

Documentation Local »

Attributes {Protected, ReadProtected}

Full Name QMRITools`TractographyTools`TractSize